QUANTUM IAS & PCS

TATTVA SERIES

UPPSC PRELIMS | FAST REVISION

General Science

PYQ-Based Revision Pointers

Chapters 1 to 29

Every pointer comes from an actual UPPSC question · Frequency marks · Year tags
Asked-Most Boxes · Trap Alerts · Quick Tables · Last-Minute 10

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How to Use This Book

Every pointer is a real question. Each pointer is the answer to a UPPSC question asked between 1990 and 2025, written as a fact you can revise in seconds, with the related fact you need for the next variation.

Frequency marks. ■ and a red count (e.g. 4×) mark a fact asked 3 or more times. • and a gold count mark facts asked once or twice. The years follow; "one-liner" means a short starred question in the source with no year given.

Asked Most box. Each chapter opens with the facts UPPSC repeats most. Revise these first.

Diagrams. Flowcharts and diagrams turn sequences (timelines, procedures, processes, river courses) into one picture.

Trap Alerts. The places where the obvious answer is wrong or two facts are commonly swapped.

Last-Minute 10. The ten lines to read on the morning of the exam.

Pairs with the Masterbook. Use this book for revision after the TATTVA PYQ Masterbook, which has the full questions, options and explanations.

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Chapter 1

Units and Measurement

Built from 27 UPPSC questions (1990–2023). Focus: SI units and unit-quantity matching, light year and parsec, small lengths, odd units such as barrel, cusec and Dobson.

ASKED MOST · REVISE FIRST

- **Unit-quantity matching**: **joule-work, newton-force, watt-power, pascal-pressure, newton second-impulse** 6× 1990, 2001, 2002, 2005, 2006, 2013
- **Light year** and **parsec** are units of **distance**, not time 6× 1997, 2010, 2010, 2013, 2023, one-liner
- **Celsius and centigrade** are temperature scales, **not units of heat** 3× 2001, 2004, 2017
- Small lengths: **1 nm = 10^{-7} cm = 10^{-9} m**; **1 micron = 10^{-4} cm**; smallest: **fermi** 4× 2005, 2011, 2013, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

1.1 Work, Power and Energy Units

- Unit of power: **watt** 1× 2010
- Unit of work: **joule** 1× 1996
- Megawatt measures power **generated** 1× 1998

1.2 Astronomical and Small Lengths

- **Light year**: distance light travels in **one year**; a unit of **distance** 3× 2010, 2013, one-liner
- **Parsec**: unit of **distance** 1× 1997
- Distance at which Earth's orbit radius subtends **1 arc second**: **1 parsec** 1× 2023
- Multiply by 0.39 to get inches: **centimetre** 1× 2010
- **1 nanometre = 10^{-7} cm** 1× 2013
- Correct: **1 metre = 10^{9} nanometre** 1× 2023
- 5 feet in nanometres: none of the options (about 1.524 x 10^{9} nm) 1× 2023
- **1 micron = 10^{-4} cm** 1× 2011
- Smallest unit of length: **fermi** (femtometre, 10^{-15} m) 1× 2005

Unit	Value	Use
Light year	9.46 x 10^{12} km	Stars, galaxies
Astronomical unit	About 1.5 x 10^{8} km	Solar system
Parsec	3.26 light years	Largest of the three
Angstrom	10^{-10} m	Wavelength, atomic size
Nanometre	10^{-9} m	Nanotechnology
Micron	10^{-6} m (10^{-4} cm)	Cells, dust
Fermi	10^{-15} m	Nuclear size

1.3 SI Units and Matching

- **Acceleration-m/s^2, force-newton, work-joule, impulse-newton second** 2× 2001, 2005
- Wrong SI pair: **pressure-dyne** (dyne is a CGS unit of force) 1× 2013
- **Watt-power, knot-speed of a ship, nautical mile-navigation, calorie-heat** 1× 2002
- **Joule-work, ampere-current, watt-power, volt-potential, calorie-heat** 1× 1990
- Wrong pair: **light year-time** (it is distance) 1× 2010
- **Mach-high speed, angstrom-wavelength, pascal-pressure, joule-energy** 1× 2006
- **Decibel**: **sound intensity** 1× 2011
- Wrong pair: **Celsius-unit of heat** 2× 2001, 2004
- Not a unit of heat: **centigrade** 1× 2017
- Wrong SI pair: **radioactivity-curie** (SI unit is the **becquerel**) 1× 2022
- Electric potential is a **scalar** quantity (only 1) 1× 2021

Quantity	SI unit	Older units
Force	Newton	Dyne, kg-wt
Work, energy, heat	Joule	Erg, calorie (4.18 J), kWh
Power	Watt	Horsepower (746 W)
Pressure	Pascal	Bar, atmosphere, mm of Hg
Temperature	Kelvin	Celsius, Fahrenheit
Radioactivity	Becquerel	Curie
Power of lens	Dioptre	-

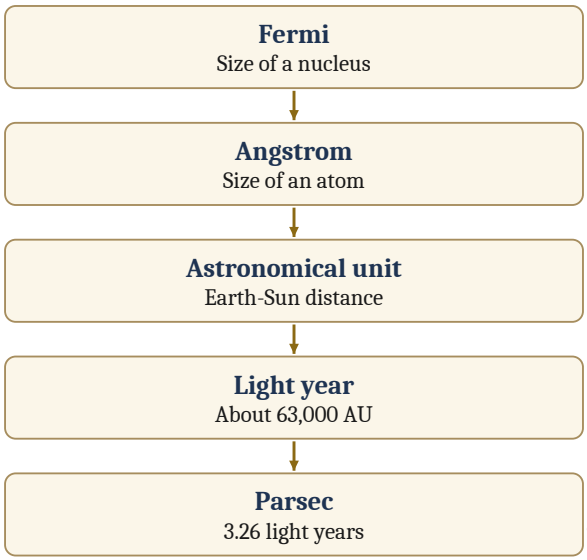
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Quantity	SI unit	Older units
Illuminance	Lux	Phot

1.4 Odd Units

- **1 barrel** of oil = **159 litres** 1× 2009
- **Cusec-rate of flow, byte-computer, Richter-earthquake, bar-pressure** 1× 2008
- **Dobson unit**: **ozone** concentration 1× 2021

DISTANCE UNITS, SMALL TO LARGE



TRAP ALERT

- **Parsec > light year > AU**; all three measure **distance**.
- **1 micron = 10^{-4} cm**, not 10^{-6} cm (that is in metres).
- SI unit of radioactivity = **becquerel**; curie is older. Dyne is force, not pressure.
- **Knot** = speed (nautical mile per hour); **nautical mile** = distance at sea.

UP LINK

UP link: **Indian Standard Time** is reckoned from **82.5 E**, which passes near **Mirzapur** (UP); the time signal itself is kept by the **National Physical Laboratory**, New Delhi.

LAST-MINUTE 10

1. Watt = power; joule = work; newton = force; pascal = pressure
2. Light year, parsec = **distance**; parsec = 3.26 light years
3. **1 nm = 10^{-9} m = 10^{-7} cm**; 1 micron = 10^{-4} cm
4. Fermi = smallest length unit
5. Impulse = **newton second**
6. Decibel = sound intensity; Mach = high speed
7. Celsius, centigrade = temperature scales
8. Becquerel = SI unit of radioactivity
9. 1 barrel = **159 litres**; cusec = rate of flow
10. Dobson unit = ozone

Chapter 2

Measuring Instruments and Scales

Built from 31 UPPSC questions (1990–2021). Focus: instrument-use matching, weather instruments, pyrometer and pyrheliometer, Richter scale and seismograph, SONAR and RADAR.

ASKED MOST · REVISE FIRST

- **Instrument matching**: **anemometer-wind, barometer-pressure, hygrometer-humidity, ammeter-current, seismograph-earthquake** 5× 1992, 2001, 2008, 2012, 2015
- **Radar**: finds objects with **radio waves**; **SONAR**: finds submerged objects with **ultrasonic** waves 7× 1996, 2000, 2001, 2004, 2008, 2016, one-liner
- **Pyrometer** = very **high temperature** 6× 1998, 2004, 2008, one-liner, one-liner, one-liner
- **Hygrometer** = **relative humidity** 3× 1995, 1996, 2014
- **Richter scale** and **seismograph** = earthquakes 6× 1990, 2004, 2014, 2015, one-liner, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

2.1 Navigation and Detection

- Locating submerged objects: **SONAR** 2× 2000, 2016

- SONAR uses **ultrasonic waves** 1× one-liner
- **Radar**: presence and location of objects with **radio waves** 4× 1996, 2001, 2004, 2008

2.2 Weather Instruments

- Wind velocity: **anemometer** 1× 2016
- Wrong pair: **tacheometer-pressure difference** (it is for surveying) 1× 1997
- **Pyrometer**: **high temperature** 4× 1998, 2004, 2008, one-liner
- Pyrometer is a **radiation thermometer**; for 2000 °C: **total radiation pyrometer** 2× one-liner, one-liner
- **Pyrheliometer**: **solar radiation** 1× 2015
- **Manometer**: **gas pressure** 1× 1990
- **Ammeter-current, hygrometer-humidity, spring balance-weight, barometer-pressure** 1× 2001
- Wrong pair: **ammeter-electric power** (ammeter measures current) 1× 2015
- Atmospheric pressure: **barometer** 1× 2010
- Relative humidity: **hygrometer** 3× 1995, 1996, 2014
- **Ringelmann scale**: density of **smoke** 1× 2021
- **Stethoscope-heart sound, sphygmomanometer-blood pressure, caratometer-gold purity, luxmeter-light** 1× 2008
- Blood pressure: **sphygmomanometer** 1× one-liner

Instrument	Measures
Anemometer	Wind speed
Barometer	Atmospheric pressure
Hygrometer	Relative humidity
Hydrometer	Density of liquids
Lactometer	Purity of milk
Pyrometer	Very high temperature
Pyrheliometer	Solar radiation
Manometer	Gas pressure
Altimeter	Height
Tacheometer	Survey distance and elevation

2.3 Temperature and Light

- No negative values: **Kelvin** scale 1× 2017
- Correct pair: **thermoresistor-electronic thermometer** 1× 2016
- Intensity of light: **luxmeter** 1× 2004

2.4 Earthquakes, Radiation and Others

- Earthquake intensity: **Richter scale** 2× 2004, 2015
- Richter scale: amplitude of waves, i.e. **magnitude** 1× one-liner
- Seismometer device: **seismograph** 3× 1990, 2014, one-liner
- **Earthquake-seismograph, height-altimeter, current-ammeter, resistance-ohm** 1× 1992
- **Anemometer-wind, seismograph-earthquakes, barograph-pressure, hygrometer-humidity** 1× 2012
- Wrong pair: **seismometer-curvature of surface** 2× 1999, 2010
- Radioactivity: **Geiger counter** 1× 2010
- **Phonometer**: sound intensity, not in the options (*) 1× 2014
- Lie detector: **polygraph** 1× 2013

Instrument	Use
Geiger-Muller counter	Radioactivity
Seismograph	Records earthquake waves
Crescograph	Plant growth (J.C. Bose)
Kymograph	Muscle movement, pulse (graph)
Polygraph	Lie detection
Sphygmomanometer	Blood pressure
Audiometer	Hearing
Luxmeter	Light intensity
Caratometer	Purity of gold
Odometer	Distance travelled by a vehicle

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TRAP ALERT

- **Hygrometer** = humidity; **hydrometer** = liquid density.
- **Pyrometer** = high temperature; **pyrheliometer** = solar radiation.
- **Barometer** = air pressure; **manometer** = gas pressure; **ammeter** = current, not power.
- Kelvin has no negative values; SONAR = sound, RADAR = radio waves.

UP LINK

UP link: IMD's **Meteorological Centre, Lucknow** runs the UP network of rain gauges, automatic weather stations and a **Doppler weather radar** (soft detail); the **seismograph** network covers UP because its Terai belt lies in seismic zone IV.

LAST-MINUTE 10

1. Anemometer = wind; barometer = air pressure
2. Hygrometer = humidity; hydrometer = liquid density
3. Pyrometer = high temperature; pyrheliometer = solar radiation
4. Manometer = gas pressure; ammeter = current
5. SONAR = ultrasonic, submerged objects; RADAR = radio waves
6. Richter = magnitude; seismograph records quakes
7. Geiger counter = radioactivity; polygraph = lie detector
8. Sphygmomanometer = BP; caratometer = gold
9. Luxmeter = light; Ringelmann = smoke
10. Kelvin scale has no negative values

Chapter 3

Mechanics and Motion under Gravity

Built from 27 UPPSC questions (1990–2024). Focus: vectors, inertia and friction, energy conversion, weight at the poles, pendulum clocks, inverse-square gravity, satellites and escape velocity.

ASKED MOST · REVISE FIRST

- ■ Weight is **maximum at the poles** (g is largest there) **3×** 2006, 2009, 2015
- • Heavy and light balls dropped together **reach the ground together** **2×** 1992, 2015
- • Pendulum clocks **lose time in summer**: the pendulum lengthens **2×** 1994, 2012
- • **Inertia of rest**: a passenger's head falls back as the train starts **1×** 2010
- • Satellite stays in orbit by **centripetal force** supplied by **Earth's gravity** **2×** 2006, 2013

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

3.1 Quantities, Motion and Inertia

- Vector quantity: **displacement** **1×** 2014
- Man inhaling deeply on a sensitive balance: reading **increases** (per key) **1×** 2013
- Acceleration: $\mathbf{a = (v - u)/t}$ **1×** 2003
- Average speed for two equal halves: $\mathbf{2v_1v_2/(v_1 + v_2)}$ **1×** 2024
- Washing machine dryer works on **centrifugation** **1×** 2004
- Head leans back as a train starts: **inertia of rest** **1×** 2010
- Hard to walk on ice: **less friction** than on a road **1×** 1994
- Harder to start a loaded cart than keep it moving: **less friction once moving** (static > kinetic) **1×** 2013

3.2 Energy

- Not from the Sun: **nuclear** (and **geothermal**) energy **1×** 2022

- Wind power converts **kinetic energy** into electricity 1× 2016
- Non-conventional sources are **renewable** 1× 2023
- **Light to electric-solar cell, electric to sound-loudspeaker, mass to heat-nuclear reactor, chemical to heat and light-fuel combustion** 1× 2002
- Kinetic energies 4 : 9 (same mass): velocities **2 : 3** 1× 2017

Device	Energy change
Solar cell	Light to electrical
Loudspeaker	Electrical to sound
Microphone	Sound to electrical
Nuclear reactor	Mass to heat
Dynamo / generator	Mechanical to electrical
Electric motor	Electrical to mechanical
Battery	Chemical to electrical

3.3 Gravity, Weight and Mass

- Leaning Tower of Pisa stands: the vertical through its **centre of gravity falls within the base** 1× 2009
- 10 kg and 1 kg balls dropped from 20 m: **land together** 2× 1992, 2015
- Weight of a body: **maximum at the poles** 3× 2006, 2009, 2015
- Apparent weight highest in **hydrogen** (least buoyancy among the options) 1× 2017
- Weight feels more in a lift **accelerating upward** 1× 1990
- Pendulum clocks lose time in summer: **longer pendulum, longer period** 2× 1994, 2012
- Earth-Sun distance doubled: gravity becomes **one-fourth** 1× 2009
- We do not feel Earth's orbital speed: our **relative speed** is zero 1× 2013
- If gravity vanished: **weight zero, mass unchanged** 1× 2012

WEIGHT VERSUS MASS



3.4 Satellites and Escape Velocity

- Astronauts cannot stand erect: key says **no gravity**; the real reason is **weightlessness** in free fall 1× 2007
- Apple released from an orbiting spaceship: **moves along with it** 1× 2010
- Geosynchronous satellite: centripetal force from **Earth's gravity** 1× 2006
- Satellite kept in orbit by **centripetal force** 1× 2013
- No atmosphere on the Moon: **escape velocity is less than** the gas molecules' rms speed 1× 2012

Fact	Value
Earth's escape velocity	11.2 km/s
Moon's escape velocity	About 2.4 km/s
Orbital speed near Earth	About 7.9 km/s
Geostationary height	About 35,786 km; period 24 h
g on the Moon	About one-sixth of Earth's

TRAP ALERT

- **Displacement, velocity, force, weight** = vectors; distance, speed, work = scalars.
- Average speed over equal distances is the **harmonic**, not simple, mean.
- Gravity follows the **inverse square**: double the distance, one-fourth the force.
- Orbit "weightlessness" is **free fall**, not zero gravity.

UP LINK

UP link: **IIT Kanpur** built and launched **Jugnu** (2011), a nano-satellite on PSLV-C18, among the first student satellites from India.

LAST-MINUTE 10

1. Displacement = vector; a = (v - u)/t
2. Inertia of rest: head goes back as the train starts
3. Washing machine = centrifugation; ice = low friction
4. Nuclear and geothermal energy are not from the Sun
5. Weight **maximum at poles**; mass never changes
6. Heavy and light balls fall together
7. Pendulum clock **loses** time in summer
8. Double distance = **1/4** gravity
9. Satellite orbit = centripetal force by gravity
10. Escape velocity **11.2 km/s**; Moon has no air

Chapter 4

Properties of Matter

Built from 23 UPPSC questions (1992–2023). Focus: surface tension, pressure in bubbles and at height, water's maximum density at 4 °C, Archimedes' principle and flotation.

ASKED MOST · REVISE FIRST

- ■ **Surface tension**: spherical raindrops, floating needle, oil spreading on water **4×** 1994, 2005, 2010, 2012
- ■ Water is **densest at 4 °C** (277 K); fish survive under ice **5×** 1997, 2005, 2008, one-liner, one-liner
- ■ A ship **rises a little** on entering the sea; sea water is denser **4×** 1992, 2002, 2004, one-liner
- • Pressure inside a soap bubble is **more than** outside **2×** 1995, 2014

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

4.1 Elasticity and Surface Tension

- Rubber's Young's modulus is **less** than steel's; coil stretching depends on **shear modulus** (only 2) **1×** 2023
- Raindrops are spherical due to **surface tension** **1×** 2005
- Oil spreads on water: its surface tension is **less** than water's **2×** 1994, 2010
- Pressure inside a soap bubble: **more than atmospheric** **2×** 1995, 2014
- Charging a soap bubble: its **radius increases** **1×** 2019
- Iron needle floats due to **surface tension** **1×** 2012

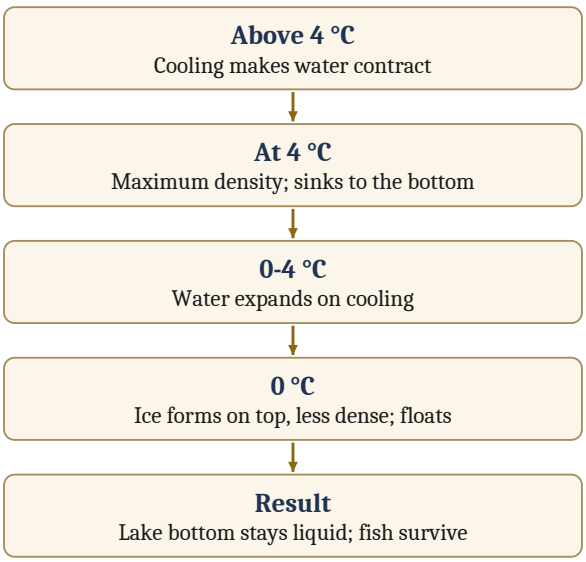
4.2 Pressure and Flow

- Pen leaks in a plane: **lower air pressure** **1×** 1992
- Sudden fall in barometer reading: **stormy** weather **1×** 1996
- Streamlined flow in a pipe: **slowest near the walls, fastest along the axis** **1×** 2003

4.3 Density and Water at 4 °C

- More of a substance does not change its **density** **1×** 1992
- Cooling water from 9 °C to 3 °C: volume **first decreases, then increases** **1×** 1997
- Maximum density of water: **4 °C** (277 K) **2×** 2008, one-liner
- Aquatic life survives freezing: water is **densest at 4 °C**, so liquid water stays below the ice **2×** 2005, one-liner

ANOMALOUS EXPANSION OF WATER



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4.4 Archimedes' Principle and Flotation

- Iron nail floats on mercury, sinks in water: iron is **denser than water, lighter than mercury** 1× 1994
- Needle sinks, iron ship floats: needle's density is **greater than the water it displaces** 1× 2004
- Bucket feels **heavier above the water surface** 1× 2009
- 200 g of water in a weightless balloon weighs **zero** in water 1× 2014
- Iceberg above the sea: about **1/9** 1× 2005
- Easier to swim in the sea: sea water is **salty** (denser) 1× 2002

■ Ship from river to sea: it **rises a little**; sea water is denser (A and R true, R explains A) 3× 1992, 2004, one-liner

Principle	Everyday example
Surface tension	Raindrops, floating needle, soap bubbles
Capillarity	Lamp wick, ink in blotting paper, sap in plants
Archimedes	Ships, submarines, hydrometer, icebergs
Bernoulli	Aircraft lift, roofs blown off, atomiser
Pascal's law	Hydraulic lift, brakes, press

TRAP ALERT

- Water's density peak is **4 °C**, not 0 °C; ice is **less** dense and floats.
- Soap bubble pressure is **higher** inside; oil spreads because its surface tension is **lower**.
- Steel is **more** elastic (higher Young's modulus) than rubber.

UP LINK

UP link: the **Ganga** at Prayagraj meets sea water nowhere, so river barges on **NW-1** float lower than they would at sea (fresh water is less dense), a direct case of Archimedes' principle.

LAST-MINUTE 10

1. Raindrops spherical = surface tension
2. Needle floats = surface tension
3. Oil spreads: lower surface tension
4. Bubble pressure **more** inside
5. Pen leaks in a plane = low pressure
6. Barometer falls fast = storm
7. Water densest at **4 °C** (277 K)
8. 1/9 of an iceberg above water
9. Ship rises in the sea; sea swimming easier
10. Iron floats on mercury, sinks in water

Chapter 5

Light and Optics

Built from 60 UPPSC questions (1990–2023). Focus: total internal reflection, scattering and red signals, mirrors in vehicles, myopia and hypermetropia, the eye's sensitivity and the speed of light.

ASKED MOST · REVISE FIRST

■ **Total internal reflection**: **mirage, diamond's sparkle, optical fibre** 11× 1995, 1996, 2008, 2010, 2012, 2012, 2013, 2014, 2015, 2017, one-liner

■ **Scattering**: blue sky, red sunset; **red** for danger as it scatters **least** 6× 2007, 2013, one-liner, one-liner, one-liner, one-liner

■ **Convex mirror** for rear view; **concave mirror** for searchlights and headlights 6× 1991, 2009, 2010, 2011, 2015, 2023

■ **Myopia** is corrected with a **concave lens**; hypermetropia with a **convex** lens 5× 2002, 2008, 2010, 2011, 2013

■ Air bubble in water = **concave lens**; parallel mirrors give **infinite** images 7× 1994, 2002, 2008, 2010, 2013, 2013, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

5.1 Nature and Speed of Light

- Visible light: about **390-780 nm** 1× 2005
- Eye most sensitive at **5500 Å** (yellow-green) 2× 2008, 2014

- Not correct: eye is most sensitive to **red** 1× 2021
- Light seems to go straight: its **wavelength is very small** 1× 2014
- Highest velocity: **light**; light is faster than sound 2× 2002, 2018
- Light is fastest in a **vacuum** 1× 2014
- Sunlight reaches Earth in about **8 minutes** (8 min 20 s) 1× 2006
- Not luminous: **Moon** (it reflects sunlight) 1× 2023

5.2 Refraction and Total Internal Reflection

- Unchanged when light changes medium: **frequency** 1× 2003
- Pool looks shallower: **refraction** 1× 2023
- Sun and Moon look oval near the horizon: **refraction** 1× 2008
- Optical fibre: **total internal reflection** 3× 2008, 2010, one-liner
- Optical fibre: high capacity, low power, **no radio interference** (1, 2, 3) 3× 2001, 2002, 2003
- **Mirage**: **total internal reflection** 4× 1995, 2012, 2013, 2014
- Diamond shines: **total internal reflection** 4× 1996, 2012, 2015, 2017
- 3-D pictures: **holography** 2× 1990, 2008

Phenomenon	Cause
Mirage, diamond sparkle, optical fibre	Total internal reflection
Shallow-looking pool, bent stick, twinkling stars	Refraction
Rainbow, prism spectrum	Dispersion (with refraction and reflection)
Blue sky, red sunset	Scattering
Colours of soap film and oil	Interference

5.3 Colour, Scattering and Dispersion

- Eclipse retinal burn: key says **ultraviolet** (visible blue light and infrared also harm) 1× 2003
- A star's colour shows its **temperature** 3× 2003, 2005, 2008
- Photosynthesis with invisible light: some **bacteria** 1× 2013
- Photosynthesis: **light to chemical** energy 1× 1998
- Diffusion of light in air: **dust particles** 1× 2021
- Colour is set by **wavelength** 2× 2012, 2014
- Shortest wavelength in the visible spectrum: **violet** 1× one-liner
- Least deviated by a prism: **red** 1× 2016
- Red traffic signals: **longest wavelength** 1× 2013
- Red for danger: it **scatters least** (A and R true, R explains A) 2× 2007, one-liner
- Blue sky and red sunset: **scattering** (short wavelengths scatter more) 3× one-liner, one-liner, one-liner
- Sky seen by an astronaut or from great height: **black** 2× 2005, one-liner
- Colours in sunlight: **7** 1× 2015
- Grass looks green: it **reflects green** light 1× 2017
- Expanding universe: **Doppler effect** (red shift) 1× 2010
- Black hole unseen: light cannot escape its gravity (A and R true, R explains A) 1× 2001
- Middle colour of a rainbow: **green**; extremes: **violet and red** 2× 2012, one-liner

VIBGYOR: WAVELENGTH AND SCATTERING



5.4 Mirrors and Lenses

- Air bubble in water acts as a **concave lens** 4× 2002, 2008, 2010, 2013

- Images between two parallel mirrors: **infinite** 3× 1994, 2013, 2015
- Mirror to see your full image: **half your height**; for 1.50 m, **0.75 m** 4× 2010, 2011, 2014, 2014
- Radius of curvature of a plane mirror: **infinity** 1× 2015
- Not correct: doctors use a **convex** mirror for ENT checks (they use **concave**) 1× 2010
- Searchlight and car headlight: **concave mirror** 2× 2009, 2023
- Rear-view mirror: **convex mirror** 4× 1991, 2010, 2011, 2015
- Power of sunglasses: **0 dioptre** 1× 1995
- Astronomical telescope image: **virtual and magnified** 1× 2013
- Equal convex and concave lenses together: a **plane glass sheet** 1× 2017

Mirror / lens	Use
Concave mirror	Searchlight, headlight, shaving, dentist and ENT, solar cooker
Convex mirror	Rear view, street-corner safety
Plane mirror	Periscope, kaleidoscope
Concave lens	Myopia
Convex lens	Hypermetropia, magnifier, camera
Cylindrical lens	Astigmatism

5.5 The Human Eye

- Image on the retina: **smaller and inverted** (real) 2× 1995, 2004
- Retina is like a camera's **film** 1× 2011
- Eye focuses by **changing the lens's convexity** (accommodation) 1× 2014
- Least distance of distinct vision: **25 cm** 1× 2008
- Not true of hypermetropia: corrected by a **concave** lens (it needs convex) 1× 2010
- Myopia corrected by a **concave lens** 4× 2002, 2008, 2011, 2013

TRAP ALERT

- Mirage is **total internal reflection** (per keys), not simple reflection.
- Frequency never changes in refraction; speed and wavelength do.
- **Convex** = rear view; **concave** = headlight; ENT doctors use **concave**.
- Myopia = **concave**; hypermetropia and old-age presbyopia = **convex**.

UP LINK

UP link: the **Aryabhatta Research Institute of Observational Sciences (ARIES)** is in Nainital (Uttarakhand, carved from UP), and the **Indian Institute of Astrophysics** runs telescopes elsewhere; in UP, **IIT Kanpur** and **BHU** run optics and photonics labs (soft detail).

LAST-MINUTE 10

1. Visible light **390-780 nm**; eye most sensitive at **555 nm** (yellow-green)
2. Light fastest in vacuum; Sun to Earth **8 min 20 s**
3. Frequency unchanged in refraction
4. Mirage, diamond, optical fibre = **TIR**
5. Blue sky = scattering; red signal = least scattered
6. Sky black from space; 7 colours; green in middle
7. Air bubble in water = **concave lens**
8. Full-length mirror = **half your height**; parallel mirrors = infinite images
9. Convex mirror = rear view; concave = headlight
10. Myopia = **concave lens**; near point 25 cm

Chapter 6

Heat and Thermodynamics

Built from 40 UPPSC questions (1993–2025). Focus: latent heat, Celsius-Fahrenheit conversion, boiling point and pressure, pressure cookers, evaporative cooling, humidity, light clothes and cloudy nights.

ASKED MOST · REVISE FIRST

- ■ **-40** is the same on Celsius and Fahrenheit; **F = (9/5)C + 32** 6× 2003, 2015, 2015, 2016, 2019, 2024
- ■ **Pressure cooker**: higher pressure raises the **boiling point** 3× 1995, 2007, 2015
- • Water boils **below 100 °C** on hills: **lower air pressure** 2× 1994, 2023
- ■ **Light/white clothes** in summer: they **reflect** heat and absorb little 3× 1993, 2007, 2016
- ■ **Earthen pitcher** cools by **evaporation**; sweat cools the body 3× 2012, 2019, 2025

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

6.1 Latent Heat and Temperature Scales

- Steam burns more than boiling water: steam carries **latent heat** 1× 1993
- **F = (9/5)C + 32** 1× 2024
- Celsius and Fahrenheit read the same at **-40** 2× 2003, 2015
- 98.4 °F = **36.89 °C** 1× 2015
- 200 °F = **93.3 °C** 1× 2019
- Averages follow the same rule: $\mathbf{F_{av} = (9/5)C_{av} + 32}$ 1× 2016
- Alcohol thermometers in cold countries: **lower freezing point** 1× 1997
- Antifreeze in car engines: **ethylene glycol** 2× 2008, 2010

Point	Celsius	Fahrenheit	Kelvin
Absolute zero	-273.15	-459.67	0
Same reading	-40	-40	233
Water freezes	0	32	273
Body temperature	37	98.6	310
Water boils (sea level)	100	212	373

6.2 Expansion and Cooling Devices

- Heating expands a body and **lowers its density** 1× 2006
- Glass rod in steam: width also **increases** 1× 1994
- Ice melting: volume **decreases** 1× 2012
- ■ **Thermostat**: keeps **temperature** steady automatically 3× 1993, 1996, one-liner
- Safest fridge temperature for food: **4 °C** 1× 2007
- Soggy biscuits turn crisp in a fridge: **low humidity** inside 1× 2009
- Ice cools a drink by taking **latent heat** (A and R true, R explains A) 1× 2009
- Ice in the freezer cuts cooling: ice is a **poor conductor** (A and R true, R explains A) 1× 2010
- Not true: AC and cooler **both keep humidity** (a cooler adds it, an AC removes it) 1× 2007

6.3 Humidity and Evaporation

- Relative humidity falls as temperature rises; evaporation raises absolute humidity (A and R true, R does not explain A) 1× 2024
- Clothes dry fastest at **20% RH and 60 °C** 1× 2008
- Earthen pitcher keeps water cold by **evaporation** 1× 2012
- Sweating depends on **temperature and humidity** both 1× 2019
- Sweat evaporation removes heat; humidity adds discomfort (A and R true, R does not explain A) 1× 2025
- Humid heat felt in **humid weather** 1× 2003
- **Xeric** condition: **low humidity** 1× 2014

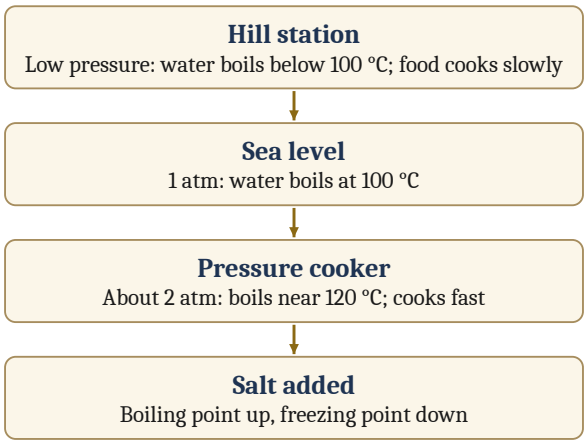
6.4 Pressure and Boiling Point

- We do not feel air pressure: **blood pressure** slightly exceeds it 1× 2010
- Water boils below 100 °C at height: **air pressure falls** 1× 1994
- Boiling point falls with height; pressure **decreases** with height (A true, R false) 1× 2023

QUANTUM IAS & PCS

- Pressure cooker cooks fast: water boils at a **higher temperature** 2× 2007, 2015
- Cooker's top temperature depends on the **vent area and weight** 1× 1995
- Salt in water: boiling point **rises**, freezing point **falls** 1× 2006
- Bottle bursts on freezing: water **expands** as it freezes 1× 1995

PRESSURE AND BOILING POINT



6.5 Conduction, Convection and Radiation

- Thermos flask: no heat loss by **conduction and convection** (both; silvered walls stop radiation) 1× 2019
- Best umbrella: **white on top, black below** 1× 1993
- White clothes in summer: they **reflect** heat 1× 2007
- Light clothes in summer: light colours absorb little heat (A and R true, R explains A) 1× 2016
- Cloudy nights are warmer: clouds **stop heat radiating** away 1× 2008
- Reflects most sunlight: **fresh snow** (highest albedo) 1× 2021
- Cold storage lengthens fruit life: **lower respiration** 1× 2021
- Wool keeps us warm, but it works by **trapping air**; the protein reason is false (A true, R false) 1× 2001

TRAP ALERT

- Steam burns more because of **latent heat**, not higher temperature alone.
- On hills water boils **below** 100 °C; in a cooker **above** 100 °C.
- Salt raises boiling point but **lowers** freezing point.
- Ice melting **shrinks** in volume; water freezing **expands**.

UP LINK

UP link: Kanpur, Lucknow and Prayagraj often cross **45 °C** in May-June, and the dry **loo** wind lowers relative humidity, which is why earthen pots (matkas) from **Khurja** and Nizamabad keep water cool by evaporation.

LAST-MINUTE 10

1. Steam burns more = latent heat
2. **F = (9/5)C + 32**; equal at **-40**
3. Antifreeze = ethylene glycol; alcohol thermometer for cold lands
4. Fridge food at **4 °C**; ice melting reduces volume
5. Matka cools by **evaporation**; clothes dry at low RH, high temp
6. Water boils below 100 °C on hills (low pressure)
7. Pressure cooker = higher boiling point
8. Salt: BP up, FP down
9. White clothes, white-top umbrella reflect heat
10. Cloudy nights warmer; fresh snow reflects most

Chapter 7

Waves and Sound

Built from 33 UPPSC questions (1990–2025). Focus: the electromagnetic spectrum, ionosphere, sound's speed in media, echo and reverberation, ultrasound, noise limits and TV remotes.

ASKED MOST · REVISE FIRST

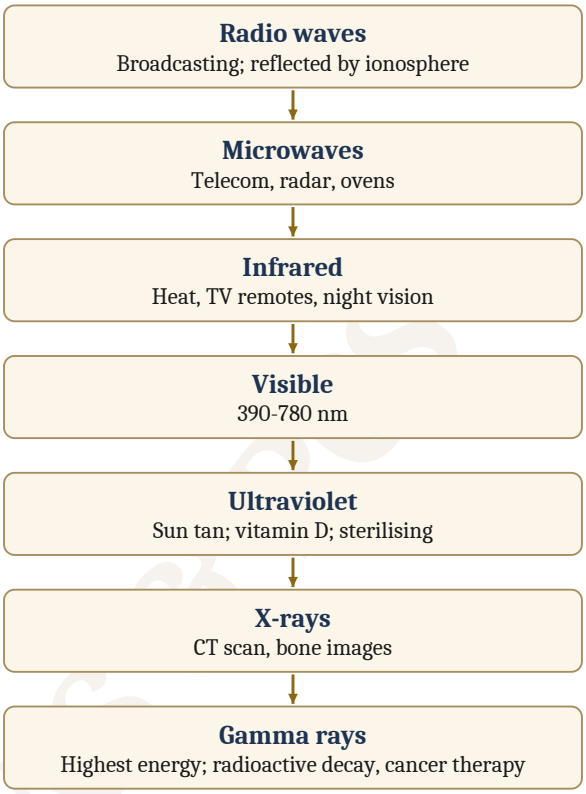
- ■ Sound is fastest in **steel** (solids); it **cannot travel in a vacuum** **8×** 1990, 2002, 2002, 2008, 2014, 2018, one-liner, one-liner
- ■ TV remote uses **infrared** **3×** 2002, 2010, 2013
- ■ **Ionosphere** reflects radio waves back to Earth **3×** 1991, 1998, one-liner
- ■ **Ultrasonic** = above **20,000 Hz**; bats navigate with it **4×** 2002, 2004, 2005, 2012
- • Hazardous noise: above **80 dB** **2×** 2008, 2013

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

7.1 The Electromagnetic Spectrum

- Increasing frequency: **radio, infrared, visible, X-rays** **1×** 2023
- Decreasing frequency: **gamma, X-rays, ultraviolet, radio** **1×** 2025
- ■ Long radio waves are reflected by the **ionosphere** **3×** 1991, 1998, one-liner
- Atmosphere is heated mainly by **longwave terrestrial radiation** **1×** 2022
- Not true of cosmic rays: they are **electromagnetic waves** (they are charged particles) **1×** 2005
- Telecommunication waves: **microwaves** **1×** 2013
- Dish signals fail in rain: rain **absorbs and scatters** radio energy (1, 2, 3 per key) **1×** 2017
- TV signals limited in range: Earth is **curved** **2×** 1994, 2010
- CT scan uses **X-rays** **1×** 2011

ELECTROMAGNETIC SPECTRUM, LOW TO HIGH FREQUENCY



7.2 Propagation and Speed of Sound

- Highest sound velocity: **steel** **2×** 2008, 2018
- Speeds: **air < water < steel** **1×** one-liner
- Sound can travel in **solids and gases** (and liquids) **1×** 2002
- Sound cannot travel in a **vacuum** **2×** 1990, 2014
- Astronauts cannot hear each other on the Moon: **no atmosphere** (no medium) **2×** 2002, one-liner
- Men's and women's voices differ: vocal cords differ in **length and thickness** (A and R true, R explains A) **1×** 2025

Medium	Speed of sound (approx.)
Air (20 °C)	343 m/s
Water	1,480 m/s
Steel	About 5,000-6,000 m/s
Vacuum	Zero (no sound)

7.3 Echo, Reverberation and Ultrasound

- Reverberation in large halls: **multiple reflections** (A and R true, R explains A) **2×** 2002, 2003
- Concert hall walls should **absorb sound** **1×** 2007

QUANTUM IAS PCS

- Clear echo needs at least **16.5 m** (17 m) 1× 2007
- Ultrasound can kill insects, clean, treat disease, run automatic doors (**all correct**) 1× 2004
- **Ultrasonic**: above **20,000 Hz** 1× 2012
- Bats hunt in the dark with **ultrasonic waves** 1× 2005
- Heart vibrates at infrasonic; **Mach number** for speed; ultrasonic above 20,000 Hz (1, 3, 4); sound is **slower** in gases 1× 2002

Range	Frequency	Example
Infrasonic	Below 20 Hz	Earthquakes, whales, heartbeats
Audible	20-20,000 Hz	Human hearing
Ultrasonic	Above 20,000 Hz	Bats, dolphins, SONAR, ultrasound scans

7.4 Noise Limits

- WHO safe city noise: **45 dB** 1× 2010
- Day limits: **residential 55, silence zone 50, industrial 75 (key option 70), commercial 65 dB** 1× 2022
- Hazardous noise: above **80 dB** 2× 2008, 2013

7.5 Musical Instruments, Remote Controls and Television

- Sitar makes **stationary transverse** waves 1× 2021
- TV remote: key says **radio waves**; most remotes use **infrared** 1× 2013
- TV remote uses **infrared** 3× 2002, 2010, 2013
- When a TV is switched on, **audio and video start together** 1× 2007
- TV audio is sent by **frequency modulation** 2× 2007, 2010

TRAP ALERT

- Sound: **steel > water > air**; light is the reverse (fastest in vacuum).
- Cosmic rays are **particles**, not EM waves.
- Remote = **infrared**; telecom = microwaves; CT = X-rays.
- Clear echo distance = **17 m** (half of 34 m at 0.1 s).

UP LINK

UP link: CPCB noise rules apply in UP cities; the UP government has enforced the **silence zones** (100 m around hospitals, schools, courts) and curbs on **loudspeakers** at religious places (2022 drive).

LAST-MINUTE 10

1. Radio, IR, visible, UV, X, gamma: rising frequency
2. **Ionosphere** reflects radio waves
3. Cosmic rays are not EM waves; CT = X-rays
4. Sound fastest in **steel**; no sound in vacuum
5. Moon: no air, so no sound
6. Clear echo needs **17 m**
7. Ultrasonic **> 20,000 Hz**; bats use it
8. WHO safe noise **45 dB**; hazard above **80 dB**
9. TV remote = **infrared**; TV audio = **FM**
10. Sitar = stationary transverse waves

Chapter 8

Electricity and Conductivity

Built from 46 UPPSC questions (1990–2025). Focus: units of electricity, bulb resistance, dynamo, motor and transformer, fuse and earthing, lamps, conductors, lightning and semiconductors.

ASKED MOST · REVISE FIRST

- **kWh sums**: units = watts x hours / 1000 (100 W for 4 h = **0.4**; 5 x 100 W for 20 h = **10**; 15 x 200 W for 24 h = **72**) 4× 2006, 2009, 2017, 2023
- **Tungsten** filament in bulbs 3× 1990, 2005, 2011
- **Dynamo** = mechanical to electrical (electromagnetic induction); **motor** = electrical to mechanical 5× 2007, 2009, 2012, 2017, one-liner
- **Fluorescent tube**: **mercury vapour and argon** at low pressure 3× 1990, 2006, 2009
- **Semiconductors**: **silicon and germanium**; resistance **falls** on heating 6× 2005, 2015, 2015, 2015, 2016, one-liner

UAITIUM

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

8.1 Electrical Energy and the Unit

- **1 kWh = 3.6 x 10^6 J** 1× 2006
- 100 W bulb for 4 hours: **0.4 unit** 1× 2009
- Five 100 W bulbs for 20 hours: **10 units** 1× 2017
- Fifteen 200 W bulbs for 24 hours (Silkyara tunnel): **72 units** 1× 2023

8.2 Resistance, Bulbs and Circuits

- Of two bulbs, the **dim one has higher resistance** 3× 2008, 2009, one-liner
- Maximum light from two identical bulbs: connect in **parallel** 1× 2021
- 15 V across a brown-green-black resistor (15 ohm): **1 ampere** 1× 2023
- No current between bodies at the **same potential** 1× 2014
- n equal charged drops merge: potential becomes **$n^{2/3}$ V** 1× 2015

8.3 Dynamo, Motor and Transformer

- Dynamo is a **converter of energy** 1× 2009
- Works on electromagnetic induction: **dynamo** 2× 2012, 2017
- Dynamo: **mechanical to electrical** 1× one-liner
- Electrical to mechanical: **electric motor** 1× 2007
- Motors burn at low voltage: they **draw more current** 1× 2008
- Transformer: **steps AC voltage up or down** 1× 2006
- Mobile charger: a **step-down transformer** 1× 2017
- AC to DC: **rectifier** 1× 2006

Device	Does
Dynamo / generator	Mechanical to electrical (induction)
Electric motor	Electrical to mechanical
Transformer	Changes AC voltage (works only on AC)
Rectifier	AC to DC
Inverter	DC to AC
Fuse	Melts on excess current

8.4 Safety: Fuse, Earthing and Plugs

- Bulb filament: **tungsten** 3× 1990, 2005, 2011
- Longest pin of a 3-pin plug: **earth** 1× 2007
- Earthing is for **safety** 1× 2002
- Fuse wire: **high resistance, low melting point** 1× 1993
- Fuse wire is lead-tin alloy; its melting point is **lower**, not higher, than copper's (A true, R false) 1× 2025

8.5 Lamps and Lighting

- **CFL** = Compact Fluorescent Lamp 1× 2015
- Gas in advertising signs: **neon** 1× 2001
- Fluorescent tubes: **mercury vapour and argon** 1× 2006
- Tube light: **argon at low pressure** with mercury vapour 1× 2009
- Gases in a fluorescent tube: **mercury and neon** (old key) 1× 1990
- Why a tube over a bulb: **none of these** (per key) 1× 2009

8.6 Solar Cells and the Earth's Magnetism

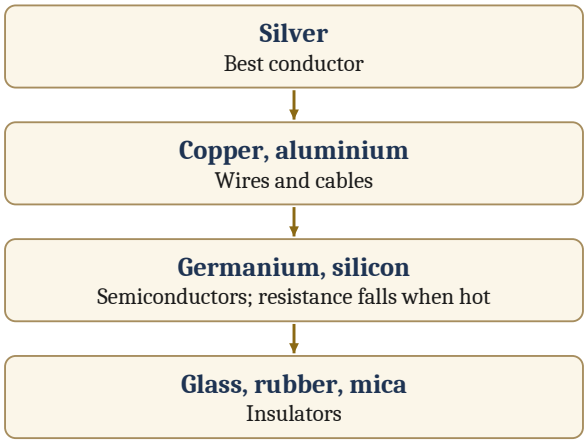
- Ascending power use: **fan, TV, electric iron, electric kettle** 1× 1992
- **Photovoltaic cells** = **solar cells**; they turn sunlight into electricity 2× 2016, 2022
- Earth's magnetic field: **currents in the core** 1× 2011

8.7 Conductors and Lightning

- Best conductor: **silver** 1× 2015
- Lightning conductors protect buildings by sending charge to earth (A and R true, R explains A) 1× 1999
- Copper rods for lightning conductors: better conductor, does not rust (A and R true, R explains A) 1× 2013
- Closed car is safe in lightning: it acts as a **hollow conductor** (A and R true, R explains A) 1× 2023

QUANTUM IAS & PCS

CONDUCTORS TO INSULATORS



8.8 Superconductors and Semiconductors

- Superconductors would pay most at **room temperature** 1× 2000
- Highest superconducting temperature: about **133-138 K** when asked (*) 1× 2013
- High-temperature superconductors: **ceramic oxides** 1× 2000
- Semiconductor resistance on heating: **decreases** 1× 2015
- Transistor component: **germanium** 1× 2016
- Commonest transistor material: **silicon** 1× 2015

■ Semiconductors: **silicon and germanium** (1 and 4) 3× 2005, 2015, one-liner

TRAP ALERT

- The **dim** bulb has the **higher** resistance ($P = V^2/R$).
- Metals' resistance rises with heat; **semiconductors' falls**.
- Fuse = **low** melting point, **high** resistance; earth pin is the **longest and thickest**.
- Transformers work only on **AC**; charger = step-down transformer + rectifier.

UP LINK

UP link: the **Silkyara tunnel** sum (2023) was set in Uttarakhand; in UP, **Jewar** (Gautam Buddha Nagar) has an approved **semiconductor** assembly unit (HCL-Foxconn, 2025), and UP's largest solar parks are in **Bundelkhand**.

LAST-MINUTE 10

1. **1 kWh = 3.6 x 10^6 J**; units = W x h / 1000
2. Dim bulb = **higher resistance**; parallel = more light
3. Tungsten filament; fuse = low melting point
4. Earth pin is longest; earthing = safety
5. Dynamo = mechanical to electrical; motor = reverse
6. Transformer = AC voltage; rectifier = AC to DC
7. Tube light = **mercury vapour + argon**; neon = signs
8. Silver = best conductor; copper lightning rods
9. Closed car = safe in lightning (hollow conductor)
10. Si, Ge = semiconductors; resistance falls on heating

Chapter 9

Nuclear Physics

Built from 45 UPPSC questions (1992–2025). Focus: fission and fusion, atom and hydrogen bombs, moderators and coolants, thorium, breeder reactors, India's nuclear plants and tests.

ASKED MOST · REVISE FIRST

- ■ **Atom bomb = fission**; **hydrogen bomb = fusion** (of deuterium); H-bomb by **Edward Teller** 6× 1994, 2007, 2015, 2016, 2021, one-liner
- ■ **Moderators**: **heavy water, light water, graphite**; heavy water also a **coolant** 4× 1993, 2011, one-liner, one-liner
- ■ **Fast Breeder Test Reactor** (Kalpakkam) coolant: **liquid sodium** 3× 1997, 2008, 2010
- ■ **Kalpakkam** = nuclear power (IGCAR, Tamil Nadu) 4× 2001, 2002, 2023, one-liner
- ■ **Plant-state matching**: **Narora-UP, Kakrapar-Gujarat, Kaiga-Karnataka, Rawatbhata-Rajasthan, Kudankulam-TN** 4× 2004, 2009, 2010, 2025

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

9.1 Discoveries and Basic Ideas

- Wrong pair: **transuranic synthesis-Enrico Fermi** (per key; Seaborg made most) **3×** 2001, 2002, 2003
- Not correct: **splitting a nucleus is fusion** (it is fission) **1×** 1999
- Rutherford doubted nuclear power ("moonshine") (A and R true, R explains A, per key) **1×** 1994
- Particle claimed to rebut relativity: **microwave photon** (per key) **1×** 2009
- Cyclotrons accelerate **protons** (charged particles) **1×** 1997
- Most energetic radiation among options: **X-rays** (gamma rays are higher still) **2×** 2005, 2018

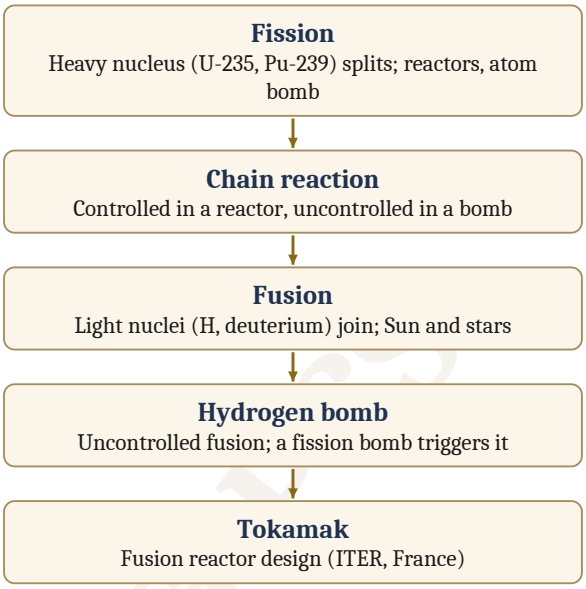
9.2 Energy of the Stars

- Stars shine by **nuclear fusion and gravitational contraction** (1 and 2) **1×** 2000
- Star energy: **hydrogen turns into helium** **1×** 2009

9.3 Reactors and Bombs

- Nuclear reactor: a "**molecular furnace**" (per key) **1×** 1992
- Fusion reactors: **tokamak** design, **very high temperature** (ii and iii); not fission **1×** 2016
- Reactor vs bomb: the reactor's chain reaction is **controlled** **1×** 1999
- Radioactivity is a **nuclear property**; H-bomb is **not** fission (only 1) **2×** 2016, 2021
- Atom bomb: **nuclear fission** **1×** 2007
- H-bomb: **fusion of deuterium**; uncontrolled fusion **2×** 1994, one-liner
- H-bomb developed by **Edward Teller** **1×** 2015
- Nuclear medicines are **radioisotopes** **1×** 1998

FISSION VERSUS FUSION



9.4 Fuel, Moderators and Coolants

- Nuclear fuel abundant in India: **thorium** **1×** 2003
- Not a nuclear fuel: **calcium** **1×** 2014
- **Enriched uranium**: natural uranium with more **U-235** **1×** 2009
- Moderator: **heavy water** (and light water); **graphite** too **2×** 2011, one-liner
- Heavy water: **coolant** and moderator; ordinary water too **2×** 1993, one-liner
- Heavy water made at **Trombay** (among options) **1×** 2015
- First heavy water plant: **Nangal** **1×** 2010
- **Breeder reactor**: makes **more fissile material than it burns** **1×** 2009
- Kalpakkam breeder statements: **1 and 3** per key **1×** 2005
- FBTR coolant: **liquid sodium** **3×** 1997, 2008, 2010

Part	Material	Job
Fuel	U-235, Pu-239, (Th-232 via U-233)	Fission
Moderator	Heavy water, light water, graphite	Slows neutrons
Control rods	Cadmium, boron	Absorb neutrons
Coolant	Heavy water, water, liquid sodium, CO2	Removes heat
Shield	Concrete, lead	Stops radiation

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9.5 India's Nuclear Map

- **Dhruva, Purnima, Cirus**: Indian **research reactors** 1× 2004
- **Kalpakkam**: nuclear power plant 3× 2001, 2002, 2023
- **IGCAR**: **Tamil Nadu** (Kalpakkam) 1× one-liner
- Wrong pair: **Harish-Chandra Research Institute-Chennai** (it is in **Prayagraj**) 1× 2010
- **Kudankulam-TN, Kakrapar-Gujarat, Kaiga-Karnataka, Rawatbhata-Rajasthan** 1× 2025
- **Kalpakkam-TN, Narora-UP, Kakrapar-Gujarat, Trombay-Maharashtra** 1× 2010
- Wrong: **Rajasthan Atomic Power-Udaipur** (it is **Rawatbhata**, near Kota) 1× 2009
- Plants at **Narora and Kalpakkam** (1 and 3) 1× 2004
- In seismic zone IV: **Narora** 1× 2010
- Recently active (2000): **Kaiga** 1× 2000
- Russia to build **6** units at Kudankulam 1× 2009
- Highest installed capacity: **Kudankulam** 1× 2014
- Nuclear share of power: about **3%** (*) 1× 2008

9.6 Nuclear Tests and Security

- Pokhran 1974 code name: **Smiling Buddha** 1× 2009
- **Operation Shakti**: Pokhran tests of **1998** 1× 1999
- ICONS-2024: India rebuilt the **Counter Nuclear Smuggling Team** (only 2) 1× 2024

TRAP ALERT

- Atom bomb = **fission**; H-bomb = **fusion**; reactors = **controlled** fission.
- Heavy water is both **moderator and coolant**; FBTR uses **liquid sodium**.
- Rajasthan plant = **Rawatbhata** (Kota), not Udaipur; Harish-Chandra Institute = **Prayagraj**.
- Gamma rays carry the most energy; X-rays only if gamma is not in the options.

UP LINK

UP link: **Narora Atomic Power Station** (Bulandshahr, 1991) lies in **seismic zone IV**; the **Harish-Chandra Research Institute** (physics and mathematics) is at **Jhunsi, Prayagraj**.

LAST-MINUTE 10

1. Fission = splitting; fusion = joining (stars: H to He)
2. Atom bomb = fission; H-bomb = fusion, **Edward Teller**
3. Reactor = controlled chain reaction; fusion reactor = tokamak
4. India rich in **thorium**; enriched U = more U-235
5. Moderators: heavy water, light water, graphite
6. FBTR coolant = **liquid sodium**; breeder makes more fuel
7. Nangal = first heavy water plant; Dhruva, Cirus = research reactors
8. Narora = UP, zone IV; Kudankulam = largest, Russian
9. Rawatbhata = Rajasthan; Kakrapar = Gujarat; Kaiga = Karnataka
10. Smiling Buddha **1974**; Operation Shakti **1998**

Chapter 10

Computer and Information Technology

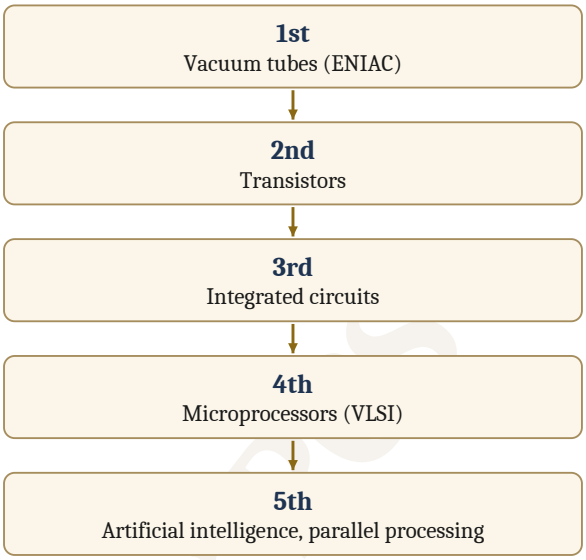
Built from 81 UPPSC questions (1993–2024). Focus: computer generations and chips, Indian supercomputers, memory and bits, operating systems and languages, the internet, modem and full forms.

ASKED MOST · REVISE FIRST

- IC chips are made of **silicon** 5× 2002, 2004, 2006, 2010, 2014
- **Modem** sends data over telephone lines 6× 2002, 2007, 2013, 2015, one-liner, one-liner
- **RAM** = Random Access Memory, **volatile**; **ROM** = permanent 6× 2002, 2003, 2012, 2013, 2024, one-liner
- Indian supercomputers: **PARAM** (first, C-DAC), **Anupam** (BARC); PARAM is not a missile 5× 1999, 2004, 2004, 2006, 2009
- **1 byte = 8 bits**; bit = **binary digit**; 1 KB = 1,024 bytes 6× 2006, 2006, 2010, 2013, one-liner, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

COMPUTER GENERATIONS



10.1 History and Supercomputers

- First computer, father of the computer: **Charles Babbage** 2× 2007, one-liner
- First mouse body (about 40 years back): **wood** 1× 2010
- Third-generation computers: **integrated circuits** 1× 2016
- Miniaturisation: **IC chips** 1× 2007
- IC chip material: **silicon** 5× 2002, 2004, 2006, 2010, 2014
- ICs are classed by the **number of transistors** 1× 2015
- First Indian supercomputer: **PARAM** 1× 2004
- Wrong pair: **PARAM 10,000-missile** (it is a supercomputer) 3× 1999, 2006, 2009
- BARC supercomputer: **Anupam** 1× 2004
- World's fastest (Dec 1996): **10^{12} operations a second** (teraflop) 1× 1998
- Fastest computer (2008): none of the options (*); **El Capitan** (USA) leads now 1× 2008
- **Magic Cube** supercomputer: **China** 1× 2015
- Smallest, fastest brain-like computer: **quantum computer** 1× 2000
- **Y2K**: problem when a year's last two digits become **00** 1× 2000

10.2 Hardware and Memory

- Most important part: **CPU** 1× 2012
- Intel Core i9: a **processor** 1× 2023
- After a double click, information goes to the **CPU** 1× 2013
- Only input devices: **mouse, keyboard, scanner** 1× 2024
- Word document, video or MP3: a **file** 1× 2013
- CPU = **Central Processing Unit** 1× one-liner
- Memory and word length are measured in **bits** 2× 2006, 2010
- **1 byte = 8 bits** 1× 2006
- **1 GB = 1,024 MB**; **1 KB = 1,024 bytes** 2× one-liner, one-liner
- **Bit** = **binary digit** 1× 2013
- MAC address: **48 bits** 1× 2023
- Self-complementing code: **5211** 1× 2013
- Permanent, non-volatile memory: **ROM** 2× 2012, 2024
- **RAM** = Random Access Memory; loses data when power goes 4× 2002, 2003, 2013, one-liner
- Fastest memory: **cache** 1× 2013
- Not true of flash memory: **cheaper than disk drives** 1× 2004
- Large-storage hardware: **both** options 1× 1999
- **Index hole**: **floppy disk**; **IRQ 6**: floppy drive controller 2× 2013, 2015

QUANTUM IAS & PCS

- Keyboard cable port: **USB** 1× 2015
- Laser printers use a **semiconductor laser** 1× 2009
- Daisy wheel printer: **impact** printer 1× 2013

Memory	Fact
Cache	Fastest; between CPU and RAM
RAM	Working memory; volatile
ROM	Permanent; holds boot program
Flash	Non-volatile; pen drives, SSD (costlier per GB than hard disks)
Hard disk	Large, cheap, slower

10.3 Software and Languages

- Controls basic functions: **operating system** 1× 2013
- Not an OS: **Microsoft Office** 1× 2023
- Running many apps at once: **multitasking** 1× 2015
- Scientific language: **FORTRAN** 2× 2002, 2003
- BASIC: a **procedural** language 1× 2013
- **Assembler**: assembly to **machine language** 1× 2005
- Computer virus: a **software program** 2× 1993, 2010
- Error in a program: **bug** 1× 2010
- **Stuxnet**: malware that sabotaged Iran's **nuclear centrifuges** 1× 2010
- **Talaash**: C-DAC's Devanagari **OCR** and portal 1× 2004
- C-DAC anti-cyber-theft software: **NAYAN** 2× 2008, 2009

10.4 Internet and Networks

- Worldwide linked computers: **Internet** 1× 2004
- Internet topology: **tree** (per key) 1× 2013
- Internet works on **packet switching** 1× 2015
- Between physical and network layers: **data link layer** 1× 2013
- Rules for data exchange: **protocols** 1× 2015
- Not IT terms: **optical storage** (per key), **Pinaka** (a rocket) 2× 2001, 2002
- IT terms: **all of the above** 1× 2011
- Finds a user's e-mail address: **Finger** 1× 2010
- **WWW** = **World Wide Web**; inventor **Tim Berners-Lee** 3× 2010, 2015, one-liner
- **Modem**: data over telephone lines 6× 2002, 2007, 2013, 2015, one-liner, one-liner
- **Bluetooth**: wireless link between devices 1× 2011
- First station with Google free Wi-Fi: **Mumbai Central** 1× 2015
- Not true of Li-Fi: uses optical fibre; India's trial by the I&B Ministry (it was **MeitY**) 1× 2017
- E-mail can attach **all** file types 2× 2013, one-liner
- **SkyDrive** (OneDrive): **Microsoft** 1× 2013

10.5 Full Forms and IT in Governance

- **SIM** = Subscriber Identity Module 1× 2012
- **ICT** = Information and Communications Technology 1× 2015
- **ATM** = Automated Teller Machine 1× 2014
- **UPS** = Uninterrupted Power Supply 1× 2015
- **Bibliometry**: an information management tool 1× 2019
- World Computer Literacy Day: **2 December** 1× 2016
- **Vidya Vahini**: **computer education** in schools 1× 2013
- e-Governance **cuts cost** and **raises citizens' input** (2 and 3) 1× 2024

TRAP ALERT

- **RAM** is volatile; **ROM** is not; **cache** is fastest.
- **IC** = 3rd generation; **microprocessor** = 4th.
- MS Office is an application, not an OS; FORTRAN = scientific, COBOL = business.
- Li-Fi uses **LED light**, not fibre.

UP LINK

UP link: **C-DAC Noida** works on language computing and cyber security; **IIT Kanpur** hosts the **PARAM Sanganak** supercomputer (National Supercomputing Mission, 2020) and the C3iHub cyber security centre.

QUANTUM IAS & PCS

LAST-MINUTE 10

1. **Charles Babbage** = father of the computer
2. IC = 3rd generation; chips = **silicon**
3. **PARAM** = first Indian supercomputer (C-DAC); Anupam = BARC
4. 1 byte = **8 bits**; 1 KB = **1,024** bytes; MAC = 48 bits
5. RAM volatile; ROM permanent; cache fastest
6. OS controls the computer; MS Office is not an OS
7. FORTRAN = scientific; assembler = assembly to machine
8. Internet = packet switching; modem = phone-line data
9. WWW = **Tim Berners-Lee**; SkyDrive = Microsoft
10. Stuxnet = Iran centrifuges; NAYAN, Talaash = C-DAC

Chapter 11

Space Technology

Built from 79 UPPSC questions (1990–2025). Focus: India's early satellites, ISRO centres, Chandrayaan, Mangalyaan and Aditya-L1, astronauts, geostationary satellites, rockets and cryogenic engines.

ASKED MOST · REVISE FIRST

- Rockets and jet engines work on **conservation of momentum** 6× 2007, 2009, 2013, 2014, 2021, 2023
- **Cryogenic engine**: **liquid hydrogen** fuel with **liquid oxygen**; used in rockets 5× 2004, 2006, 2006, 2010, one-liner
- Satellites run on **solar cells** 3× 2000, 2006, 2014
- **Black box** of an aircraft is **orange** 3× 1996, 2004, 2009
- **ISRO places**: **Thumba-rocket launching (Kerala), Sriharikota-satellite launches (near Pulicat lake), SAC-Ahmedabad, Hassan-Master Control** 5× 1998, 2003, 2004, 2010, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

11.1 India's Early Satellites and Programmes

- First Indian satellite: **Aryabhata** (1975) 2× 2006, 2009
- Order: **SITE, STEP, APPLE, IGMDP** 1× 1998
- India's telecom is satellite-linked; not for "star war" (A true, R false) 1× 1990
- **IRS-1A** launched from **Baikonur** 1× 1994
- **Kalpana-1** = renamed **METSAT** (September 2002) 2× 2004, one-liner
- Not a satellite: **SLV-3** (a launch vehicle) 2× 2004, 2010
- GSLV-D2 is a launch vehicle, not a satellite (*) 1× 2001
- **EDUSAT**: launched **20 September 2004** 1× 2004
- PSLV-C12 (2009): RISAT-2 for border watch and **ANUSAT** by Anna University (all four correct) 1× 2008
- Launched 22 December 2005: **INSAT-4A** 1× 2004
- **INSAT-3C** and **GSAT-8** launched from **Kourou** 2× 2002, 2010
- INSAT-4C (failed) weighed **2,168 kg** 1× 2005
- Launched 29 September 2012: **GSAT-10** 1× 2012
- Launch vehicle order: **SLV, ASLV, PSLV, GSLV** 1× 2004
- **BHUVAN**: ISRO's **3-D geoportal** 1× 2008

INDIAN LAUNCH VEHICLES

SLV-3
1980; Rohini; first Indian launch

↓

ASLV
1990s; augmented SLV

↓

PSLV
1994 onwards; workhorse; 104 satellites in 2017

↓

GSLV / LVM3
Cryogenic upper stage; Chandrayaan-2, 3

↓

SSLV
Small satellites, 2022-23

11.2 ISRO and Its Centres

TUMLAS PCS

- Master Control Facility (Hassan): **Karnataka** 1× 2010
- **IIRS**: **Dehradun** 1× 2006
- **ISRO-Bengaluru, IUCAA-Pune, IUAC-New Delhi, VSSC-Thiruvananthapuram** 1× 2010
- Wrong: **ISRO Satellite Centre-Sriharikota** (it is Bengaluru) 1× 2010
- **Space Applications Centre**: **Ahmedabad** 1× 1998
- **Thumba**: **Kerala** 1× 2004
- **Sriharikota** lies near **Pulicat lake** 1× 2003
- **Kakrapar-nuclear plant, Pokhran-bomb test, Thumba-rocket launching, Sriharikota-satellite launching** 1× 2016
- VSSC works on **propellants and launch vehicles** (1 and 4) 1× 2004
- **PraVaHa** CFD software: a VSSC feat for aerodynamic design (both) 1× 2024
- Tenth ISRO Chairman (January 2022): **S. Somanath** 1× 2021

Centre	Place	Work
ISRO HQ, URSC	Bengaluru	Satellites
VSSC	Thiruvananthapuram	Launch vehicles
SDSC-SHAR	Sriharikota (AP)	Launch pad
SAC	Ahmedabad	Payloads, applications
MCF	Hassan, Bhopal	Satellite control
IIRS	Dehradun	Remote sensing

11.3 Moon, Mars and Sun Missions

- **Chandrayaan-3: 14 July 2023; Aditya-L1: 2 September 2023; INSAT-3DS: 17 February 2024; RLV-LEX-01: 2 April 2023** 1× 2025
- Chandrayaan-2 project director: **M. Annadurai** (per key) 1× 2010
- Chandrayaan-2 launcher: **GSLV Mk III-M1** 1× one-liner
- Chandrayaan-3 soft landing made India the **fourth** country to land on the Moon 1× 2023
- Chandrayaan-3 payload for soil elements: **APXS** 1× 2023
- LUPEX: India-Japan Moon mission; JAXA builds the **rover**, ISRO the **lander** (only 1) 1× 2024
- Luna-25 aimed at Moon access; Russia-China lead the ILRS (A and R true, R explains A) 1× 2023
- **Mangalyaan**: launched **November 2013** (both) 1× 2023
- Evidence of life on Mars: **water, gypsum, methane** 1× 2009
- Planet that may suit life: **Mars** 1× 2018
- Methane in the atmosphere of **Mars** (and Jupiter) 2× 2010, 2016
- Saturn's atmosphere has much **methane** (per key) 1× 2022
- Water on the Moon confirmed by NASA's **SOFIA** 1× 2016
- Underground lake (2018): **Mars** 1× 2018
- **DAVINCI+** and **VERITAS**: **Venus** 1× 2021
- **NISAR**: **ISRO and NASA** 1× 2023
- **Kepler** telescope: **Earth-like planets** 1× 2008

11.4 Astronauts and Human Spaceflight

- First woman in space: **Valentina Tereshkova** 2× 2001, 2004
- "I am a citizen of the Milky Way": **Kalpana Chawla** 1× 2010
- Longest single NASA flight: none of the options (Frank Rubio, 371 days) 1× 2022
- Wilmore and Williams returned (March 2025) on SpaceX **Dragon** 1× 2023
- **Tim Peake**: astronaut (UK) 1× 2015
- NASA's last space shuttle: **Atlantis** 1× 2009
- Not correct of Shenzhou-7: **four** astronauts (it had three) 1× 2008
- **Micius**: China's **quantum satellite** communication system 1× 2016
- **Gravity Tractor**: craft to deflect **asteroids** 1× 2009
- Apollo 8 (21 December 1968): no option correct (*) 1× 1999

11.5 Satellites, Black Holes and Comets

- **Geostationary**: moves with Earth's rotation and stays fixed over one place; period **24 hours** 2× 2005, one-liner
- Satellites tracked by the **Doppler effect** 1× 2017

Quantum IAS & PCS

- Radio blackout during re-entry: heat ionises the air (A and R true, R explains A) **2×** 2002, 2003
- Communication satellites lie in the **exosphere** **1×** 2004
- Satellite power: **solar cells** **3×** 2000, 2006, 2014
- No candle flame in space: flames need **gravity** (convection) (A and R true, R explains A) **1×** 2009
- Black hole idea: **S. Chandrasekhar** **1×** 2011
- Black hole: a **collapsed star** **1×** 2019
- Comet's tail points **away from the Sun** **1×** 2011

11.6 Rockets, Jets and Cryogenics

- Jet engine uses air's oxygen; a rocket carries **liquid** oxidiser, not gas (only 1) **2×** 2007, 2009
- Jet engine: **conservation of linear momentum** **1×** 2013
- Rocket: **conservation of momentum** **3×** 2014, 2021, 2023
- Jets fly in the lower stratosphere: **no clouds, no vertical winds** (both) **1×** 2013
- Wing is convex above; faster air on top gives **lower** pressure there (A true, R false) **1×** 2013
- **Black box**: **orange** **3×** 1996, 2004, 2009
- **Stratellite**: high-altitude **airship** in the stratosphere **1×** 2016
- Liquid oxygen and hydrogen technology: **cryogenics** **1×** 2010
- Cryogenic engine: **rocket** technology; LOX oxidiser, LH2 fuel **3×** 2004, 2006, one-liner
- Rocket propellants: all options are used (*) **1×** 2010
- Discovery's tank (2006): **liquid hydrogen and oxygen** **1×** 2006
- **Curiosity-Mars rover, Messenger-Mercury probe, Rustom-1-DRDO UAV, Aakash-2-tablet** **1×** 2013

TRAP ALERT

- **SLV-3, PSLV, GSLV** are rockets, not satellites.
- ISRO Satellite Centre = **Bengaluru**; Sriharikota = launches; Thumba = sounding rockets.
- Black box is **orange**, not black.
- Chandrayaan-3 made India the **4th** to land and **1st** near the lunar south pole.

UP LINK

UP link: **Shubhanshu Shukla** of **Lucknow** flew on **Axiom-4** (June-July 2025), the first Indian on the **International Space Station**; ISRO's space technology incubation centre and Gaganyaan training links include **IIT Kanpur** (soft detail).

LAST-MINUTE 10

1. Aryabhata = first satellite (1975); Kalpana-1 = METSAT
2. SLV, ASLV, PSLV, GSLV in order
3. Thumba = Kerala; Sriharikota = Pulicat lake; SAC = Ahmedabad
4. Chandrayaan-3: 14 July 2023; India **4th** on the Moon; APXS
5. Aditya-L1: 2 September 2023; NISAR = ISRO + NASA
6. Tereshkova = first woman in space; Atlantis = last shuttle
7. Geostationary period **24 h**; satellites run on solar cells
8. Comet tail away from the Sun; Chandrasekhar = black holes
9. Rocket = conservation of momentum; cryogenic = LH2 + LOX
10. Black box = **orange**

Chapter 12

Defence Technology

Built from 49 UPPSC questions (1995–2025). Focus: IGMDP missiles and their types, Agni ranges, BrahMos, Israeli systems, Tejas and drones, ships and submarines, joint exercises.

QUANTUM

ASKED MOST · REVISE FIRST

- **Missile matching**: **Prithvi-surface to surface, Trishul-surface to air, Agni-intermediate range, Nag-anti-tank, Pinaka-rocket launcher, Nishant-UAV** 6× 2001, 2002, 2007, 2012, 2013, one-liner
- **BrahMos** = **India and Russia** (Brahmaputra + Moskva) 5× 2003, 2004, 2016, 2018, one-liner
- **Israel** partner: **Phalcon radar, Barak-8, Iron Dome**, naval pilotless helicopter 4× 2004, 2008, 2016, 2025
- **Astra** = **air-to-air** missile 3× 2004, 2010, one-liner
- **Agni-I** about **700 km**; **Agni-II** above 2,000 km 4× 2003, 2006, 2008, 2011

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

12.1 The Integrated Guided Missile Programme

- **Prithvi-surface to surface, Trishul-surface to air, Pinaka-multi-barrel rocket launcher, Nishant-remote battlefield vehicle** 1× 2002
- Correct pair: **Nag-anti-tank missile** 2× 2012, 2013
- Surface-to-surface: **Prithvi** 1× one-liner
- **Trishul-short-range SAM, Prithvi-surface to surface, Agni-intermediate range, Nag-anti-tank** 2× 2001, 2007
- Short-range missile (key): **Nag** 1× 1996
- Not correct of Astra: kills planes up to **90 km** (per key) 1× 2010
- Air-to-air missile tested in Balasore (May 2003): **Astra** 2× 2004, one-liner
- **Agni-I** range: **700-2,000 km** band; about **700 km** (March 2010) 2× 2003, 2008
- **Dhanush** range: **350 km** 1× 2008
- Long-range missile above 2,000 km: **Agni-II** (an IRBM, not a true ICBM) 2× 2006, 2011
- Agni launch statements (2002): no clean answer (*) 1× 2002
- Agni-II launched from **Chandipur** (per key) 1× 2004
- Not correct of Agni-5: "**not exclusively indigenous**" (it is indigenous) 1× 2015
- **Nirbhay** (7 November 2017): nuclear-capable, **1,000 km**, land-attack cruise missile, counters Babur (1, 2, 3); it is **subsonic** 1× 2017
- Tested 27 January 1996 at Chandipur: **Prithvi-II** 2× 1995, 2006
- Not correct of **Shaurya**: range **900 km** (it is about 700 km); a surface-to-surface missile 2× 2011, one-liner

Missile	Type	Note
Prithvi	Surface to surface	Short range; first IGMDP missile
Agni series	Ballistic	Agni-V about 5,000 km; Agni-P new generation
Akash	Surface to air	Medium range
Trishul	Surface to air	Short range (closed)
Nag	Anti-tank	Fire and forget
Astra	Air to air	Beyond visual range
BrahMos	Supersonic cruise	India-Russia
Nirbhay	Subsonic cruise	1,000 km

12.2 BrahMos

- Developed by **India and Russia** 4× 2003, 2004, 2018, one-liner
- Test of 18 October 2020: **BrahMos naval version** 1× 2016
- Not correct of BrahMos: options (c) and (d) per key; base range was **290 km** 2× 2008, 2009
- Tested March 2009 at **Pokhran** 1× 2007
- Delivered to the **Philippines** (April 2024); its range is **290 km**, not 490 (A true, R false) 1× 2024

12.3 Air Defence and Foreign Partners

- **Iron Dome**: **Israel** (only 3) 1× 2025
- **RudraM-II**: an **air-to-surface** missile tested off Odisha, not Rajasthan (neither) 1× 2024
- **Phalcon radar**: from **Israel** 1× 2004
- **Barak-8**: with **Israel** 1× 2016
- Navy's pilotless helicopter: with **Israel** 1× 2008

12.4 Aircraft and Drones

- **Hansa-2**: a **trainer aircraft** 1× 1996

QUANTUM IAS

- **Tejas**: indigenous **light combat aircraft** 2× 2004, one-liner
- **INDRA**: a **radar** 2× 2003, 2004
- **FWD-200B**: India's first indigenous **bomber UAV**; 100 kg payload (both) 1× 2024
- **NETRA**: a mini **UAV** that flies up to 300 m 1× 2009
- Not true of F-18 Super Hornet: used by **eight** air forces 1× 2002
- Not an aircraft: **Sarath (BMP-II)**, an infantry combat vehicle 1× 2009

12.5 Ships and Submarines

- **Arihant-nuclear submarine, AWACS-Israeli Phalcon, Atlas Centaur-US cryogenic rocket, Nag-anti-tank** 1× 2010
- **Admiral Gorshkov**: aircraft carrier (now **INS Vikramaditya**) 1× 2012
- Scorpene submarine commissioned January 2023: **INS Vagir** 1× 2023
- **Sarthak** (2020): Coast Guard **offshore patrol vessel** 1× 2020
- Coast Guard OPV (March 2010): **Vishwast** 1× 2008
- **INS Shakti**: a fleet tanker built in Italy, so no option fits (*) 1× 2010
- Incorrect: a virus **cannot** reduce pollution 1× 2004

12.6 Exercises and Operations

- **SLINEX II**: India-**Sri Lanka** naval exercise 1× 2011
- **Operation Pawan**: **IPKF** in Jaffna 1× 1998
- **KHANJAR-XII** (March 2025): India-**Kyrgyzstan** 1× 2025
- 2024 dates: **Dustlik April 15-28, Shakti May 13-26, Tiger Triumph March 18-25, MILAN February 19-27** 1× 2024

Exercise	With
SLINEX	Sri Lanka
Khanjar	Kyrgyzstan
Dustlik	Uzbekistan
Shakti	France
Tiger Triumph	USA (tri-service)
Malabar	USA, Japan, Australia

TRAP ALERT

- **Trishul** = surface to air; **Prithvi** = surface to surface; **Nag** = anti-tank; **Astra** = air to air.
- BrahMos = **India-Russia**; Phalcon, Barak, Iron Dome = **Israel**.
- Nirbhay is **subsonic**; BrahMos is **supersonic**.
- INDRA = radar; Tejas = LCA; Sarath = infantry vehicle, not aircraft.

UP LINK

UP link: the **UP Defence Industrial Corridor** has nodes at Lucknow, Kanpur, Agra, Aligarh, Jhansi and Chitrakoot; the **BrahMos integration and testing facility** at **Lucknow** opened in May 2025, and HAL Lucknow and Kanpur make aircraft parts.

LAST-MINUTE 10

1. Prithvi = SSM; Trishul = SAM; Nag = anti-tank; Astra = AAM
2. Agni-I 700 km; Agni-II above 2,000 km; Dhanush 350 km
3. Shaurya = surface to surface, about 700 km
4. Nirbhay = subsonic cruise, 1,000 km
5. BrahMos = India + Russia; to Philippines (2024)
6. Israel: Phalcon, Barak-8, Iron Dome
7. Tejas = LCA; Hansa = trainer; INDRA = radar; NETRA = UAV
8. Arihant = nuclear submarine; Gorshkov = Vikramaditya
9. Vagir = Scorpene (2023)
10. Khanjar = Kyrgyzstan; SLINEX = Sri Lanka; Pawan = IPKF

Chapter 13

Scientists, Inventions and Physics Miscellany

Built from 66 UPPSC questions (1990–2024). Focus: inventor-invention matching, Indian scientists, lasers, nanotechnology and MRI, magnets and cells, labs and polar stations.

ASKED MOST · REVISE FIRST

- **National Science Day (28 February)** marks the **Raman effect** (1928) 5× 1990, 1993, 2002, 2003, one-liner
- **Discoverer matching**: **radium-Curie, penicillin-Fleming, X-ray-Roentgen, dynamite-Nobel, evolution-Darwin, revolver-Colt** 5× 1992, 2005, 2009, 2010, 2011
- Antarctic stations: **Dakshin Gangotri** (first), **Bharati** (latest) 3× 2006, 2009, 2010
- **Aluminium** is not attracted by an ordinary magnet; copper is non-magnetic 3× 1990, 2002, one-liner
- **J.C. Bose** worked in biology and physics; he won **no Nobel** 2× 2006, 2008

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

13.1 Inventions and Discoveries

- Machine gun: **James Puckle** 1× 2014
- Telescope (astronomical use): **Galileo** 1× 1994
- Biggest "telescope": **IceCube** (a neutrino detector at the South Pole) 1× 2010
- X-ray machine: **Wilhelm Roentgen** 1× 2005
- Gas engine: **Daimler** 1× 1995
- Wrong pair: **J. Perkins-penicillin** (Fleming found penicillin) 1× 2011
- **Radium-Curie, penicillin-Fleming, X-ray-Roentgen, measles vaccine** (smallpox: Jenner) 1× 1992
- Radioactivity: **Becquerel** 1× 2003
- Order: **diode, transistor, IC, SMD** 1× 2016
- **Nobel-dynamite, Fleming-penicillin, Darwin-evolution, Curie-radium** 1× 2010
- **Revolver-Colt, dynamite-Nobel, law of cooling-Newton, law of pressure-Pascal** 1× 2009
- Laws of planetary motion: **Kepler** 1× 2010
- Photoelectric effect discovered by **Hertz** 1× 2003
- National Science Day: **Raman effect** 4× 1990, 1993, 2002, 2003

Discovery	Scientist
Radioactivity	Henri Becquerel
Radium, polonium	Marie Curie
X-rays	Wilhelm Roentgen
Penicillin	Alexander Fleming
Smallpox vaccine	Edward Jenner
Dynamite	Alfred Nobel
Planetary laws	Johannes Kepler
Photoelectric effect	Hertz (found), Einstein (explained)

13.2 Indian Scientists

- Birth centenary in 1988: **C.V. Raman** 1× 1991
- New theory of relativity (Hoyle-Narlikar): **J.V. Narlikar** 1× 2016
- **Homi Bhabha Award**: **nuclear energy** 1× 2012
- Biology and physics both: **J.C. Bose** 1× 2008
- Designed Aryabhata: **U.R. Rao** 1× 2017
- **Vinod Dham** (Padma Bhushan 2025): father of the **Pentium chip** 1× 2023
- India's atom bomb: **Raja Ramanna** 2× 2002, 2004
- Not a Nobel winner: **J.C. Bose** 1× 2006
- **Thermal ionisation**: **M.N. Saha** 1× 2014
- Einstein's Nobel: **photoelectric effect** 1× one-liner
- National Science Day: **28 February** 1× one-liner

13.3 Modern Physics and Devices

- "God particle": **Higgs boson** 1× 2013
- Laser gives **stimulated** radiation 1× 2012
- Lasers treat cancer, heart, eye and kidney problems (*) 1× 2002
- First ruby laser gave **red** pulses (only 2) 1× 2023
- Miniaturisation: **nanotechnology** 1× 2004
- **Nano plug**: a tiny **hearing aid** 1× 2015
- Carbon nanotubes: **Sumio Iijima** 1× 2008

- **Nano Hummingbird**: pocket US **spy drone** 1× 2010
- Space-based solar power as a national goal: supply is nearly uninterrupted (A and R true, R explains A) 1× 2010
- Most accurate clock (1 s in 300 million years): **strontium** atoms 1× 2008
- Carbon microphone: **telephone** 1× 2009
- **MRI**: **nuclear magnetic resonance** 1× 2008

13.4 Magnetism, Cells and Everyday Devices

- Not attracted by a magnet: **aluminium** (weakly paramagnetic) 2× 2002, one-liner
- Non-magnetic element among options: **copper** 1× 1990
- Tape-recorder tape: coated with **ferromagnetic powder** 1× 1998
- Torch cell terminals: **zinc and carbon** 1× 2010
- Automatic watches run on **hand movements** 1× 2007
- Correct: liquid crystals in displays, modem on phone lines (1 and 2); NIO is in **Goa** 1× 2009
- Liquid crystals: **all of the above** uses 1× 1996
- **Soleckshaw**: a **solar rickshaw** 1× 2008
- **Cytotron**: makes **artificial climate** 1× 2016
- Water jet technology: **drilling mines** 2× 1998, 2004
- Engine cooling: **radiator** 1× 2014

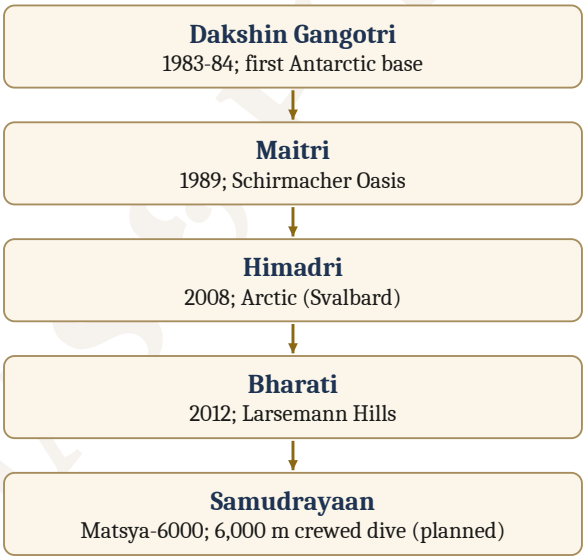
13.5 Laboratories and Institutions

- **National Physical Laboratory**: **New Delhi** 1× 2009
- **National Chemical Laboratory**: **Pune** 1× 2015
- Not a science body: **ICSSR** (social science) 1× 2008
- Wrong pair: **Antarctic Study Centre-Bangalore** (NCPOR is in **Goa**) 1× 2004
- Naturopathy and Yogic Science institute: **Bengaluru** 1× 2002
- **Hi-Tec City-Hyderabad, Science City-Kolkata, rocket centre-Thumba, CDRI-Lucknow** 1× 1999
- "Nuclear Reactor Time Bomb": **Takashi Hirose** 1× 2011

13.6 Polar and Ocean Missions

- First Antarctic station: **Dakshin Gangotri** 1× 2006
- Latest Antarctic station: **Bharati** 2× 2009, 2010
- First manned deep-ocean mission: **Samudrayaan** 1× 2023
- Wrong: **isohaline-snowfall** (isohalines join equal salinity) 1× 2001
- Correct: **pycnocline-sea water density** 1× 2024
- Salinity gradient: **halocline** 1× one-liner

INDIA'S POLAR AND OCEAN STATIONS



13.7 Technology and Society

- **Dry ice-solid CO2, gene therapy-blood diseases, cryonics-freezing bodies, cobalt-60-cancer** 2× 1997, 2004
- iPhone 4S: 8 hours 3G talk time and **Siri** (2 and 3) 1× 2009
- Mobiles and vehicles changed rural **mobility and connectivity** (1 and 2) 1× 2010
- **GPS** = Global Positioning System 1× 2015

TRAP ALERT

- Galileo improved the telescope; **Hertz found** and **Einstein explained** the photoelectric effect.
- **Jenner** = smallpox vaccine; Fleming = penicillin; Becquerel = radioactivity.
- **J.C. Bose** has no Nobel; Raman (1930) does.
- NCPOR (polar research) and NIO are both in **Goa**.

QUANTUM IAS & PCS

UP LINK

UP link: the **Central Drug Research Institute (CDRI)**, **NBRI**, **CIMAP** and **IITR** (toxicology) are CSIR labs in **Lucknow**; **Shanti Swarup Bhatnagar**, the first CSIR director-general, was born in Bhera (now Pakistan) but studied in Lahore, while **Birbal Sahni Institute of Palaeosciences** is also at Lucknow.

LAST-MINUTE 10

1. National Science Day **28 February** = Raman effect
2. Roentgen = X-ray; Becquerel = radioactivity; Curie = radium
3. Fleming = penicillin; Nobel = dynamite; Colt = revolver
4. Kepler = planetary laws; Hertz = photoelectric effect
5. J.C. Bose = biology + physics, no Nobel; M.N. Saha = thermal ionisation
6. Raja Ramanna = atom bomb; U.R. Rao = Aryabhata; Vinod Dham = Pentium
7. Higgs boson = God particle; MRI = NMR
8. Aluminium not attracted by magnets; torch cell = zinc-carbon
9. NPL = Delhi; NCL = Pune; CDRI = **Lucknow**
10. Dakshin Gangotri first, Bharati latest; Samudrayaan = deep ocean

Chapter 14

Atomic Structure, Matter and Solutions

Built from 20 UPPSC questions (1992–2025). Focus: particles of the atom, isotopes, physical and chemical change, colloids and emulsions, osmosis and sublimation.

ASKED MOST · REVISE FIRST

- Nucleus = **protons and neutrons**; neutron found by **Chadwick** 3× 1995, 1996, 1996
- **Isotopes**: same **atomic number**, different mass; protium, deuterium, tritium are isotopes of **hydrogen** 3× 2009, 2025, one-liner
- **Milk** = an **emulsion**; **honey** is not a colloid 3× 2002, 2024, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

14.1 Particles of the Atom

- Nucleus holds **protons and neutrons** 1× 1996
- Order of discovery: **electron, proton, neutron, pi meson** 1× 2023
- Neutron discovered by **James Chadwick** (1932) 2× 1995, 1996
- Not a sub-atomic particle: **deuteron** (a deuterium nucleus) 1× 2007
- Particle-antiparticle pair: **electron-positron** 1× 2005
- Zero charge: **neutrino** 1× 2021

Particle	Discoverer	Charge
Electron	J.J. Thomson (1897)	Negative
Proton	Rutherford (1919)	Positive
Neutron	Chadwick (1932)	None
Positron	Anderson (1932)	Positive (antielectron)
Neutrino	Predicted by Pauli (1930)	None

14.2 Isotopes and Isobars

- Isotopes: same atomic number, different mass; H isotopes are **not** of carbon (only 1) 1× 2025
- Same protons, different neutrons: **isotopes** 2× 2009, one-liner

Term	Same	Different	Example
Isotopes	Atomic number	Mass number	Protium, deuterium, tritium
Isobars	Mass number	Atomic number	Argon-40, calcium-40

QUANTUM IAS & PCS

Term	Same	Different	Example
Isotones	Neutron number	Protons	C-14, N-15

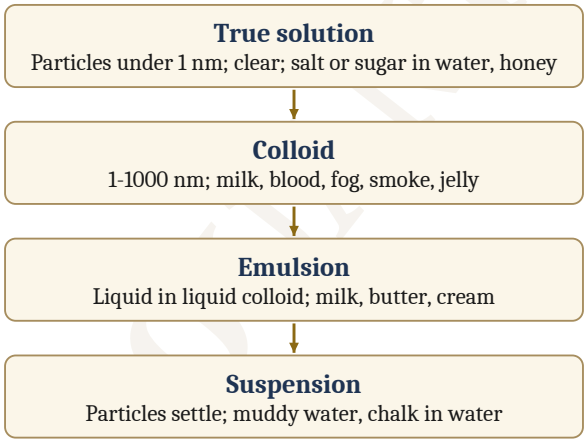
14.3 Physical and Chemical Changes

- Water to vapour: a **physical** change 1× 1992
- Wrong pair: **pasteurisation-tea** (it is for **milk**) 1× 2019

14.4 Mixtures, Colloids and Solutions

- Salt solution is homogeneous; a homogeneous mix has the **same** composition throughout (A true, R false) 1× 2022
- Salt-ice mix goes below 0 °C; salt **lowers**, not raises, the freezing point (A true, R false) 1× 1998
- Milk: an **emulsion** 2× 2024, one-liner
- Not a colloid: **honey** (a true solution) 1× 2002
- Dissolving 100 g sugar in half a litre of water: **volume** stays the same (per key) 1× 2017
- Osmosis: solvent flows from **pure solvent to solution**; reversing it is reverse osmosis (only 1) 1× 2023
- Not used to purify solid impurities: **distillation** (it is for liquids) 1× 2016
- Camphor, naphthalene, dry ice vanish: **sublimation** 1× 2019

KINDS OF MIXTURES



TRAP ALERT

- Deuteron is a **nucleus**, not a fundamental sub-atomic particle.
- Salt **lowers** freezing point (salt-ice freezing mixtures).
- Honey is a true solution; milk is an emulsion.
- Sublimation = solid straight to vapour (camphor, naphthalene, dry ice, iodine).

UP LINK

UP link: **pasteurised milk** from UP's dairy cooperative **Parag** (PCDF, Lucknow) and the large private dairies of western UP; UP is India's top milk-producing state.

LAST-MINUTE 10

1. Nucleus = protons + neutrons; Chadwick = neutron
2. Electron, proton, neutron, pi meson: order of discovery
3. Electron-positron = particle-antiparticle; neutrino = no charge
4. Isotopes = same atomic number; isobars = same mass number
5. H isotopes: protium, deuterium, tritium
6. Water to vapour = physical change; pasteurise milk
7. Milk = emulsion; honey not a colloid
8. Salt lowers freezing point
9. Osmosis: solvent to solution
10. Camphor, naphthalene, dry ice = sublimation

Chapter 15

Metals, Ores and Alloys

Built from 35 UPPSC questions (1993–2025). Focus: special metals (tungsten, sodium, mercury, uranium), ores and minerals, rusting and tarnishing, alloy composition.

ASKED MOST · REVISE FIRST

- **Alloys**: **solder = tin + lead; bronze = copper + tin; brass = copper + zinc**; German silver has **no silver** 7× 1996, 2003, 2005, 2015, 2016, one-liner, one-liner
- **Sodium**: kept in **kerosene**; yellow **street lamps** 3× 2000, 2007, 2017
- **Mercury**: liquid metal, best liquid conductor of heat, used in thermometers 5× 2003, 2005, 2014, 2020, one-liner
- **Rusting** needs **oxygen and moisture** and **increases** iron's weight; galvanised (zinc) iron does not rust 3× 2002, 2008, 2016
- **Stainless steel** = iron + **chromium** (+ nickel); **silicon** is not added 4× 2004, 2017, one-liner, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

15.1 Properties of Metals

- Hardest metal among options: **tungsten** 1× 1996
- Heaviest natural element: **uranium** 1× 2004
- Costliest metal "discovered recently": no fit (*); californium and rhodium are the costliest 1× 2016
- Stored in **kerosene**: **sodium** 1× 2017
- Yellow street lamps: **sodium** 1× 2000
- Sodium lamp light is **monochromatic** and does not split in fog 1× 2007
- Best liquid conductor of heat: **mercury** 2× 2005, 2014
- Mercury in thermometers: it **expands uniformly** (per key: "high circulation power") 1× 2003
- Liquid metal at room temperature: **mercury** 1× one-liner
- **Sodium-cut with a knife, mercury-liquid, silver-good conductor, lead-poor conductor of heat** 1× 2020
- **Silver-best conductor, aluminium-most abundant metal, gold-most malleable, lead-least heat conducting** 1× 2002

15.2 Ores and Minerals

- Found in native state: **gold** 1× 2016
- Contains calcium: **gypsum** 1× 2019
- Limestone: **calcium carbonate** 1× 1993
- Doctors and sculptors use **Plaster of Paris** (calcium sulphate) 1× 2004
- **Mica**: **electrical** industry 1× 2010
- **Bauxite**: ore of **aluminium** 2× 2006, 2015
- Wrong pair: **copper-cinnabar** (cinnabar is mercury's ore) 1× 2017
- **Diamond-carbon, marble-calcium, sand-silicon, ruby-aluminium** 1× 2010

Metal	Main ore
Aluminium	Bauxite
Iron	Haematite, magnetite
Copper	Chalcopyrite
Mercury	Cinnabar
Zinc	Zinc blende
Lead	Galena
Tin	Cassiterite

15.3 Corrosion

- Galvanised iron does not rust: **zinc** oxidises first (A and R true, R explains A) 1× 2002
- Rusting needs **oxygen and moisture** 1× 2016
- Rusting **increases** iron's weight 1× 2008
- Rust is brown iron oxide, not **tannin** (A true, R false) 1× 1998
- Silver turns black from **silver sulphide** (both, 2025); the 2008 key is misprinted (*) 2× 2025, 2008

15.4 Alloys

- **Amalgam**: alloy with **mercury** (mercury-zinc) 1× 2012
- **Solder**: **tin and lead** 2× 1996, 2005
- **Bronze**: **copper and tin** 2× 2015, 2016

QUANTUM IAS & PCS

- **German silver**: **0% silver** (copper, zinc, nickel) 1× 2003
- Common to brass, bronze, German silver: **copper**; brass = **copper and zinc** 2× one-liner, one-liner
- True: brass is copper-zinc; amalgams contain mercury (i and iii); magnetite is an **iron** ore 1× 2001
- Not used for stainless steel: **silicon** 2× 2004, 2017
- Stainless steel: iron with **chromium**, nickel, manganese, carbon 2× one-liner, one-liner

Alloy	Made of
Brass	Copper + zinc
Bronze	Copper + tin
German silver	Copper + zinc + nickel
Solder	Tin + lead
Stainless steel	Iron + chromium + nickel + carbon
Amalgam	Mercury + another metal
Duralumin	Aluminium + copper + magnesium

TRAP ALERT

- **Cinnabar** = mercury; **magnetite** = iron; **bauxite** = aluminium.
- German silver has **no silver**; bronze has **tin**, brass has **zinc**.
- Rust **adds** weight (iron oxide); galvanising = **zinc** coat.
- Hardest metal: tungsten among options (chromium is hardest pure metal); densest: osmium.

UP LINK

UP link: **Moradabad** ("Peetal Nagri") is famous for **brassware** (a GI product); **Renukoot** (Sonbhadra) has Hindalco's aluminium smelter, fed by bauxite from Jharkhand and Chhattisgarh.

LAST-MINUTE 10

1. Tungsten = hard, high melting point (bulb filament)
2. Uranium = heaviest natural element
3. Sodium in kerosene; sodium lamps = yellow street lights
4. Mercury = liquid metal; best liquid heat conductor
5. Gold occurs native; bauxite = aluminium; cinnabar = mercury
6. Limestone = CaCO3; Plaster of Paris = calcium sulphate
7. Rust needs O2 + moisture; weight increases
8. Galvanised = zinc; silver tarnish = silver sulphide
9. Solder = Sn + Pb; bronze = Cu + Sn; brass = Cu + Zn
10. German silver = no silver; stainless steel = chromium

Chapter 16

Non-Metals, Gases, Acids, Bases and Salts

Built from 53 UPPSC questions (1993–2025). Focus: carbon's allotropes and fullerene, dry ice, heavy water, gases for divers, balloons and tyres, laughing gas, pH and aqua regia, common salts.

ASKED MOST · REVISE FIRST

- **Fullerene** is a carbon cage (C-60), **not** a fluorine compound 6× 2001, 2002, 2003, 2006, 2010, 2017
- **Dry ice** = **solid carbon dioxide** 3× 2009, 2013, 2014
- **Heavy water**: **deuterium** in place of hydrogen; molecular weight **20** 4× 1993, 2007, 2015, 2016
- **Divers** breathe **oxygen and helium** 3× 2004, 2014, 2016
- **Hypo (sodium thiosulphate)** = photographic fixer; plates coated with **silver bromide** 5× 2010, 2014, 2015, one-liner, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

16.1 Carbon and Its Allotropes

- Most compounds formed by **carbon** 1× 2016
- Most stable form of carbon: **graphite** 1× 2022
- Made of carbon only: **graphene**, a carbon **nanostructure** and the **strongest** material 3× 2015, 2015, 2018
- Not a carbon allotrope: **oxocarbon** 1× 2010
- Fundamental element among options: **diamond** (pure carbon) 1× 1995
- Hardest natural substance: **diamond** 1× 2016
- Pencil "lead": **graphite** 2× 1994, 2006
- Not in the fullerene Nobel team: **"Faimen"**; winners were **Kroto, Curl, Smalley** 1× 2009
- **Buckminsterfullerene**: **60 carbon atoms** in pentagons and hexagons 1× 2010
- Wrong pair: **fullerene-fluorine compound** 5× 2001, 2002, 2003, 2006, 2017
- First made in labs: **Kevlar, fullerene, graphene, black gold** 1× 2019
- No carbon: **sand** (silica) 1× 2013
- Contains carbon: **lignite**; highest-carbon coal: **anthracite** 2× 1993, one-liner
- Most carbon among options: **pig iron** 1× 2014
- **1 carat = 200 mg** 1× 2013
- True: calcium sulphate makes water hard; diamond is harder than copper and iron (1 and 2) 1× 2001
- **Dry ice**: **solid CO2** 3× 2009, 2013, 2014

Form of carbon	Note
Diamond	Hardest natural substance; poor conductor
Graphite	Most stable; pencil lead, lubricant, moderator
Fullerene (C-60)	Football cage; Kroto, Curl, Smalley (Nobel 1996)
Graphene	One-atom sheet; strongest material (Geim, Novoselov, 2010 Nobel)
Carbon nanotube	Rolled graphene (Iijima, 1991)

16.2 Hydrogen and Water

- Commonest element in the universe: **hydrogen** 1× 2007
- Cleanest fuel: **hydrogen** 1× 2015
- **Heavy water**: **deuterium** in place of hydrogen 3× 1993, 2007, 2016
- Heavy water molecular weight: **20** 1× 2015
- Salty to pure water: **reverse osmosis** 1× 2013
- Vanaspati ghee: oil hardened with **hydrogen** 1× 2013

16.3 Noble Gases and Nitrogen

- Not a noble gas: **actinium** 1× 2023
- Divers' breathing gas: **oxygen and helium** 3× 2004, 2014, 2016
- Very soluble in water: **ammonia** 1× 1993
- Laughing gas: **nitrous oxide** (N2O) 3× 2010, 2011, one-liner
- Not correct: **nitric oxide is laughing gas** 1× 2010
- Nitrogen cycle keeps nitrogen in land and air **unchanged** 1× 2008
- Aircraft tyres: **nitrogen** 1× 2014
- Helium over hydrogen in balloons: **no explosive mix** with air 1× 2011
- Lighter than air: **helium** only (HCl and SO2 are heavier) 1× 2025
- Liquid non-metal at room temperature: **bromine** 1× 2017

Gas	Use
Helium	Balloons, divers' mix (with oxygen), MRI cooling
Neon	Advertising signs
Argon	Bulbs, tube lights, welding
Nitrogen	Aircraft tyres, food packing
Nitrous oxide	Laughing gas, anaesthetic
Hydrogen	Vanaspati, cleanest fuel

16.4 Acids, Bases and pH

- Acids have pH **below 7**, bases **above 7** (neither statement right) 1× 2025
- **Aqua regia**: concentrated **HCl and HNO3** (3:1); dissolves gold 2× 2014, 2023

16.5 Common Salts and Their Uses

- **Washing soda-sodium carbonate, caustic soda-sodium hydroxide, neela thotha-copper sulphate, hypo-sodium thiosulphate** 1× 2002
- **Bleaching powder-calcium oxychloride, baking soda-sodium bicarbonate, washing soda-sodium carbonate, Plaster of Paris-calcium sulphate hemihydrate** 2× 2016, 2021
- Baking soda: **sodium hydrogen carbonate**, used in **fire extinguishers** (both) 1× 2025
- Baking soda formula: **$NaHCO_3$** 1× 2024
- Bleaching powder: **calcium oxychloride** 1× 2010
- Salt not for the dining table: all options are edible (*) 1× 2014
- Photography and antichlor: **sodium thiosulphate** 1× 2010
- Photographic fixer: **sodium thiosulphate** (hypo) 2× 2014, one-liner
- Photographic plate: **silver bromide** 2× 2015, one-liner
- Wrong pair: **silver bromide-artificial photosynthesis** 1× 2019

Common name	Chemical
Baking soda	Sodium bicarbonate (NaHCO3)
Washing soda	Sodium carbonate
Caustic soda	Sodium hydroxide
Bleaching powder	Calcium oxychloride
Plaster of Paris	Calcium sulphate hemihydrate
Neela thotha	Copper sulphate
Hypo	Sodium thiosulphate

TRAP ALERT

- **Nitrous** oxide (N2O) = laughing gas; nitric oxide (NO) is not.
- Fullerene is **pure carbon**, not fluorine.
- Acid pH < 7; base pH > 7.
- **Baking soda** = bicarbonate; **washing soda** = carbonate.

UP LINK

UP link: **Phulpur** (IFFCO) and other UP fertiliser plants make **ammonia** from natural gas; **Rae Bareli** (Unchahar) and other UP plants use nitrogen and hydrogen gases in industry (soft detail).

LAST-MINUTE 10

1. Carbon forms most compounds; graphite = most stable
2. Graphene = strongest; diamond = hardest natural
3. Fullerene = C-60; Kroto, Curl, Smalley
4. 1 carat = **200 mg**; anthracite = most carbon
5. Dry ice = solid **CO2**; heavy water = **D2O**, weight 20
6. Divers: **O2 + He**; aircraft tyres: **N2**
7. Laughing gas = **N2O**; helium safer than hydrogen
8. Bromine = liquid non-metal; aqua regia = HCl + HNO3
9. Bleaching powder = **calcium oxychloride**
10. Hypo = fixer; silver bromide = photographic plate

Chapter 17

Organic Chemistry, Polymers, Acids and Explosives

Built from 42 UPPSC questions (1992–2025). Focus: fruit-ripening chemicals, natural and synthetic polymers, bullet-proof polymers, Teflon and Bakelite, natural acids and their sources, explosives.

ASKED MOST · REVISE FIRST

- **Lactic acid**: muscle fatigue, sour milk, curd **5×** 1992, 2004, 2008, 2010, one-liner
- Not an explosive: **nitrochloroform**; RDX (cyclonite) is one **7×** 2001, 2002, 2003, 2014, 2015, one-liner, one-liner
- **Bullet-proof**: **Kevlar (polyamide)** vests, **Lexan (polycarbonate)** windows **5×** 2005, 2010, 2014, 2015, 2017
- **Fruit ripening**: **ethylene, ethephon, calcium carbide** **4×** 2005, 2010, 2011, 2014
- **Acid-source matching**: **acetic-vinegar/pickle, tartaric-grapes/tamarind, lactic-curd, malic-apple, carbonic-soda, methanoic-nettle sting** **7×** 2003, 2008, 2009, 2015, 2017, 2023, 2025

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

17.1 Basic Organic Compounds

- Basis of all organic compounds: **carbon** **2×** 2001, 2015
- Methane source: **paddy fields** **1×** 2014
- Ripening chemicals: **ethephon**, **calcium carbide** (banned), **ethylene** **4×** 2005, 2010, 2011, 2014
- **Wood spirit**: **methyl alcohol** **1×** 2015
- Fermented germinated barley: **beer** **1×** 2016

17.2 Polymers

- Natural rubber: polymer of **isoprene** **1×** 1992
- Not a natural polymer: **nylon** **2×** 2001, 2015
- Natural silk contains **nitrogen** (a protein) **1×** 2023
- Natural polymer: **silk** **1×** one-liner
- Not a natural polymer: **ghee** (a fat) **1×** 2009
- Not synthetic: **fibroin** (silk protein) **1×** 2020
- Polythene from **ethylene** **1×** 2021
- Not a thermoplastic: **neoprene** (an elastomer) **1×** 2020
- Bullet-proof vests: **polyamides** (Kevlar) **3×** 2005, 2010, 2014
- Bullet-proof windows: **polycarbonates** **1×** 2015
- Bullet-proof materials: **Kevlar and Lexan** (I and III) **1×** 2017
- Teflon, Dacron are polymers; neoprene is synthetic rubber; polythene is polyethylene (1, 2, 3); natural rubber is **isoprene** **2×** 2002, 2003
- Non-stick pans: **Teflon** **2×** 2015, one-liner
- Pottery glazing: **zinc oxide** (per key) **1×** 2015
- Phenol makes **Bakelite** **1×** 2010
- Not biodegradable: **PVC** **1×** 2017

Polymer	Made from / use
Natural rubber	Isoprene
Neoprene	Chloroprene; synthetic rubber
Polythene	Ethylene
PVC	Vinyl chloride; pipes; not biodegradable
Teflon	Tetrafluoroethylene; non-stick pans
Bakelite	Phenol + formaldehyde; switches
Nylon	Polyamide; first synthetic fibre
Kevlar	Aramid (polyamide); bullet-proof vests
Lexan	Polycarbonate; bullet-proof glass
Dacron / terylene	Polyester fibre

17.3 Organic Acids in Nature

- Muscle fatigue: **lactic acid** **3×** 1992, 2008, 2010
- Milk turns sour: **lactic acid** **2×** 2004, one-liner
- Wrong pair: **acetic acid-curd** (curd has lactic acid) **1×** 2015
- Wrong pair: **acetic acid-potato** (only 3) **1×** 2025
- **Pickle-acetic, sour milk-lactic, apple-malic, soda water-carbonic** **1×** 2003
- **Vinegar-acetic, tamarind-tartaric, curd-lactic, nettle sting-methanoic (formic)** **1×** 2023
- In grapes, tamarind, banana: **tartaric acid** **2×** 2008, 2017
- Photography acid: **oxalic acid** **1×** 2003
- Baking powder: **tartaric acid** **1×** 2012
- Vinegar: **acetic acid** **1×** 2009

Acid	Found in
Acetic	Vinegar, pickles
Citric	Lemon, orange

QUANTUM IAS & PCS

Acid	Found in
Tartaric	Grapes, tamarind; baking powder
Lactic	Curd, sour milk, tired muscles
Malic	Apple
Oxalic	Tomato, spinach
Formic (methanoic)	Ant and nettle sting
Carbonic	Soda water, cold drinks

17.4 Explosives

- Not an explosive: **nitrochloroform** 5× 2001, 2002, 2003, 2014, 2015
- **RDX** invented by **Henning** (1898); RDX = **cyclonite**, an explosive 3× 2016, one-liner, one-liner
- No nitroglycerine: **amatol** (TNT + ammonium nitrate) 1× 2010
- Explosives made from **glycerol** (nitroglycerine) 1× 2016
- "Nobel oil": **TNG** (nitroglycerine) 1× one-liner

TRAP ALERT

- Curd = **lactic** acid, not acetic; vinegar = **acetic**.
- Natural rubber = **isoprene**; neoprene = **chloroprene**.
- Bullet-proof vest = **Kevlar**; window = **polycarbonate**.
- **Calcium carbide** ripening is banned in India (it gives acetylene and arsenic traces).

UP LINK

UP link: **mango** ripening with banned **calcium carbide** is a recurring FSDA (UP food safety) target in Lucknow, Malihabad and Saharanpur mandis; ethephon and ethylene chambers are the approved methods.

LAST-MINUTE 10

1. Carbon = basis of organic compounds; paddy = methane
2. Ethylene, ethephon ripen fruit; carbide banned
3. Wood spirit = methyl alcohol; barley = beer
4. Natural rubber = isoprene; nylon not natural
5. Kevlar vests; Lexan windows; Teflon non-stick
6. Phenol = Bakelite; PVC not biodegradable
7. Lactic = curd, fatigue; tartaric = grapes, tamarind, baking powder
8. Acetic = vinegar; formic = nettle, ant; carbonic = soda
9. RDX = cyclonite (Henning); TNG = Nobel oil
10. Nitrochloroform is not an explosive

Chapter 18

Fuels

Built from 28 UPPSC questions (1990–2025). Focus: natural gas and CNG, octane and cetane numbers, anti-knock agents, biogas, LPG, gasohol and hydrogen storage.

ASKED MOST · REVISE FIRST

- **Methane** = main part of **natural gas, CNG and biogas** 7× 2005, 2010, 2016, 2025, one-liner, one-liner, one-liner
- **Biogas** = **methane + carbon dioxide**, made by **fermentation** 5× 2009, 2010, 2013, 2016, 2025
- **LPG** = **propane and butane** (key adds ethane) 3× 2013, 2021, one-liner
- **Octane** number rates **petrol**; **cetane** rates **diesel** 2× 2009, 2015
- **Gasohol** = **petrol + ethanol** (not benzene) 2× 2009, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

18.1 Natural and Fossil Fuels

- Natural fuel: **petroleum** 1× 1990
- Cleanest fossil fuel: **natural gas** 1× 2014
- Fossil fuel among options: **natural gas** 1× 2016
- Natural gas is mainly **methane** 1× one-liner

18.2 Engines, Octane and Cetane

- **Octane number**: quality of **petrol** 1× 2009
- **Cetane number**: quality of **diesel** 1× 2015
- Diesel engine burns **diesel vapour and air** (no spark; compression ignition) 1× 1994
- **Carburettor**: in a **petrol** engine 1× 2022
- Anti-knock agent: **tetraethyl lead** (now banned) 2× 1998, one-liner
- Diesel in heavy vehicles: **more power and economy** 1× 1992

PETROL VERSUS DIESEL ENGINES

Petrol engine
Carburettor or injector; spark plug; octane number

↓

Diesel engine
No spark; compression heats air; cetane number

↓

Knocking
Early burning in petrol engines; TEL once stopped it

↓

Unleaded petrol
India from 2000; TEL is now banned

18.3 Biogas, LPG and CNG

- Fuel gas from a biogas plant: **methane** 1× 2016
- Biogas: mainly **methane and CO2** (1 and 3) 2× 2010, 2025
- Gobar gas: mainly **methane** 1× one-liner
- Biogas plant works by **fermentation** (anaerobic) 1× 2013
- Gobar gas plant: **S.V. Desai** (IARI); printed option did not match (*) 1× 2015
- LPG: **propane and butane** (key: ethane, propane, butane) 3× 2013, 2021, one-liner
- **CNG** = **Compressed Natural Gas**; mainly **CH4** 3× 2005, 2015, one-liner

Fuel	Main component
Natural gas, CNG	Methane
Biogas, gobar gas	Methane + CO2
LPG	Propane + butane (ethyl mercaptan added for smell)
Coal gas	Hydrogen + methane + CO
Water gas	CO + hydrogen
Producer gas	CO + nitrogen

18.4 Alcohol Fuels and Hydrogen

- **Gasohol**: **gasoline and ethanol** 1× 2016
- Mandatory ethanol blending: **Brazil** 1× 2015
- True: fossil fuels, geothermal as non-conventional, gobar gas = methane (1, 3, 4); gasohol is **not** benzene-alcohol 1× 2009
- Hydrogen for vehicles stored in a **hydride** 1× 2008

18.5 Petroleum Refining

- Refining by-product: **asphalt** (bitumen) 1× 2013
- **Hydrocarbon Vision 2025**: **storage of petroleum products** (energy security) 1× 2000

TRAP ALERT

- Octane = **petrol**; cetane = **diesel**.
- Carburettor is in **petrol** engines only.
- Gasohol has **ethanol**, not benzene; biogas is not hydrogen-rich.
- LPG smells because **ethyl mercaptan** is added; LPG itself is odourless.

UP LINK

UP link: UP is India's top **ethanol** (sugarcane molasses) producer, feeding the **E20** petrol programme; **CNG** buses run in Lucknow, Agra and Kanpur, and **compressed biogas** plants have come up at Gorakhpur and Budaun (soft detail).

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LAST-MINUTE 10

1. Natural gas = **methane**; cleanest fossil fuel
2. Octane = petrol; cetane = diesel
3. Carburettor = petrol engine; diesel = compression ignition
4. Tetraethyl lead = old anti-knock (banned)
5. Biogas = CH4 + CO2; made by fermentation
6. Gobar gas = S.V. Desai (IARI)
7. LPG = propane + butane; CNG = CH4
8. Gasohol = petrol + **ethanol**; Brazil blends by law
9. Hydrogen stored as **hydride**
10. Asphalt = refining by-product

Chapter 19

Food Chemistry, Fertilisers and Chemistry Miscellany

Built from 65 UPPSC questions (1992–2021). Focus: food preservatives, urea and DAP, pesticides and rat poison, silver iodide for artificial rain, chemical-use matching, phosphorus in matches, battery acid.

ASKED MOST · REVISE FIRST

- **Preservatives**: **sodium benzoate**, benzoic acid; chips packed in **nitrogen** 6× 1992, 1996, 2010, 2013, 2016, one-liner
- **Silver iodide** makes **artificial rain** 4× 2008, 2013, 2020, one-liner
- **Zinc phosphide** = **rat poison** 4× 2011, 2012, 2019, 2020
- **Chemical-use matching**: **acetone-nail polish remover, CCl4-fire extinguisher, freon-refrigerant, TEL-anti-knock, BHC-insecticide, eosin-red ink, blue vitriol-fungicide** 6× 2010, 2011, 2013, 2013, 2014, 2020
- **Urea**: about **46% nitrogen** (amide form); best for **foliar spray**; 2.2 kg gives 1 kg N 6× 2015, 2017, 2017, 2019, one-liner, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

19.1 Food Preservation

- Preservative: **sodium benzoate** (fruit juice) 3× 2010, 2013, one-liner
- Preservative: **benzoic acid** 2× 1992, 1996
- Potato chips packed in **nitrogen** 1× 2016
- Refrigeration **slows biochemical reactions** 2× 2011, 2013
- Decolouring raw sugar: **animal charcoal** 1× 1998

19.2 Sugars and Foods

- Main sugar in honey: **fructose** 1× 2002
- Invert sugar is sweeter than sucrose; it comes from hydrolysing sucrose (A and R true, R does not explain A) 1× 2020
- Best fruit for jelly: **guava** (rich in pectin) 1× 2013
- Organic food: grown **without artificial fertilisers and pesticides** 1× 2014

19.3 Chemicals in Medicine and Daily Use

- Sedative: **potassium bromide** 1× 2010
- Irreversible complex with haemoglobin: **carbon monoxide** 1× 1996
- Clove oil extract: **eugenol** 1× 2010
- Detergents clean well, but they **lower** surface tension (A true, R false) 1× 1997

19.4 Fertilisers

- Plants need chemical fertilisers for phosphorus; manure gives little (A and R true, R explains A) 1× 1998
- Element not supplied by fertilisers: **hydrogen** 1× 2007
- Leaves most acidity in soil: **ammonium sulphate** 1× 1996
- **Urea**: a nitrogen-containing **organic** compound 1× 2017
- Urea nitrogen: about **46-47%**, in **amide** form 2× one-liner, one-liner
- Most soluble at 20 °C among options: **urea** 1× 2017
- Urea for 1 kg nitrogen: **2.2 kg** 1× 2019
- **DAP**: **18% N, 46% P2O5** 1× 2008
- Commonest foliar fertiliser: **urea** 1× 2015

Fertiliser	Nutrient content
Urea	46% N (amide)
DAP	18% N, 46% P_2O_5
Ammonium sulphate	21% N; makes soil acidic
SSP	16% P_2O_5
MOP	60% K_2O
Gypsum	Reclaims alkaline soil

19.5 Pesticides and Agricultural Chemicals

- Fire alarm: **photocell** 1× 2020
- Fireproof clothing: **aluminium sulphate** (per key) 1× 2015
- Mosquito repellent: **pyrethrum** 1× 2009
- Pyrethrin comes from a **seed plant** (chrysanthemum) 1× 2005
- Rat poison: **zinc phosphide** 1× 2012
- Aluminium phosphide: a grain **fumigant** (several right uses, *) 1× 2008
- Herbicide: **sodium chlorate** 1× 2019
- Correct: **ECG-heart, nicotine-tobacco** (2 and 4) 1× 2002
- Alkaline soil reclaimed by **calcium sulphate** (gypsum) 1× 2008
- Soil-related: **edaphic** 1× 2015
- **Humus**: an **organic colloid** 1× 2014

19.6 Uses of Chemicals

- Artificial clouds and rain: **silver iodide** 3× 2008, 2013, one-liner
- **Blue vitriol-fungicide, eosin-red ink, silver iodide-artificial rain, zinc phosphide-rodenticide** 1× 2020
- **Philosopher's wool**: **zinc oxide** 1× 2007
- Correct pair: **zinc phosphide-rat poison** 1× 2011
- **Iron-haemoglobin, lead-storage battery, silver-photography, copper-lightning conductor** 1× 2010
- **Morphine-analgesic, sodium-kerosene, boric acid-antiseptic, German silver-alloy** 1× 2011
- "Metal of the future": **titanium** 1× 2010
- Wrong pair: **silver-electric bulb** (bulbs use tungsten) 1× 2002
- Red ink: **eosin** 1× 2014
- Sodium chromate gives **red** (per key; it is yellow) 1× 2016
- **Acetone-nail polish remover, CCl_4-fire extinguisher, H_2O_2-wound dressing, liquid ammonia-refrigerant** 1× 2013
- **Freon-refrigerant, TEL-anti-knock, BHC-insecticide, CCl_4-fire extinguisher** 1× 2013
- **Kevlar-synthetic fibre, Taxol-anticancer, zinc phosphide-rodenticide, nitrocellulose-explosive** 1× 2019

19.7 Glass, Batteries and Matches

- Glass that cuts UV: **Crookes glass** 1× 2021
- Wrong pair: **HCl-accumulator** (car batteries use **sulphuric acid**) 1× 1998
- Safety matches: **red phosphorus** 2× 2016, one-liner
- Glows in air and darkness: **white phosphorus** 1× 2016
- Dry cell cannot be recharged; its reaction is **irreversible** (A true, R false) 1× 2004
- Car battery acid: **sulphuric acid** 2× 2015, 2016

19.8 Miscellany

- Ozone kills microbes; it is not radioactive (A and R true, R does not explain A) 1× 2013
- **Radioactive dating**: age of **rocks** 1× 2008
- Age of the Earth: **uranium** method 1× 1992
- Desalination plant: **Lakshadweep** (Kavaratti) and Chennai 1× 2007
- **Mustard gas**: **chemical warfare** 1× 2008
- Smoke screens: **titanium oxide** particles 1× 2009
- Incorrect: **nuclear splitting is fusion** 2× 2004, 2010
- Correct pair: **ozone depletion-skin cancer** 1× 2015
- **Technology Day 11 May, Photography Day 19 August, Sports Day 29 August, Nagasaki Day 9 August** 1× 2003

QUANTUM IAS & PCS

TRAP ALERT

- Car battery = **sulphuric acid**; dry cell cannot be recharged.
- Safety matches = **red** phosphorus; white phosphorus glows and is stored in water.
- Zinc phosphide = rat poison; aluminium phosphide = grain fumigant.
- Gypsum fixes **alkaline** soil; lime fixes **acidic** soil.

UP LINK

UP link: UP is the largest user of **urea**; plants at **Phulpur** (IFFCO), **Gorakhpur** (HURL, restarted 2021), **Babrala**, **Aonla** and **Shahjahanpur** make it; IIT Kanpur ran **cloud-seeding** (silver iodide) trials in 2023.

LAST-MINUTE 10

1. Sodium benzoate = preservative; chips in **nitrogen**
2. Honey = fructose; guava = jelly; eugenol = clove oil
3. CO binds haemoglobin; KBr = sedative
4. Urea 46% N; DAP 18-46; ammonium sulphate = acidity
5. Pyrethrum = mosquito coil; zinc phosphide = rat poison
6. Silver iodide = artificial rain; eosin = red ink
7. Philosopher's wool = **ZnO**; titanium = metal of future
8. Freon = refrigerant; CCl4 = fire extinguisher; acetone = nail polish remover
9. Red phosphorus = safety matches; H2SO4 = car battery
10. Technology Day 11 May; Sports Day 29 August

Chapter 20

Branches of Biology, Evolution and Classification

Built from 82 UPPSC questions (1990–2025). Focus: branches and "-cultures", evolution, odd animals (seahorse, bat, echidna, spider), insectivorous plants, edible plant parts, plant pigments.

ASKED MOST · REVISE FIRST

- **Edible parts**: potato = **stem** (tuber), turmeric and ginger = **rhizome** (stem), sweet potato = **root**, clove = **flower bud** **11×** 2002, 2003, 2005, 2006, 2010, 2011, 2013, 2015, 2021, 2021, one-liner
- **Nepenthes** = **insectivorous** pitcher plant (Meghalaya); traps insects for **nitrogen** **6×** 2011, 2013, 2016, 2017, 2023, one-liner
- **Gerontology** = study of **ageing**; osteology = bones; demography = population **5×** 1990, 1994, 2008, 2014, 2017
- **Spider is not an insect**; bat is not a bird; seahorse **is** a fish **4×** 2008, 2008, 2010, 2011
- **Pigments**: apple red = **anthocyanin**; tomato = **lycopene**; papaya = **caricaxanthin**; chilli heat = **capsaicin** **5×** 2002, 2008, 2015, 2016, one-liner

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

20.1 Branches of Biology

- Zoology: **living and dead** animals **1×** 1995
- **Birds-ornithology, inheritance-genetics, environment-ecology, fossils-palaeobiology** **1×** 1998
- Population study: **demography** **2×** 1994, 2017
- Wrong pair: **vegetable growing-horticulture** (that is **olericulture**) **1×** 2015
- Wrong pair: **pisciculture-lac insects** (pisciculture = fish) **1×** 2004
- **Viticulture**: **grapes** **1×** 2003
- **Vermiculture**: **earthworms** **1×** 2005
- Ageing: **gerontology** **2×** 2008, 2014

- Bones: **osteology** 1× 1990

Culture	Rearing / growing of
Apiculture	Honeybees
Sericulture	Silkworms
Lac culture	Lac insects
Pisciculture	Fish
Viticulture	Grapes
Olericulture	Vegetables
Floriculture	Flowers
Vermiculture	Earthworms

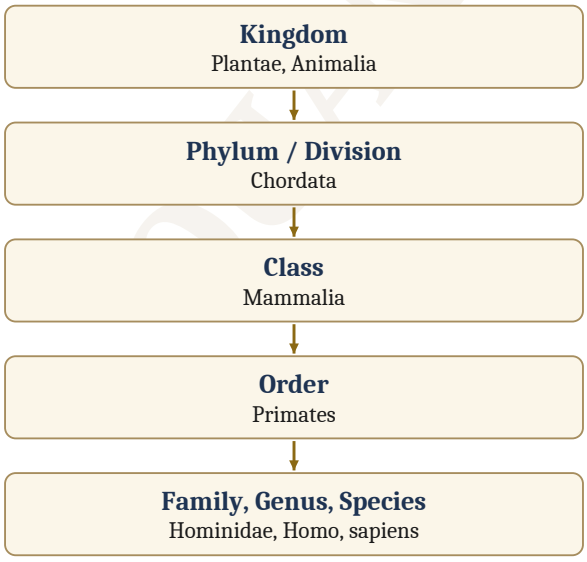
20.2 Origin and Evolution of Life

- Absent in the early atmosphere: **oxygen** 1× 2016
- Oldest organism: **blue-green algae** (cyanobacteria) 1× 2008
- **Mutation theory**: **Hugo de Vries** 1× 2005
- Dinosaurs: **Mesozoic** reptiles 2× 1998, 2001

20.3 Classification and Microbes

- Order from higher to lower: **kingdom, phylum, class, order** 1× 2024
- Bacteria seen with a **compound microscope** 1× 1995
- Some bacteria make food, but they lack **chloroplasts** (A true, R false) 1× 2000
- Statements about bacteria: **none** correct 1× 1995

TAXONOMIC RANKS



20.4 The Animal World

- A fish: **seahorse** 1× 2008
- Largest toothed whale: **sperm whale** 1× 2015
- Dolphins and whales breathe through **blowholes**; earthworms through **skin** (both) 1× 2025
- Earthworms are farmers' friends; they have **five pairs** of hearts, not two (only 1) 1× 2025
- Not a bird: **bat** (a mammal) 1× 2011
- Amphibians tolerate mixed blood; their heart has **three** chambers (A true, R false) 1× 2023
- Egg-laying mammal: **echidna** (and platypus) 1× 2022
- Smallest ape: **gibbon** 1× 2016
- **Nilgai**: key says deer family (it is an **antelope**, Bovidae) 1× 2006
- No hooves: **jackal** 1× 2017
- **Octopus**: a **mollusc** 1× 2004
- Largest invertebrate: **squid** 1× 2004

20.5 Insects and Silk

- Not an insect: **spider** (8 legs) 2× 2008, 2010
- White (colourless) blood: **cockroach** 1× 2019
- Spider silk: **gossamer** 2× 2017, 2021
- Not from insects: **pearl** (from oysters) 1× 2018
- Not a beneficial insect: **Helicoverpa armigera** (pod borer) 1× 2023
- Only **female** mosquitoes suck blood and have **bigger wings** (2 and 5) 1× 2016

20.6 Animals of Special Habitats

- Desert animals: **xericoles** 1× 2023
- Longest without water: **kangaroo rat**; near-solid urine 2× 2018, one-liner
- Snake fangs: modified **maxillary teeth** 1× 2016

20.7 Special Plants

- Wrong pairs: **Nepenthes-parasite**, **Nepenthes-hydrophyte** 2× 2016, 2017
- Insectivorous plant: **Nepenthes** 1× 2023
- **Nepenthes khasiana**: **Meghalaya** 1× 2011
- Insectivorous plants make up for **nitrogen** 2× 2013, one-liner
- **Mycorrhiza**: **fungi and plant roots** 1× 2019
- **Leghaemoglobin**: **legume root nodules** 1× 2005

20.8 Edible Parts of Plants

- **Lychee**: a **drupe** (per key) 1× 2010
- Turmeric: **stem**, a **rhizome** 2× 2006, 2021
- Modified root: **sweet potato** 2× 2005, 2010
- A stem: **ginger** 1× 2015
- Wrong pair: **potato-root** (it is a stem tuber) 2× 2011, 2013
- **Clove**: dried **flower buds** 4× 2002, 2003, 2021, one-liner
- Wrong pair: **saffron-seed** (saffron = stigma) 2× 2001, one-liner
- Garlic smell: a **sulphur compound** (allicin) 1× 2008
- Onion tears: **sulfenic acid** (syn-propanethial oxide) 1× 2015
- Chilli heat: **capsaicin** 2× 2008, 2016
- **Mulberry**: a **sorosis** fruit 1× 2006
- Not a stem fibre: **cotton** (a seed hair) 1× 2002
- Man-made cereal: **triticale** (wheat x rye) 1× 2017

Plant	Edible part
Potato	Stem tuber
Ginger, turmeric	Rhizome (stem)
Onion, garlic	Bulb (leaf bases)
Carrot, radish, sweet potato	Root
Clove	Flower bud
Saffron	Stigma
Cauliflower	Inflorescence

20.9 Ecological Terms and Pigments

- Tolerate wide temperature ranges: **eurythermal** 1× 2023
- Desert roots are long to **seek water**, not because of hot soil (A true, R false) 2× 1998, one-liner
- Desert plants: **succulent** 1× 2013
- Water absorbed mostly by **root hairs** 1× 2007
- **Hydrophyte**: aquatic plant 1× 1994
- Land dwellers: **terrestrial** 1× 2024
- Wrong pair: **herbivore-secondary consumer** (it is primary) 1× 2000
- Fish survive under ice: **bottom water does not freeze** 1× 2010
- Red apple: **anthocyanin**; red tomato: **lycopene** 2× 2002, one-liner
- Yellow papaya: **caricaxanthin** 1× 2015
- **Yeast** and **mushroom**: **fungi** 2× 2008, 2013
- Cotton is mostly **cellulose** 1× 2009

TRAP ALERT

- Potato = **stem**; sweet potato = **root**; turmeric = rhizome (stem).
- Spider (arachnid) is not an insect; bat is a mammal; seahorse **is** a fish.
- Nepenthes = insectivorous, not parasitic or aquatic.
- Saffron = **stigma**, not seed; clove = flower bud.

UP LINK

UP link: **NBRI Lucknow** keeps a large botanic garden and herbarium; **Nilgai** (blue bull) raids crops across UP; **Bhadohi-Mirzapur** carpets and **Varanasi** silk link to sericulture and wool, and **Malihabad** is famous for mango horticulture.

LAST-MINUTE 10

1. Gerontology = ageing; osteology = bones; viticulture = grapes
2. Blue-green algae = oldest; de Vries = mutation; dinosaurs = Mesozoic
3. Seahorse = fish; bat = mammal; echidna lays eggs
4. Spider not an insect; cockroach = white blood
5. Only female mosquitoes bite
6. Nepenthes = insectivorous (nitrogen); mycorrhiza = fungus + root
7. Potato = stem; sweet potato = root; clove = bud; saffron = stigma
8. Capsaicin = chilli; anthocyanin = apple; lycopene = tomato
9. Triticale = man-made cereal; cotton = cellulose
10. Kangaroo rat = never drinks; desert plants = succulent

Chapter 21

Cell, Heredity and Biofertilisers

Built from 30 UPPSC questions (1991–2024). Focus: cell organelles, Mendel and DNA, chromosome number, Azolla and blue-green algae, Rhizobium and other nitrogen fixers.

ASKED MOST · REVISE FIRST

- ■ **Azolla** (aquatic fern, with blue-green algae) = biofertiliser for **paddy** 6× 2008, 2009, 2011, 2023, one-liner, one-liner
- ■ **Blue-green algae** fix nitrogen, chiefly in **paddy** fields 4× 2006, 2010, 2019, one-liner
- ■ **Watson and Crick**: **double helix** of DNA; DNA sugar = **deoxyribose** 4× 2012, 2016, 2018, one-liner
- Humans have **46** chromosomes 2× 1991, 1992
- **Mitochondria** = power house; **cell wall** marks a plant cell 2× 1991, 2013

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

21.1 The Cell

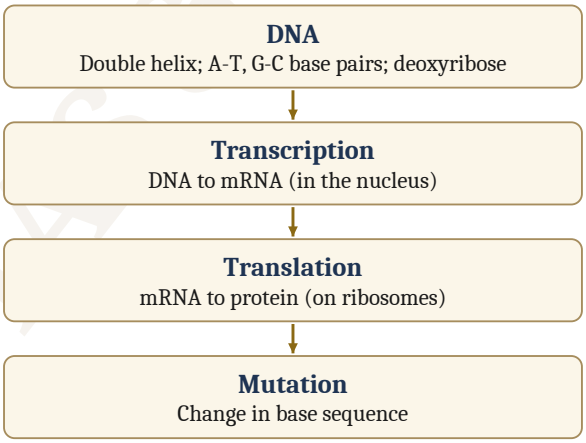
- Plant cell differs from animal cell by its **cell wall** 1× 1991
- Power house of the cell: **mitochondria** 1× 2013
- Plasma membrane: **lipids and proteins** (both) 1× 2008
- Not true: the nucleus has **no role** in reproduction (it does) 1× 2024

Organelle	Job
Nucleus	Controls the cell; holds DNA
Mitochondria	Respiration; ATP (power house)
Chloroplast	Photosynthesis (plants)
Ribosome	Protein synthesis
Lysosome	Digestion ("suicide bags")
Golgi body	Packaging and secretion
Cell wall	Plants only; cellulose

21.2 Heredity and DNA

- Mendel's laws: worked on the **garden pea** 1× 2002
- **Genome mapping**: mapping **genes** 1× 2002
- Change in a gene's base sequence: **mutation** 1× 2020
- Unit of heredity: **gene** 1× 2010
- Not a part of DNA among options: **tyrosine** (an amino acid) 1× 1994
- DNA sugar: **deoxyribose** 1× 2016
- ■ Double helix: **Watson and Crick** (1953) 3× 2012, 2018, one-liner
- Human chromosomes: **46** (23 pairs) 2× 1991, 1992
- No sex chromosomes: **lizard** (per key) 1× 2004

FROM GENE TO PROTEIN



21.3 Biofertilisers

- **Biofertilisers**: **cultures of micro-organisms** 1× 2023
- Azolla and blue-green algae are biofertilisers; alfalfa is a crop (*) 2× 2001, 2002
- Not a nitrogen fixer: **mycorrhiza** 1× 2017
- Nitrogen-fixing bacteria live in roots of **pulses** 1× 2019
- Wrong pair: **Azotobacter-bioinsecticide** (it fixes nitrogen) 2× 2002, 2003
- Blue-green algae: biofertiliser for **paddy** 2× 2006, 2010
- Biological nitrogen fixation: **blue-green algae** 1× 2019
- Azolla works because of its partner **blue-green alga** (Anabaena) 1× 2011
- Azolla is an **aquatic fern** 2× one-liner, one-liner
- Biofertiliser microbes: **cyanobacteria** 1× one-liner

QUANTUM IAS & PCS

- Water-logged rice fields: **Azolla** 1× 2023
- Biofertiliser source: **Azolla** 2× 2008, 2009
- Nitrogen-use efficiency improved by **all of the above** 1× 2009
- Phosphate biofertiliser: **VAM** (vesicular arbuscular mycorrhiza) 1× 2019
- Soybean nitrogen fixer: **Rhizobium japonicum** 1× 2006

Biofertiliser	Use
Rhizobium	Legume root nodules (pulses, soybean)
Azotobacter	Free-living nitrogen fixer
Blue-green algae (Anabaena, Nostoc)	Paddy fields
Azolla + Anabaena	Paddy fields
VAM / mycorrhiza	Phosphorus uptake

TRAP ALERT

- Mycorrhiza helps with **phosphorus**, not nitrogen fixation.
- Azotobacter is a nitrogen fixer, not a bioinsecticide.
- DNA sugar = deoxyribose; RNA sugar = ribose.
- Cell wall is in **plants**; centrioles are in **animals**.

UP LINK

UP link: UP's paddy belt in the east (Gorakhpur, Basti, Deoria) is promoted for **Azolla** and blue-green algae; the **Indian Institute of Pulses Research** at **Kanpur** works on Rhizobium-rich pulse crops.

LAST-MINUTE 10

1. Cell wall = plant cell; mitochondria = power house
2. Mendel = garden pea; gene = unit of heredity
3. Watson-Crick = double helix (1953); DNA sugar = deoxyribose
4. Humans = **46** chromosomes
5. Mutation = change in base sequence
6. Biofertilisers = microbe cultures
7. Azolla = aquatic fern + Anabaena; paddy
8. Blue-green algae = paddy nitrogen
9. Rhizobium japonicum = soybean
10. VAM = phosphate; mycorrhiza does not fix N

Chapter 22

Human Body Systems

Built from 90 UPPSC questions (1991–2025). Focus: bones and body make-up, digestion and enzymes, heart and blood, haemoglobin and blood groups, kidneys, brain parts, hormones and glands.

ASKED MOST · REVISE FIRST

- **Adrenaline** = "**fight or flight**" hormone 5× 2001, 2002, 2003, 2004, 2010
- **Haemoglobin**: has **iron**, makes blood **red**, carries **oxygen**; gives no immunity 6× 2001, 2004, 2005, 2016, 2019, one-liner
- **O** group = **universal donor** (O negative in emergencies); blood groups found by **Landsteiner** 4× 2003, 2006, 2016, one-liner
- **Enzymes are proteins**; protein digestion starts in the **stomach** (pepsin) 5× 1996, 2004, 2013, 2020, 2025
- **206** bones; smallest = **stapes** (ear) 4× 2004, 2008, 2010, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

22.1 Composition and Skeleton

- Water in an adult body: about **65%** 1× 2007
- Most abundant element: **oxygen** 2× 2013, 2014

- Increasing amount in a 70 kg body: **copper, iron, sodium, potassium** 1× one-liner
- Bones in an adult: **206** 1× 2004
- Smallest bone: **stapes**, in the **ear** 3× 2008, 2010, one-liner
- Head-neck joint: **pivot** joint 1× 2024
- Main salt in bones: **calcium phosphate** 1× 2015
- **Arthroplasty**: **hip-joint replacement** 2× 2001, 2002
- Bones joined by **ligaments** 1× 2008
- Dental plaque: **food, saliva, acids and bacteria** 1× 2004
- Bones and teeth need **calcium and phosphorus** 1× 2011
- Not needed for bones and teeth: **iodine** 1× 2015
- Strongest muscle: **jaw** (masseter) 1× 2008

22.2 Digestion

- Most digestion: **small intestine** 1× 1991
- Enzymes are **proteins** 3× 1996, 2004, 2013
- Fat digestion needs **bile and lipase** (1 and 2) 1× 2016
- Protein digestion begins in the **stomach** 1× 2020
- Protein-digesting enzyme: **trypsin** (among options) 1× 2010
- Released in the stomach: **pepsin** only (only 3) 1× 2025
- Humans cannot digest cellulose; diastase digests starch (A and R true, R does not explain A) 1× 2017
- Extra glucose stored as **glycogen** in the **liver** 1× 2004
- Fasting over 10 days: liver **glucose falls** 1× 2021
- **Bile** is made in the **liver** 1× 1999
- Gallstones affect **fat** digestion 1× 2017
- Proteins to amino acids: **pepsin** (per key) 1× 2017
- **Vitamin-carotene, enzyme-pepsin, hormone-testosterone, protein-keratin** 2× 1998, 2021

Enzyme	Where	Acts on
Salivary amylase (ptyalin)	Mouth	Starch
Pepsin	Stomach	Proteins
Trypsin	Pancreas (small intestine)	Proteins
Lipase	Pancreas	Fats (with bile)
Bile (not an enzyme)	Liver	Emulsifies fats

22.3 Respiration

- Gas exchange in the lungs: **alveoli** 1× 2021
- Less oxygen in blood: breathing **increases** 1× 2000
- Most energy: **ATP to ADP** 1× 2015
- Energy is stored as **ATP** 1× 2016

22.4 Heart and Blood

- Human heart: **four** chambers 1× 2023
- Only vein with oxygenated blood: **pulmonary vein** 1× 2024
- Heart rests **between two beats** 2× 2004, 2008
- Wrist pulse beats **at the heart's rate** 2× 2008, 2010
- Blood in a 70 kg adult: about **5-5.6 litres** (*) 1× 2017
- Blood: plasma and cells; **haemoglobin** makes it red (both) 2× 2025, one-liner
- Blood is a **connective tissue** 1× 2023
- Not correct of haemoglobin: "**slightly acidic**" (per key) 1× 2019
- Haemoglobin: **iron, red colour, oxygen carrier** (1, 2, 4) 4× 2001, 2004, 2005, 2016
- Contains iron: **myoglobin** 1× 2020
- Average BP: **120/80** mm Hg; **Hg = mercury** 3× 1991, 1993, 2015
- Universal donor: **O** 2× 2003, 2016
- Emergency transfusion: **O negative** 1× 2006
- Blood groups: **Karl Landsteiner** 1× one-liner
- Blood pH: **7.35-7.45** 1× 2007
- RBCs destroyed (and some WBCs formed) in the **spleen** 1× 2008
- Blood sugar in **mg per decilitre** 1× 2021
- **Thrombin**: **blood clotting** 1× 2011
- Alum stops bleeding: Al ions coagulate colloidal blood (A and R true, R explains A) 1× 2010
- "Plastic blood": developed in **Britain** 1× 2006

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Blood group	Can give to	Can receive from
O negative	All (universal donor)	O negative only
AB positive	AB positive only	All (universal recipient)
A	A, AB	A, O
B	B, AB	B, O

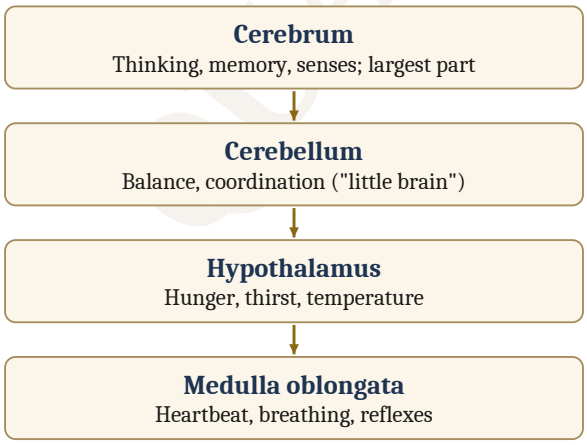
22.5 Excretion

- Detoxifies alcohol: **liver** 1× 2010
- Urine a day: about **1.5 litres** 1× 2004
- Correct: **all the blood** is filtered through the kidneys 1× 1994
- Urea removed from blood in the **kidney** 1× 2002
- **Dialyser**: does the **kidney's** work 1× 2002
- Not a kidney job: **blood sugar regulation** 1× 2011
- Kidney stones: **calcium oxalate** crystals 2× 2015, 2023

22.6 Nervous System

- Brain controls **all three** (heart rate, balance, thought) 2× 2016, one-liner
- "Little brain": **cerebellum** 1× 2023
- Feels no pain from a needle: **brain** tissue 1× 2017
- Body temperature **stays the same** in winter and summer 1× 2015
- Hunger and thirst centres: **hypothalamus** 1× 2010
- Normal temperature: **98.6 °F (37 °C)** 1× one-liner

PARTS OF THE BRAIN



22.7 Endocrine Glands and Hormones

- Urine volume: **ADH** (vasopressin) 1× 2010
- **Progesterone-ovaries, testosterone-testes, thyroxine-thyroid, insulin-pancreas** 1× 2010
- Both endocrine and exocrine: **pancreas** 1× 2019
- Second-largest gland, alkaline complete digestive juice: **pancreas** 1× 2008
- Stimulates pancreatic juice: **secretin** 1× 1996

■ **Insulin**: a **protein**; from **beta cells** of the islets of Langerhans 3× 1993, 2015, one-liner

- Wrong pair: **gastrin-blood pressure** (gastrin makes stomach acid) 1× 2017

■ Mother's hug and kiss release **oxytocin**; it also induces milk letdown in cattle 3× 2002, 2003, one-liner

- Not a protein (per key): **oxytocin** (a peptide) 1× 2008

■ **Thyroxine** and **estrogen** are **hormones** 3× 2000, 2003, 2004

■ **Fight or flight**: **adrenaline** 5× 2001, 2002, 2003, 2004, 2010

- Iodine deficiency: **goitre** 1× one-liner
- After menopause: **progesterone** stops; no option exact (*) 1× 2014
- Female sex hormone: **estrone** 1× 2017
- **Hormone-testosterone, enzyme-lipase, phospholipid-lecithin, polymer-polythene** 1× 1997
- Bright light is bad for sleep: light **lowers** melatonin (A true, R false) 1× 2014
- Smallest endocrine gland: **pituitary** per key (pineal is smaller) 1× 1996
- Sperm stay fertile for **one to two days** 1× 2010

TRAP ALERT

- **Pulmonary vein** carries oxygenated blood; pulmonary artery carries deoxygenated.
- Bile is made in the **liver**, stored in the gall bladder; it is not an enzyme.
- Melatonin rises in **darkness**, not light.
- **O** = universal donor; **AB** = universal recipient.

QUANTUM IAS & PCS

UP LINK

UP link: **SGPGIMS** and **KGMU** (Lucknow) run UP's major kidney transplant and dialysis units; the **PM National Dialysis Programme** gives free dialysis at district hospitals across UP.

LAST-MINUTE 10

1. Body water **65%**; oxygen = most abundant element
2. **206** bones; stapes smallest; jaw = strongest muscle
3. Enzymes = proteins; pepsin in the stomach
4. Bile from the liver; gallstones hit fat digestion
5. Alveoli = gas exchange; ATP = energy currency
6. Pulmonary vein = oxygenated; BP 120/80
7. O = universal donor; Landsteiner = blood groups; pH 7.35-7.45
8. Spleen = RBC graveyard; kidney stones = calcium oxalate
9. Cerebellum = balance; hypothalamus = hunger, thirst
10. Adrenaline = fight or flight; ADH = urine; oxytocin = milk letdown

Chapter 23

Vitamins and Nutrition

Built from 92 UPPSC questions (1990–2024). Focus: water- and fat-soluble vitamins, vitamin names and deficiency diseases, rich sources, proteins and amino acids, minerals, milk.

ASKED MOST · REVISE FIRST

- **Deficiency matching**: **A-night blindness, B1-beriberi, C-scurvy, D-rickets, B12-pernicious anaemia, folic acid-anaemia** **9×** 1999, 2002, 2010, 2015, 2016, 2017, 2023, 2023, 2024
- **Water-soluble**: **B and C**; fat-soluble: A, D, E, K **4×** 2002, 2019, 2021, 2023
- **Vitamin K** = **blood clotting** **4×** 1991, 2008, 2012, 2020
- **B12** has **cobalt** **6×** 2000, 2001, 2004, 2006, 2014, one-liner
- Highest protein: **soybean**; best all-amino-acid food: **egg** **4×** 2004, 2008, 2014, 2021

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

23.1 What Vitamins Are

- **Nutraceuticals**: nutrient plus medicinal effect **1×** 2009
- Vitamins named by **Casimir Funk** **1×** 2012
- "Accessory dietary factors": **vitamins** **1×** 2014
- Vitamins are needed in small amounts; they do **not kill bacteria** (A true, R false) **1×** 2002
- Not made in our body: **vitamin A** (made from carotene) **2×** 2011, 2013
- Made by gut bacteria: **vitamin K** and **B12** **1×** 2010
- Water-soluble: **B and C** **3×** 2002, 2021, 2023
- Not correct: **vitamin K is water-soluble** **1×** 2019
- B12 is stored in the liver for years; being water-soluble does not explain it (A and R true, R does not explain A) **1×** 2017
- Most vitamins are not made in the body (A true, R false) **1×** 2019
- The body does **not** make all vitamins; vitamins are essential (A false, R true) **2×** 2016, 2021

23.2 The B Vitamins

- **B1-thiamine, B2-riboflavin, B6-pyridoxine, B12-cobalamin** **2×** 2010, one-liner
- Wrong pair: **riboflavin-beriberi** (riboflavin lack causes **cheilosis**) **2×** 2002, one-liner
- Polished rice causes **beriberi** **2×** 2019, 2020
- Wrong pair: **B2-cod-liver oil** **1×** 2019

QUANTUM IAS & PCS

- Cobalt-containing vitamin: **B12** 4× 2001, 2004, 2006, one-liner
- **Thiamine** = **B1** 1× 1999
- Helps remember dreams: **B6** (per key) 2× 2010, 2013

23.3 Vitamin C

- Most vitamin C among vegetables: **chilli** 1× 2016
- Citrus vitamin for healthy skin: **C** 1× 1999
- Best source: **amla** 2× 1995, 2018
- Wound healing: **vitamin C** 1× one-liner
- Vitamin C helps absorb **iron** 1× 2010
- Not stored in the body: **vitamin C** 1× 2014

23.4 Vitamins A, D, E and K

- Vitamin D source: **sunlight**; made in the skin 2× 1993, one-liner
- Wrong pair: **vitamin D-antisterility** (that is E) 1× 2017
- Vitamin D: **calciferol** (cholecalciferol) 2× 1997, one-liner
- Night blindness: lack of **vitamin A** 1× 2006
- Wrong pair: **vitamin A-orange** (orange is vitamin C) 1× 2021
- Vitamin A stored in the **liver**; **retinol** for vision 2× 1997, one-liner
- Quitting greens alone need not cause night blindness (A false, R true) 1× 2011
- Rich in vitamin A: **carrot**; richest among options: **mango** 2× 2004, 2015
- Not a natural vitamin A source (per key): **milk** 1× 2009
- Vitamin E: **wheat germ oil** 1× 2016
- Clotting: **vitamin K** 4× 1991, 2008, 2012, 2020
- Wrong pair: **vitamin K-sterility** 1× 2008
- **K-alfalfa, D-cod-liver oil, E-wheat germ oil, C-lemon** 1× 2016

Vitamin	Name	Deficiency	Source
A	Retinol	Night blindness, xerophthalmia	Carrot, mango, liver
B1	Thiamine	Beriberi	Whole grains (not polished rice)
B2	Riboflavin	Cheilosis	Milk, eggs
B6	Pyridoxine	Anaemia, nerve problems	Grains, meat
B12	Cobalamin (cobalt)	Pernicious anaemia	Animal foods
C	Ascorbic acid	Scurvy	Amla, citrus, chilli
D	Calciferol	Rickets	Sunlight, cod-liver oil
E	Tocopherol	Infertility	Wheat germ oil
K	Phylloquinone	Poor clotting	Green leaves, alfalfa

23.5 Deficiency Diseases

- **A-blindness, B-beriberi, C-scurvy, D-rickets** 1× 2024
- **Night blindness-A, rickets-D, scurvy-C, beriberi-B** 3× 2010, 2015, 2023
- **C-scurvy, folic acid-anaemia, A-night blindness, B1-beriberi** 1× 2023
- Wrong pairs: **vitamin D-colour blindness, vitamin A-rheumatism** 2× 2016, 2017
- **Retinol-conjunctivitis, tocopherol-sterility, cyanocobalamin-pernicious anaemia, pyridoxine-mental illness** 2× 1999, 2002
- Not a vitamin: **thyroxine** 1× 2017

23.6 Energy Foods

- Most energy: **carbohydrate, fat, protein** together 1× 2021
- Most calories per gram: **fats** (9 kcal) 2× 1999, one-liner
- Apples help heart patients: rich in **potassium** 1× 2014

QUANTUM IAS & PCS

- Not a micronutrient: **sulphur** (a secondary macronutrient) 1× 2014
- Banana: about **116 kcal** per 100 g (per key) 1× 1998
- A glass of water: **zero** calories 1× 2015
- Essential diet constituent: **carbohydrate** 1× 2014
- No nitrogen in **carbohydrates** 1× one-liner
- Basmati grains lengthen on cooking: high **amylose** 1× 2015

23.7 Proteins and Amino Acids

- About 80% of living material (dry): **protein** 1× 1994
- Protein need: **higher in the old** (per key) 1× 2013
- Most protein: **soybean** 3× 2004, 2008, 2014
- Rich in protein and fat: **groundnut** 1× 2008
- Not correct of tyrosine: it guards against ageing free radicals 1× 2010
- Bread quality: **gluten** 1× 2010
- Tissues are built from **protein** 1× 1990
- **Alpha-keratin**: **skin**; hair and nails have **keratin** 2× 2003, one-liner
- Not a protein: **cotton** (cellulose) 1× 2019
- Does not denature protein: **infrared** rays 1× 2016
- Enzymes: **biocatalysts**, act where produced (both, per key) 1× 2019
- Not stored in the body: **amino acids** 1× 2015
- All essential amino acids: milk, egg, soybean (*); best: **egg** 2× 2016, 2021
- Sunflower oil is good; it is rich in **polyunsaturated fats** (A and R true, R explains A) 1× 2019

23.8 Minerals

- Acid-alkali balance: **phosphorus** (per key) 1× 2004
- Iron-rich fruit: **karonda**; iron-rich: **spinach** 2× 2006, 2008
- **B12-cobalt, haemoglobin-iron, chlorophyll-magnesium, brass-copper** 2× 2000, 2014
- Wrong pair: **washing soda-potassium** (it is sodium) 1× 2014

23.9 Milk

- Sweetness of milk: **lactose** 2× 2007, one-liner
- Cow milk is yellowish: **carotene** with casein; white colour: **casein** 3× 2014, one-liner, one-liner
- **Casein**: milk **protein** 1× 2008
- Wrong pair: **saccharin-carbohydrate** (it is an artificial sweetener) 1× 2017
- Lactating mother needs about **70 g** protein a day (per key) 1× 2008
- Mother's milk lacks **iron** 1× 2019

TRAP ALERT

- Water-soluble = **B and C** only; K is **fat**-soluble.
- B12 = **cobalt**; chlorophyll = **magnesium**; haemoglobin = **iron**.
- Vitamin E = fertility; vitamin K = clotting; vitamin D = rickets.
- Riboflavin (B2) lack = cheilosis, not beriberi.

UP LINK

UP link: UP's **Poshan Abhiyan** drives fight child **anaemia and stunting**; fortified rice (with iron, folic acid, B12) is now given through the PDS and mid-day meals in UP, and **Anganwadi** centres give take-home rations.

LAST-MINUTE 10

1. Funk named vitamins; vitamins = accessory factors
2. Water-soluble **B, C**; fat-soluble **A, D, E, K**
3. B1 = beriberi (polished rice); B2 = cheilosis
4. B12 = cobalt, pernicious anaemia
5. C = scurvy; amla best; helps iron absorption
6. D = sunlight, calciferol, rickets
7. A = retinol, night blindness, stored in liver
8. E = wheat germ oil, fertility; K = clotting
9. Fats = most calories; soybean = most protein; egg = all amino acids
10. Milk: lactose sweet, casein white, carotene yellow; lacks iron

Chapter 24

Plant Physiology

Built from 38 UPPSC questions (1996–2023). Focus: photosynthesis inputs and outputs, xylem and phloem, plant nutrients, plant hormones and their roles, crop practices and hydroponics.

ASKED MOST · REVISE FIRST

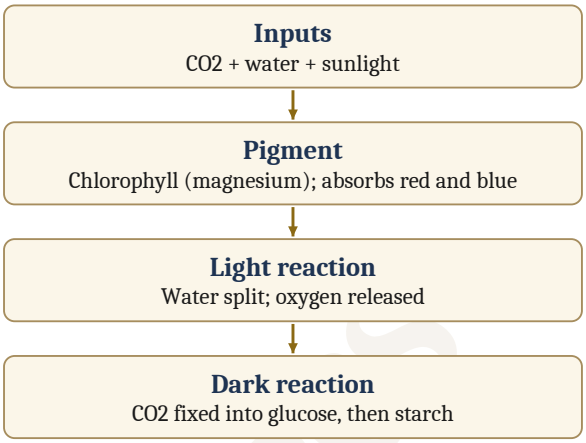
- ■ **Ethylene** ripens fruit (ethrel releases it) 6× 1996, 2007, 2008, 2017, 2023, one-liner
- ■ **Chlorophyll** has **magnesium**; photosynthesis needs **CO2**, releases O2 **from water** 5× 2011, 2012, 2013, 2014, one-liner
- ■ **Xylem** carries **water**; **phloem** carries **food** 4× 2012, 2015, 2015, 2023
- ■ **Plant hormones**: **auxin-apical dominance, gibberellin-elongation, cytokinin-cell division, abscisic acid-drought and dormancy**; insulin is not one 6× 2011, 2013, 2015, 2016, 2016, 2021
- • **Hydroponics** = growing plants **without soil** 2× 2009, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

24.1 Photosynthesis

- Element in chlorophyll: **magnesium** 2× 2012, 2013
- Plants make food by **photosynthesis** 1× 2013
- Sunlight becomes chemical energy; glucose turns into starch (A and R true, R explains A) 1× one-liner
- Essential gas: **CO2**; **oxygen** is not needed 2× 2011, one-liner
- Oxygen released comes from **water** 1× 2014
- Most of Earth's oxygen: **algae** (phytoplankton) 1× 2015
- Most effective light: **red** 1× 2014
- Artificial light **can** drive photosynthesis 1× 2008
- Best solar-to-chemical converter: **Chlorella** 1× 2014
- Does not add CO2 to the carbon cycle: **photosynthesis** 2× 2012, 2013

PHOTOSYNTHESIS IN ONE LINE



24.2 Transport in Plants

- **Phloem**: transports **food** (organic nutrients) 2× 2012, 2015
- **Xylem**: transports **water** 2× 2015, 2023

24.3 Mineral Nutrition

- Not a micronutrient: **magnesium** (a macronutrient) 1× 2019
- Plants take nitrogen as **nitrate** 2× 2016, 2021
- Yellowing and poor growth: lack of **nitrogen** 1× 2023
- More oil in mustard: **sulphur** 1× 2021
- Nodule formation in legumes: **boron** (per key) 1× 2019
- **Cuscuta** (amarbel) is a parasite; it feeds on the host's **stem** via haustoria, not leaves (A true, R false) 1× 2016
- **Crop logging**: plant analysis to judge **nutrient needs** 1× 2006

Nutrient	Role / deficiency
Nitrogen	Leaf growth; lack = yellowing (chlorosis)
Phosphorus	Roots, flowering
Potassium	Disease resistance, quality
Sulphur	Oil content (mustard)
Magnesium	Chlorophyll centre
Zinc	Khaira disease of rice
Boron	Flowering, nodules

24.4 Plant Hormones

QUANTUM IAS & PCS

- A plant hormone: **cytokinin** 3× 2015, 2016, 2021
- Not a plant hormone: **insulin** 1× 2011
- **Auxin-apical dominance, gibberellin-cell elongation, cytokinin-cell division, abscisic acid-growth inhibition** 1× 2016
- Drought tolerance: **abscisic acid** 1× 2013
- Correct (per key): weeping jelly from excess acid; **cytokinin** breaks dormancy (1 and 4) 1× 2013
- Ripening gas: **ethylene** (the key prints "ethane") 4× 1996, 2007, 2008, one-liner
- Hastens ripening: **ethrel** 1× 2023
- **Ripening-ethylene, baking soda-sodium bicarbonate, soft drink-CO2, quartz-sodium silicate** (per key) 1× 2017

24.5 Crop Practices

- **Gootee**: **vegetative propagation** (air layering) 1× 2021
- Seed changed every year: **hybrid seed** 1× 2010
- Crop rotation: a legume should be followed by a **non-legume** (only 2) 1× 2021
- Censer (pepper-pot) seed dispersal: **poppy** 1× 2010
- Clonal selection: **potato** 1× 2009
- Not correctly matched: key says **maize-tasseling** (*) 1× 2021
- **Hydroponics**: plants **without soil** 2× 2009, 2023

TRAP ALERT

- Oxygen from photosynthesis comes from **water**, not CO2.
- Magnesium is a **macro**nutrient; boron and zinc are micro.
- Abscisic acid = stress and dormancy; **gibberellin** is the main dormancy breaker.
- Fruit ripening gas is ethyl**ene**, not ethane.

UP LINK

UP link: **CIMAP** (Lucknow) and **NBRI** study medicinal and aromatic plant physiology; UP's **mustard** belt (Agra, Mathura, Etawah) is advised to use **sulphur** for higher oil content.

LAST-MINUTE 10

1. Chlorophyll = **magnesium**; red light most effective
2. O2 from **water**; algae make most O2
3. Chlorella = best solar converter
4. Xylem = water; phloem = food
5. Nitrate form of N; N lack = yellowing
6. Sulphur = mustard oil; boron = nodules
7. Cuscuta = parasite (amarbel)
8. Auxin = apical dominance; cytokinin = division; ABA = drought
9. Ethylene / ethrel = ripening
10. Hydroponics = no soil; gootee = air layering

Chapter 25

Economic Importance of Plants

Built from 35 UPPSC questions (2000–2023). Focus: improved crop varieties, biodiesel plants (Jatropha, Pongamia), caffeine and iodine, plant products and their parts, quinine and opium.

ASKED MOST · REVISE FIRST

- **Jatropha** = **biodiesel** plant; **Pongamia** (karanj) = leguminous petro-plant 4× 2004, 2010, 2021, 2021
- **Quinine** from **cinchona bark**; replaced by synthetic **chloroquine** 4× 2000, 2016, 2017, one-liner
- **Opium**: **latex of the unripe poppy fruit**; main constituent **morphine**; heroin and noscapine also from poppy 5× 2005, 2011, 2012, 2014, 2016
- **Seaweeds** are a major source of **iodine** 3× 2015, 2017, 2023
- **Caffeine** is in **tea and coffee** 2× 2016, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

25.1 Improved Varieties

- **Kanchan**: an improved variety of **amla** (Indian gooseberry) 1× 2015
- **Kashi Lalima**: red **okra** (IIVR, Varanasi) 1× 2021

- **Co 1148**: **sugarcane** 1× 2017
- First dwarf rice of India: **Jaya** 1× 2010
- Wrong pair: **linseed-Surya** 1× 2017

25.2 Oils, Biofuels and Industrial Products

- Green-synthesised nanoparticles for cancer research: **Parthenium** extract 1× 2004
- Peppermint oil is highest in the **leaf** 1× 2009
- **Litmus**: from **lichens** 1× 2010
- Biodiesel plant: **Jatropha** 3× 2004, 2021, 2021
- Leguminous petro-plant: **Pongam** (Pongamia) 1× 2010
- Petro-crops are rich in **hydrocarbons and lipids** (2 and 4) 1× 2009
- First biodiesel plant: **Andhra Pradesh** 1× 2004
- **Chilgoza**: seed of the **pine** 1× 2009
- Caffeine: **tea and coffee** both; tea's main stimulant 2× 2016, 2017
- Seaweeds: **iodine** 3× 2015, 2017, 2023

25.3 Spices and Plant Parts

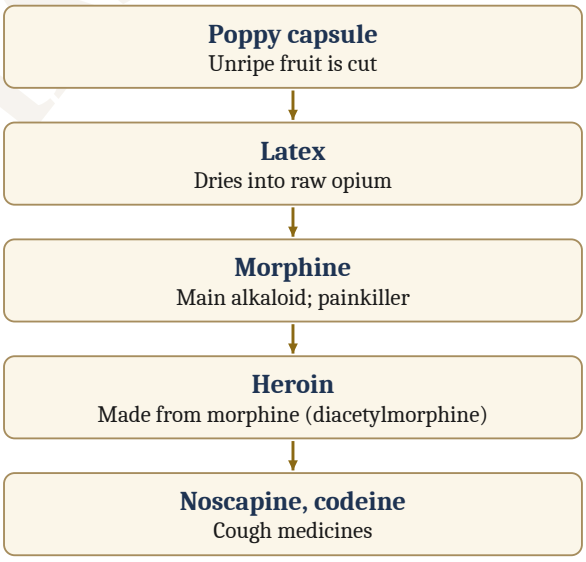
- **Opium-fruit, hing-root, rubber-stem, quinine-bark** 1× 2017
- **Asafoetida (hing)**: resin from the **stem base and roots** of Ferula 2× 2016, one-liner
- Main mosquito repellent plant: **neem** 1× 2004
- Tulsi's medicinal value: **phenols and flavonoids** 2× 2019, 2021
- Wrong pair: **rayon-modified starch** (rayon is regenerated **cellulose**) 2× 2019, 2022
- **Tapa cloth**: bark of plants of the **Moraceae** (mulberry) family 1× 2021
- **Beet-sugar, honey-glucose and fructose, cotton-cellulose, milk-lactose** (all four) 1× 2021

Product	Plant part
Opium	Latex of unripe poppy fruit
Hing (asafoetida)	Resin from root and stem base
Rubber	Latex from stem (bark)
Quinine	Cinchona bark
Clove	Flower bud
Saffron	Stigma
Cinnamon	Bark
Chilgoza	Pine seed

25.4 Medicinal and Drug Plants

- **Quinine**: **cinchona bark** 2× 2016, one-liner
- **Arteether** (antimalarial): from a **seed plant** (Artemisia annua) 1× 2005
- Synthetic drug that replaced quinine: **chloroquine** 1× 2000
- Hydroxychloroquine is fully **synthetic**; no plant fits (*) 1× 2016
- **Noscapine**: from **poppy** 1× 2012
- **Heroin**: from **opium poppy** 1× 2005
- Opium: **latex of the unripe fruit** 2× 2011, 2014
- Chief constituent of opium: **morphine** 1× 2016

FROM POPPY TO DRUGS



TRAP ALERT

- Opium comes from the **fruit**, not the seed or flower.
- Rayon is made from **cellulose**, not starch.
- Hing is a **resin** from roots and stem base, not a seed.
- Litmus = lichen; caffeine = tea **and** coffee.

QUANTUM IAS & PCS

UP LINK

UP link: **Kashi Lalima** (red okra) comes from **IIVR, Varanasi**; **Kanchan** amla was bred at **Narendra Deva Agricultural University** (Kumarganj, Ayodhya); **Barabanki** and **Ghazipur** are licensed **opium** poppy areas, and the **Government Opium Factory** is at **Ghazipur**.

LAST-MINUTE 10

1. Kanchan = amla; Kashi Lalima = okra; Co 1148 = sugarcane
2. Jaya = first dwarf rice
3. Jatropha = biodiesel; Pongamia = leguminous petro-plant
4. Litmus = lichen; chilgoza = pine seed
5. Caffeine = tea and coffee; seaweed = iodine
6. Hing = root/stem resin; neem = mosquito repellent
7. Rayon = cellulose; tapa cloth = Moraceae
8. Quinine = cinchona bark; chloroquine replaced it
9. Arteether = Artemisia; HCQ = synthetic
10. Opium = unripe fruit latex; morphine main; heroin from it

Chapter 26

Viral, Bacterial and Parasitic Diseases

Built from 70 UPPSC questions (1990–2024). Focus: viral diseases (dengue, polio, AIDS, hepatitis, Nipah, Ebola), bacterial diseases (TB, tetanus, cholera), vaccines, malaria and other parasites.

ASKED MOST · REVISE FIRST

- **Polio** vaccine: **Jonas Salk**; polio virus spreads through **polluted food and water** 6× 1991, 1993, 1995, one-liner, one-liner, one-liner
- **AIDS**: a virus (**retrovirus**, RNA + protein); tested by **ELISA**; India's strain **HIV-1C** 7× 2004, 2004, 2007, 2008, 2010, one-liner, one-liner
- **DPT** = **diphtheria, pertussis (whooping cough), tetanus**; not polio 4× 1994, 2004, 2013, one-liner
- **Tuberculosis** is **bacterial**: **BCG** vaccine, **Mantoux** test 7× 1990, 2010, 2015, 2016, 2017, one-liner, one-liner
- **Malaria** = **protozoan** (Plasmodium; **P. vivax** commonest in India) 5× 1998, 2008, 2008, 2012, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

26.1 Basic Terms and COVID-19

- Worldwide outbreak: **pandemic** 1× 2015
- Disease-causing microbe: **pathogen** 1× one-liner
- COVID-19 pause caused by **SARS-CoV-2** 1× 2024
- mRNA vaccine: **Moderna** 1× 2023
- **Sinovac**: **whole (inactivated) virus** vaccine 1× 2022

26.2 Mosquito-borne Viral Diseases

- Dengue lowers **platelets** 1× 2012
- Dengue mosquito: **Aedes** 1× 2022
- Not spread by tiger mosquitoes (Aedes): **Japanese encephalitis** (spread by **Culex**) 1× 2013
- Japanese encephalitis: a **virus** 1× one-liner

Mosquito	Disease
Aedes (tiger mosquito)	Dengue, chikungunya, Zika, yellow fever
Anopheles (female)	Malaria
Culex	Filaria, Japanese encephalitis

26.3 Other Viral Diseases

- **Hydrophobia** (rabies): **virus** 1× 2008
- Swelling of brain and spinal membranes: **meningitis** 1× 2008
- Infant diarrhoea deaths: **rotavirus** 1× 2010

QUANTUM IAS

- Polio virus enters through **polluted food and water** 2× 1993, one-liner
- Polio vaccine: **Jonas Salk** (injected); Sabin = oral 4× 1991, 1995, one-liner, one-liner
- Not viral: **tuberculosis** 1× 2017
- Common cold: **rhinovirus** 1× 1996
- **Hepatitis B**: a **DNA virus**; hits the **liver** 2× 2011, one-liner
- AIDS: caused by a **virus** (Acquired Immunodeficiency Syndrome) 2× 2010, one-liner
- HIV: **RNA + protein**; a **retrovirus** 2× 2008, one-liner
- Dominant strain in India: **HIV-1C** 1× 2004
- HTLV-III (HIV) discovered by **Robert Gallo** 1× 1999
- **ELISA** test: **AIDS** 2× 2004, 2007
- **Nipah** outbreak (2018): **Kerala** 1× 2018
- Smallpox is viral; isolation helps (A and R true, R explains A) 1× 1990
- Infectious cattle diseases: **FMD, anthrax, black quarter, cowpox** (all four) 1× 2003
- 2011 marked by the eradication of **rinderpest** 1× 2012
- Bird flu virus: **H5N1** 1× 2004
- **Ebola**: a deadly virus named after a **river** (Congo) 2× 2013, 2014

26.4 Bacterial Diseases

- Femur is the longest bone; cholera is bacterial (1 and 2); athlete's foot is **fungal** 1× 2008
- **Botulism** (food poisoning): **Clostridium** 2× 2013, one-liner
- **Haemophilus influenzae** (Hib): a **meningitis** in young children 1× 2022
- Bacterial disease: **tuberculosis** 2× 2010, one-liner
- Wrong pair: **genetic disease-tuberculosis** 1× 2016
- TB test: **Mantoux** test 1× 2015
- **Tetanus** = **lockjaw** 1× 2008
- Not from mosquitoes: **plague** (rat flea) 1× 1997
- Correct pair: **pneumonia-lungs** 1× 1996
- TB : lungs :: typhoid : **intestine** 1× 2023
- **Air-TB, water-cholera, contact-syphilis, wound-tetanus** 1× 1999
- Waterborne: **cholera** 1× 2015
- **BCG** vaccine: **TB**; give right **after birth** 2× 1990, one-liner
- Leprosy: **bacteria** 1× 2010
- Not viral: **typhoid** 1× 2008

Disease	Germ	Organ / note
Tuberculosis	Mycobacterium tuberculosis	Lungs; BCG, Mantoux
Typhoid	Salmonella typhi	Intestine; Widal test
Cholera	Vibrio cholerae	Intestine; waterborne
Tetanus	Clostridium tetani	Lockjaw; wounds
Diphtheria	Corynebacterium	Throat; toxoid vaccine
Leprosy	Mycobacterium leprae	Skin, nerves
Plague	Yersinia pestis	Rat flea

26.5 Vaccines and Antibiotics

- **DPT**: **diphtheria, whooping cough, tetanus** 3× 1994, 2004, 2013
- DPT does not protect against **polio** 1× one-liner
- Diphtheria vaccine: a **toxoid** 1× 2023
- **Sulpha drugs**: **antibacterial**; e.g. **Cibazole** 2× 2017, 2022
- Antibiotics are mostly from **bacteria** (actinomycetes) and fungi (per key) 1× 2021

26.6 Protozoan, Worm and Fungal Diseases

- Not common to malaria and dengue: **mosquito species** 1× 1998
- Protozoan disease: **malaria** 1× 2008
- About 65% of India's malaria: **P. vivax** (per key) 1× 2008
- **R21/Matrix-M**: malaria **vaccine** made by the **Serum Institute** 1× 2024
- Nematode disease: **filaria** 1× 1999
- Fungal (per key): **baldness** 1× 2016
- **Plague-bacteria, AIDS-virus, baldness-fungus, malaria-protozoa** 1× 2012
- **Peptic ulcer-bacteria, dengue-virus, elephant foot-worm, malaria-protozoa** 1× 2023

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- **Sleeping sickness**: **Trypanosoma** (tsetse fly), not mosquitoes **2×** 2007, one-liner
- Leander Paes's brain problem (2003): a **cyst** (neurocysticercosis) **1×** 2005

TRAP ALERT

- **Tetanus** = lockjaw; **typhoid** = intestine; **TB** = lungs (all bacterial).
- **Japanese encephalitis** and filaria = Culex; dengue = **Aedes**; malaria = Anopheles.
- Plague spreads by **rat fleas**, not mosquitoes.
- Salk = injected polio vaccine; Sabin = oral.

UP LINK

UP link: **Japanese encephalitis/AES** hit **Gorakhpur** and eastern UP for decades; JE vaccination and the **"Dastak"** campaign (from 2017) cut deaths sharply; UP was also the last big polio reservoir before India was declared **polio-free (2014)**.

LAST-MINUTE 10

1. Pandemic = worldwide; SARS-CoV-2 = COVID-19; Moderna = mRNA
2. Dengue = Aedes, low platelets; JE = Culex (Gorakhpur)
3. Polio = Salk; spreads via food and water
4. Hepatitis B = DNA virus, liver; common cold = rhinovirus
5. AIDS = retrovirus; ELISA; HIV-1C in India; Gallo
6. H5N1 = bird flu; Ebola = river name; Nipah = Kerala 2018
7. TB = BCG, Mantoux; tetanus = lockjaw; typhoid = intestine
8. DPT = diphtheria, pertussis, tetanus; diphtheria = toxoid
9. Malaria = protozoa (P. vivax); R21 = Serum Institute vaccine
10. Filaria = nematode; sleeping sickness = Trypanosoma

Chapter 27

Genetic, Deficiency and Other Diseases; Plant Diseases

Built from 92 UPPSC questions (1991–2024). Focus: haemophilia and colour blindness, deficiency diseases, pollution diseases (Minamata, Itai-itai, blue baby), radioisotopes in medicine, eye and heart terms, crop diseases.

ASKED MOST · REVISE FIRST

- **Radioisotope matching**: **arsenic-74-tumour, cobalt-60-cancer (gamma rays), iodine-131-thyroid, sodium-24-blood flow, phosphorus-32-leukaemia** **9×** 2001, 2002, 2003, 2010, 2010, 2012, 2012, 2013, 2016
- **Pollution diseases**: **Minamata-mercury, Itai-itai-cadmium, blue baby-nitrate (methaemoglobin), knock-knee-fluoride** **7×** 2008, 2021, 2023, 2023, one-liner, one-liner, one-liner
- **Goitre** = **iodine** lack; **anaemia** = iron; **khaira** (rice) = zinc; **scurvy** = vitamin C **5×** 2005, 2007, 2007, 2011, 2015
- **Potassium** regulates the heartbeat; its lack gives **low blood pressure** (per key) **5×** 2001, 2002, 2003, 2004, 2008
- **Haemophilia** = sex-linked **hereditary** disease; night blindness is not genetic **4×** 2015, 2017, one-liner, one-liner

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

27.1 Genetic Disorders

- Not genetic: **night blindness** **1×** 2017
- Hereditary: **haemophilia** ("royal disease", sex-linked) **3×** 2015, one-liner, one-liner
- **Bubble baby disease** (SCID): baby kept in a **germ-free plastic bubble** **1×** 1997
- Wrong pair: **vitamin A-colour blindness** (colour blindness is genetic) **1×** 2016
- Colour-blind man and normal woman: the trait appears in the **daughters' sons** **1×** 2009

27.2 Deficiency and Lifestyle Diseases

- Arthritis (gout): **uric acid** deposits **2×** 2013, one-liner
- Epidemic dropsy is from **argemone oil** in mustard oil, not rice husk **1×** 1998
- Hysteria (per key): **young women** **1×** 1991
- **Goitre**: lack of **iodine**; iodine is given for goitre **2×** 2007, 2007
- Polished rice: **beriberi** **3×** 2010, 2019, 2020
- **BMD** (Bone Mineral Density) test: **osteoporosis** **2×** 2014, one-liner
- Not knowing when to stop eating: **bulimia** **1×** 2016
- **Anosmia**: loss of **smell** **1×** 2011
- **Sleep apnoea**: breathing stops briefly and repeatedly in sleep **1×** 2014
- Heartbeat regulated by **potassium** **2×** 2004, 2008
- Potassium deficiency (per key): **low blood pressure** **3×** 2001, 2002, 2003
- Albumin in urine: **kidney** failure; prolonged fasting hurts the **kidneys** **2×** 1998, one-liner
- Scurvy treated with **gooseberry** (amla) **1×** 2011
- **Khaira-zinc, anaemia-iron, goitre-iodine, scurvy-vitamin C** **1×** 2005
- **Marasmus-starvation, kwashiorkor-protein lack, TB-bacteria, hepatitis B-virus** **2×** 2010, one-liner
- Iron deficiency: **anaemia** **1×** 2015
- Heavy drinking: **liver cirrhosis** **1×** 2008
- **Oral submucous fibrosis**: **gutka** **1×** 2014
- **Bright's disease**: **kidney** **1×** 1991
- Not controlled by vaccines: **diabetes** **1×** 2005
- Diabetes: **pancreas**; tracked by **HbA1c** **3×** 2019, 2024, one-liner
- Curry leaf claimed to help **diabetes** **1×** 2004
- **Cerebral palsy**: **small children** **1×** 2014
- Alzheimer's: **amyloid precursor protein** **1×** 2013
- Mad cow disease: **prions** **1×** 2009

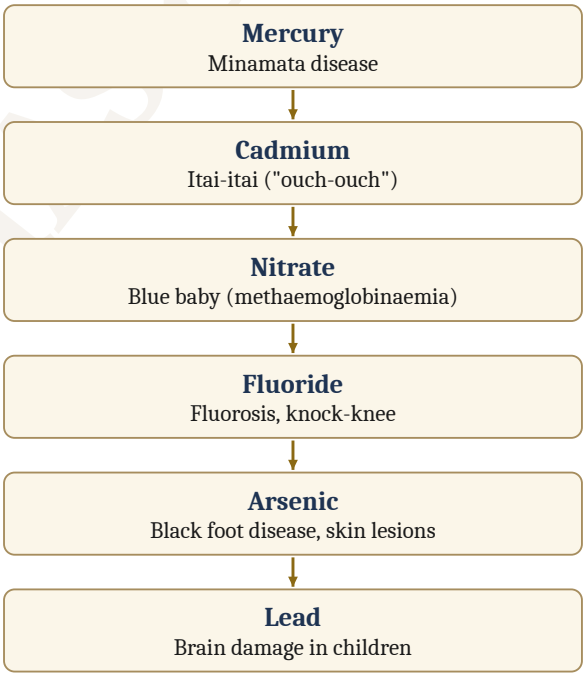
Disease	Cause
Goitre	Iodine deficiency
Anaemia	Iron deficiency
Kwashiorkor	Protein deficiency
Marasmus	Protein-energy starvation

Disease	Cause
Gout	Uric acid
Diabetes	Insulin lack (pancreas)
Cirrhosis	Alcohol
Mad cow disease	Prions

27.3 Pollution Diseases

- **Blue baby syndrome**: **methaemoglobin**, from **nitrate** in water **2×** 2021, one-liner
- **Knock-knee**: **fluoride** pollution **1×** 2023
- **Minamata**: **mercury** in fish and water (Japan, 1953) **3×** 2023, one-liner, one-liner
- **Itai-itai**: **cadmium** **1×** 2008

METAL AND CHEMICAL DISEASES



27.4 Radioisotopes and Cancer

- **As-74-tumour, Co-60-cancer, I-131-thyroid, Na-24-blood disorder** **3×** 2001, 2002, 2003
- Wrong pair: **gold-198-retinal disorders** **1×** 2010
- Speed of blood flow: **radio-sodium** **1×** 2012
- Cobalt-60 therapy: it emits **gamma rays** **1×** 2010
- Tumour detection: **arsenic-74** **2×** 2010, 2013
- Leukaemia control: **phosphorus-32** **2×** 2012, 2016
- **Leukaemia**: abnormal rise in **white blood cells** **1×** 2016
- Latest cancer therapy: **IMRT** **1×** 2017

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- **Interferon**: against **cancer** (and viruses) 1× 2010

27.5 Brain, Eyes and Medical Terms

- Least harmed by radiation: **brain** 2× 1999, 2011
- Brain diseases: **EEG** 2× 1993, one-liner
- **MRI** = Magnetic Resonance Imaging 2× 2008, 2012
- **Malaria-blood cells, filaria-lymph nodes, encephalitis-brain, leukaemia-bone marrow** 2× 1998, 2010
- **Astigmatism**: eye 1× 2004
- Myopic person sees **near** objects clearly 1× 2015
- Myopia corrected by a **concave lens** 1× 2003
- Long-sightedness: image forms **behind the retina** 1× 2017
- Blind spot and yellow spot: **retina** 1× 2008
- Light becomes nerve impulses at the **retina** 1× 1997
- Cataract: none of the options (it affects the **lens**) 1× 2017
- Pupil dilated with **atropine** 1× 2014
- Biological resurfacing: more than one joint fits (*) 1× 2014
- Cu-T common side effect: **bleeding** 1× 2014
- **Emphysema**: **lungs** 1× 2008
- Wrong pair: **aspirin-anaesthetic** (it is a painkiller) 1× 2017
- **Golden hour**: **heart attack** (and trauma) 1× 2014
- **Beta-blocker**: for **heart attack** 1× 2014
- Bypass graft vessel: **any of these** 1× 2014
- **Cath lab**: **cardiology** 1× 2014
- **CPR** = Cardiopulmonary Resuscitation 1× 2024
- Capsules: made of **gelatine** or **starch** 1× 2001

27.6 Plant Diseases

- Citrus canker = **bacteria**, red rot = **fungus**, potato black heart = **oxygen lack**; no code fits (*) 1× 2022
- **Rice-brown leaf spot, wheat-black rust, sugarcane-red rot, pigeon pea-wilt** 1× one-liner
- Wrong pair: **sugarcane-powdery mildew** 1× 2009
- Wrong pair: **rice-green hair** (per key) 1× 2015
- Maize chlorosis: **zinc** deficiency (white leaf tips) 2× 2006, one-liner
- **Karnal bunt**: **wheat** 1× 2016
- Rice **tungro** virus: **green leafhoppers** 1× 2009
- Not true: **false smut in barley** (it is a **rice** disease) 1× 2017
- Potato **black heart**: **oxygen deficiency** 1× 2008
- Okra **yellow vein mosaic**: spread by **whitefly** 2× 2002, one-liner
- Wrong pair: **paddy-wilt** 1× 2017
- **Spongy tissue**: **Alphonso** mango 1× 2002

Crop	Disease	Cause
Wheat	Black/brown rust, Karnal bunt	Fungus
Rice	Blast, brown spot, false smut	Fungus
Rice	Tungro	Virus (leafhopper)
Rice	Khaira	Zinc deficiency
Sugarcane	Red rot	Fungus
Potato	Late blight	Phytophthora
Potato	Black heart	Oxygen lack
Okra	Yellow vein mosaic	Virus (whitefly)
Citrus	Canker	Bacteria

TRAP ALERT

- **Minamata = mercury**; **Itai-itai = cadmium**; blue baby = **nitrate**.
- Cobalt-60 = **gamma** radiotherapy; I-131 = thyroid; P-32 = leukaemia.
- Cataract hits the **lens**; myopia sees **near** things; hypermetropia image is behind the retina.
- Khaira = **zinc**; maize white tips = zinc; potato black heart = **oxygen** lack.

UP LINK

UP link: **Ballia** and other eastern UP districts have **arsenic** in groundwater; **Unnao, Sonbhadra** and **Mathura-Agra** report **fluoride**; UP's sugarcane belt fights **red rot** (variety Co 0238 was hit hard), and **SGPGI Lucknow** runs nuclear medicine with I-131.

LAST-MINUTE 10

1. Haemophilia = sex-linked; colour blindness passes to daughters' sons
2. Gout = uric acid; goitre = iodine; khaira = zinc
3. Kwashiorkor = protein; marasmus = starvation; bulimia = overeating
4. Diabetes = pancreas, HbA1c; mad cow = prions
5. Minamata = Hg; Itai-itai = Cd; blue baby = nitrate; knock-knee = fluoride
6. Co-60 = gamma; I-131 = thyroid; P-32 = leukaemia; As-74 = tumour
7. EEG = brain; atropine dilates pupils; retina = blind and yellow spots
8. Golden hour = heart attack; cath lab = cardiology; CPR
9. Karnal bunt = wheat; tungro = leafhopper; red rot = sugarcane
10. Yellow vein mosaic = okra (whitefly); spongy tissue = Alphonso

Chapter 28

Genetic Engineering and Biotechnology

Built from 31 UPPSC questions (1999–2017). Focus: cloned animals, plasmids and phages, the Human Genome Project, stem cells, Bt cotton, golden rice and terminator seeds, DNA fingerprinting.

ASKED MOST · REVISE FIRST

- **Dolly** the **sheep** (1996) = first cloned mammal **2×** 2002, one-liner
- ■ **DNA fingerprinting**: paternity tests and crime; samples from blood, bone, hair, saliva **5×** 2002, 2004, 2006, 2008, 2010
- ■ **Golden rice** = transgenic rice rich in **beta-carotene** (vitamin A) **4×** 2000, 2012, one-liner, one-liner
- **Terminator technology**: seeds whose next generation is **sterile** **2×** 1999, one-liner
- **Bt cotton** carries a gene from a **bacterium** (Bacillus thuringiensis) **1×** 2010

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

28.1 Cloning

- First cloned organism: **Dolly**, a **sheep** **2×** 2002, one-liner
- **Garima II**: a **cloned buffalo** (NDRI Karnal) **1×** 2009
- First all-green glowing transgenic pigs: **Taiwan** **1×** 2010
- **Injaz**: first cloned **camel** **1×** 2008
- Tried with some success: **disease resistance, growth promotion, animal cloning** (1, 2, 3); not human cloning **2×** 2001, 2003
- Genetic engineering vector: **plasmid** **1×** 2015
- **Bacteriophages** infect bacteria and are used in genetic engineering (both) **1×** 2017

Clone	Animal	Where / year
Dolly	Sheep	Scotland, 1996
Samrupa, Garima	Buffalo	NDRI Karnal, 2009-10
Injaz	Camel	Dubai, 2009
Noori	Pashmina goat	Kashmir, 2012

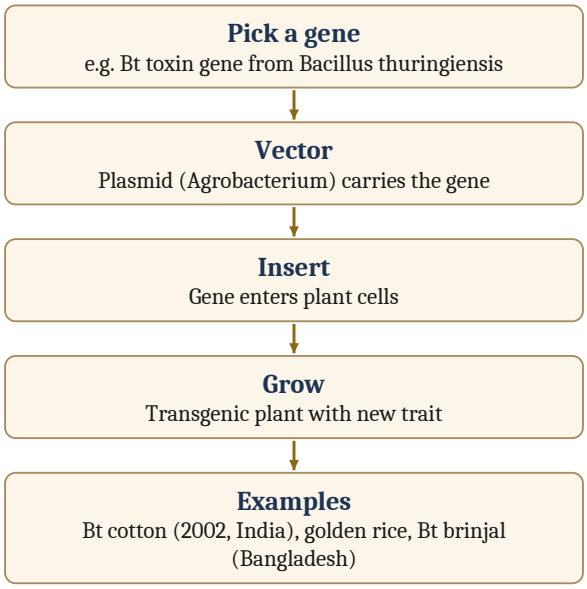
28.2 Genomes and Stem Cells

- **Human Genome Project**: identifying and **mapping human genes** and their sequence **1×** 2001
- Cells that become any cell type: **stem cells** **1×** 2006
- Non-controversial stem cells: **bone marrow** stem cells **1×** 2008
- **Plant Field Gene Bank, Banthara** (Lucknow): looks after **all the above** **2×** 2001, 2003

28.3 Transgenic Crops

- Bt cotton: gene from a **bacterium** **1×** 2010
- **Terminator technology**: seeds that give **sterile** next-generation seed **2×** 1999, one-liner
- **Orobanche** weed: **tobacco** fields **1×** 2014
- Transgenic plant: **golden rice** **1×** 2012
- ■ Golden rice helps consumers and farmers; its yellow comes from **beta-carotene** (A and R true, R explains A) **3×** 2000, one-liner, one-liner
- **Super rice**: **G.S. Khush** **1×** 2007

HOW A TRANSGENIC CROP IS MADE



28.4 Diagnosis and DNA Fingerprinting

- **Amniocentesis**: used to find **foetal sex** (now banned for that in India) 1× 2011
- **Genico** technology: advance **information on genetic diseases** 1× 1999
- DNA samples for crime: **blood, bone, hair, saliva** (all) 1× 2008
- **Biochip**: **RNA, DNA and protein** 1× 2008
- Paternity: **DNA fingerprinting** 2× 2002, 2010
- First crime solved by DNA (England): **1983** per key (the Pitchfork case, solved 1987-88) 1× 2006
- DNA fingerprinting used in **all** the areas 1× 2004
- Fingerprints on multi-coloured surfaces: **fluorescent powder** 1× 2013

28.5 Transgenics and Gene Therapy

- Not achieved by transgenics: **cloned animals** 1× 2000
- Gene therapy: several ways work (*) 1× 2004

TRAP ALERT

- Dolly = **sheep**; Injaz = camel; Garima = buffalo.
- Golden rice = **vitamin A** (beta-carotene), not iron.
- Bt gene comes from a **bacterium**, not a virus or fungus.
- Cloning is not a transgenic technique.

UP LINK

UP link: NBRI's **Plant Field Gene Bank** at **Banthara** (Lucknow) conserves plant genetic resources; the **National Bureau of Fish Genetic Resources** is also in **Lucknow**, and **IVRI Izatnagar** (Bareilly) works on animal biotechnology.

LAST-MINUTE 10

1. Dolly = first cloned mammal (sheep, 1996)
2. Garima II = buffalo; Injaz = camel; glowing pigs = Taiwan
3. Plasmid = vector; phages infect bacteria
4. Human Genome Project = gene mapping
5. Bone marrow = non-controversial stem cells
6. Bt cotton = bacterial gene
7. Terminator = sterile next-generation seed
8. Golden rice = beta-carotene; super rice = G.S. Khush
9. DNA fingerprinting = paternity, crime
10. Banthara = Plant Field Gene Bank (Lucknow)

Chapter 29

Biology Miscellaneous

Built from 133 UPPSC questions (1990–2024). Focus: health days and programmes, body chemistry, microbes in industry, medical tests and devices, animals and breeds, herbicides, and research institutes (many in UP).

ASKED MOST · REVISE FIRST

- **UP institutes: CDRI, CIMAP-Lucknow; IVRI-Bareilly; vegetable research-Varanasi; goat research-Makhdoom (Mathura); ALIMCO-Kanpur** 9× 1991, 2001, 2013, 2016, 2017, 2017, 2017, 2021, 2021
- **Health days: World AIDS Day 1 December, TB Day 24 March, Anti-Leprosy Day 30-31 January, Environment Day 5 June, Food Day 16 October** 5× 1992, 1999, 1999, 2002, 2009
- **Medical tests: EEG-brain, ECG-heart, EOG-eyes, EMG-muscle, audiogram-ear, mammogram-breast** 2× 2009, 2012
- **Endoscope** works by **total internal reflection** 5× 2008, 2009, 2010, 2014, 2014
- **Rayon** is regenerated **cellulose**, not modified starch 2× 2015, 2022
- **Three-chambered heart**: **frog** (amphibians) 2× 2013, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

29.1 Science News and Scientists

- Nobel Medicine 2024 (Ambros, Ruvkun): **microRNAs** 1× 2023
- **Birbal Sahni-palaeobotany, K.C. Mehta-plant pathology, P. Maheshwari-embryology, M.S. Swaminathan-Green Revolution** 1× 2021
- Science and Technology Policy 2003 aims: **all four** 1× 2008
- Not a science (Narlikar): **astrology** 1× 2018
- Science needs commitment to ideals and **honesty** (both) 1× 2024

29.2 Health Programmes and Days

- **Red Ribbon Express**: **AIDS awareness** 1× 2010
- **Mission Indradhanush**: immunising children against **seven** preventable diseases (now 12) 3× 2014, 2015, 2017
- **SANRAKSHA**: AIDS research and control project (Bengaluru) 1× 2014
- **NACO**: **AIDS** 1× 2006
- WHO Global Traditional Medicine Centre: **Jamnagar** 1× 2022
- Indian Biological Data Centre: **Haryana** (Faridabad) 1× 2023
- **1 December**: **World AIDS Day** 1× 2009
- Calendar order: **Anti-Leprosy (30 Jan), TB (24 Mar), Alzheimer's (21 Sep), AIDS (1 Dec)** 1× 2002
- **5 June-Environment, 1 December-AIDS, 5 September-Teachers, 16 October-Food** 1× 1992
- **TB-24 March, AIDS-1 December, Anti-Leprosy-30/31 January, Ozone-16 September** 1× 1999
- Minority Rights, AIDS, Animal, Food days: dates printed did not fit (*) 1× 1999
- Order: **Maritime (5 Apr), Health (7 Apr), Athletics (7 May), Red Cross (8 May)** 1× 2024
- Order: **Heritage (18 Apr), Light (16 May), Thyroid (25 May), No Tobacco (31 May)** 1× 2024

Day	Date
Anti-Leprosy Day	30 January
World TB Day	24 March
World Health Day	7 April
World Environment Day	5 June
Ozone Day	16 September
World Food Day	16 October
World AIDS Day	1 December

29.3 Body Chemistry

- **Cholesterol**: a **steroid** 2× 2005, 2021
- Trans fat lowers **HDL** (good cholesterol) 1× 2012
- Wrong pair: **mustard oil-lubricant** 1× 2023
- Yellow urine: **urochrome** 1× 2008
- **Antioxidants** protect cells from **free radicals** 1× 2012
- Silk, hair, nails, claws: **keratin** (per key; silk is fibroin) 1× 2009
- Lost in dehydration: **sodium chloride** 2× 1998, 2009
- Dehydrated person must not drink **sea water** 1× 2015
- Discomfort in humid heat: **sweat cannot evaporate** 1× 1995
- Most abundant organic compound: **cellulose** 2× 2012, 2014
- Fruit sweetness: **fructose** 1× 2017

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- Wrong pair: **pasteurisation-tea** 1× 2019
- **Acetic-vinegar, lactic-milk, butyric-butter, citric-lemon** 1× 2013
- **Bones-calcium phosphate, vinegar-acetic acid, souring milk-lactic acid, ripening-ethylene** 1× 2016
- **Vinegar-acetic, sour milk-lactic, bones-calcium phosphate, gastric juice-HCl** 1× 2016
- Wrong pair: **lactose-sour milk** (that is lactic acid) 1× 2015
- **Skin cancer-UV, noise-decibel, global warming-CO2, ozone hole-CFCs** 1× 2015
- Wrong pair: **antibiotic-quinine** (quinine is an antimalarial) 1× 2010

29.4 Microbes and Fermentation

- **Lactobacillus-curd, Rhizobium-nitrogen fixation, yeast-alcohol, Aspergillus-citric acid** 1× 2003
- Toddy fermentation: **Saccharomyces** (yeast) 3× 2010, one-liner, one-liner
- Microbe-made nanoparticle diode (India): **yeast** 1× 2002

29.5 Medical Devices and Medicine

- **EEG-brain, ECG-heart, EOG-eyes, EMG-muscle** 1× 2009
- **Audiogram-ear, ECG-heart, EEG-brain, mammogram-breast** 1× 2012
- Internal organs examined by **endoscope** 1× 2014
- Endoscope: **total internal reflection** 3× 2008, 2009, 2010
- What an endoscope is: **none of these** (per key) 1× 2014
- **Tetraplegia**: paralysis of **both arms and both legs** 1× 2015
- Electron beam therapy: **certain cancers** 1× 2008
- Dentist's mirror: **concave** 2× 2009, one-liner
- Foetal growth monitored by **ultrasonics** 1× 2006
- First heart transplant: **Christiaan Barnard** (1967) 2× 2004, 2021
- **Jarvik-7**: **artificial heart** 1× 2007
- **Pacemaker**: regulates heartbeat; natural one = **SA node** 2× 1999, 2007
- **Biojector**: painless **needleless injector** 1× 1997
- Wrong pairs: **antiseptic-aspirin, antiseptic-phenolphthalein** 2× 2003, 2006
- **Myoglobin-muscle, sarpagandha-tranquiliser, carcinoma-radiotherapy, haemoglobin-oxygen transport** 1× 2004
- **Bulimia-eating disorder, cholesterol-egg yolk, atropine-alkaloid, insulin-pancreas** 1× 2004
- Anaesthetics: **chloroform and nitrous oxide** (1 and 2) 2× 2002, one-liner
- **Cardiologist-heart, nephrologist-kidney, urologist-urinary tract, oculist-eye** 1× 2010
- Correct pair: **neurology-nervous system** 1× 2015
- Not banned for athletes: **colostrum** 1× 2003
- **Testosterone-hormone, codeine-sedative, caoutchouc-rubber, eugenol-clove oil** 1× 2008
- **Anthrax-biowarfare toxin, thalassaemia-defective gene, surrogacy-womb renting, transgenics-altering genomes** 1× 2003
- **Saheli**: CDRI's non-steroidal weekly **contraceptive pill** 1× 2017

29.6 Animals

- Dog that cannot bark: **dingo** (per key; basenji also) 1× 2010
- **Pheromones**: mostly **insects** 1× 2015
- Iodine from **Laminaria** (kelp) 2× 2004, one-liner
- **Cuckoo-no nest, lizard-sheds tail, blue whale-largest, chimpanzee-most intelligent** 1× 2004
- Stem fibre: **sunn hemp** 1× 2010
- Frog metamorphosis: **intestine shortens** 1× 2010
- Bull semen stored in **liquid nitrogen** 1× 2007
- High-yield cow: **Holstein** 1× 2006
- Wrong pairs: **sheep-Gir**, **sheep-Tharparkar** (both are cattle breeds) 2× 2021, 2021
- Three-chambered heart: **frog** (amphibians) 2× 2013, 2016
- Animals without a gall bladder: several fit (*) 1× 2004

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- First animal in space (key): a **dog** (Laika, 1957) 1× 2010
- **Angora wool**: **rabbit** 1× 2010
- Street-performing bear: **sloth bear** 1× 2008
- Largest brain for body size: **ant** 1× 2017
- Landmine detection: **bees** 1× 2002
- Cannot digest fibre well: **pig** 1× 1991
- Insect secretion: **lac** 1× 1996
- Light from living bodies: **bioluminescence** 1× 2014
- Mosquito-eating fish: **Gambusia** 1× 2011
- **Prallethrin**: mosquito repellent 1× 2017

29.7 Plants and Agriculture

- Herbicides spare crops as uptake rates differ (A and R true, R explains A) 1× 2002
- **2,4-D**: a **herbicide**; used in paddy 2× 1999, 2015
- Herbicide: **butachlor** 1× 2017
- Largest flower: **Rafflesia** 1× 1998
- Not needed for germination: **light** 1× 1990
- Mustard seed adulterant (per key): **poppy seeds**; argemone seeds cause dropsy 1× 2008
- Used in World War I: **mustard gas** 1× 2009
- Wax emulsion on fruit: **longer storage life** 1× 2010
- Clay holds most water, but its pores are **small** (A true, R false) 1× 2017
- **Eutrophication**: nutrient enrichment that **cuts dissolved oxygen** (both) 2× 2019, 2020
- Biodegradable: **wool** 1× 2022
- **Auxanometer**: rate of **growth** 1× 2010
- Tree age: **annual rings** 1× 2000
- Green Revolution seeds needed **more fertiliser and water** 1× 2001
- Paddy fields emit **methane** 1× 2006
- **Dapog** rice nursery: **Philippines** 1× 2010
- **Destructive Insects and Pests Act**: **1914** 1× 2023
- **Silicon carbide-artificial diamond, carbon fibre-aircraft, CO2-photosynthesis, CFC-refrigerator** 1× 2009
- Wrong pair: **rayon-modified starch** 2× 2015, 2022
- Wrong pair: **biological clock-biomass** 1× 2010

29.8 Research Institutes

- Indian Institute of Vegetable Research: **Varanasi**; not Gorakhpur 2× 2017, 2021
- **IVRI**: **Bareilly** (Izatnagar) 2× 2013, 2016
- **CDRI**: **Lucknow** 1× 1991
- **CIMAP**: **Lucknow** 1× 2021
- Wrong pair: **CECRI-Kolkata** (it is Karaikudi) 1× 2004
- High Security Animal Disease Laboratory: **Bhopal** 1× 2005
- Non-CSIR Lucknow institute: **BSIP** (DST) 1× 2005
- Institute for the mentally handicapped: **Hyderabad** (Secunderabad) 1× 2013
- **CDRI-Lucknow, Central Leprosy Institute-Agra, MNNIT-Allahabad, Sugar Technology-Kanpur** 1× 2001
- Wrong pair: **subtropical horticulture-Srinagar** (it is Lucknow) 1× 2017
- **ALIMCO** (artificial limbs): **Kanpur** 1× 2017
- **CDFD** (DNA fingerprinting): **Hyderabad** 1× 2013
- **FAO-Rome, Agmark lab-Nagpur, rice institute-Cuttack, potato institute-Shimla** 1× 2009
- International Potato Centre: **Lima, Peru** 1× 2021
- Mismatch: **NBRI-Kolkata** (it is Lucknow) 1× 2005
- **NBPGR**: **New Delhi** 1× 2013
- Goat research: **Makhdoom, Mathura** 1× 2017
- First dinosaurium: **Hyderabad** (per key) 1× 2004

Institute	Location (UP)
CDRI, CIMAP, NBRI, IITR, BSIP	Lucknow
IVRI	Izatnagar, Bareilly
Indian Institute of Vegetable Research	Varanasi
Central Institute for Research on Goats	Makhdoom, Mathura
ALIMCO	Kanpur
National Sugar Institute	Kanpur
Central JALMA (leprosy) Institute	Agra

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TRAP ALERT

- **Gir** and **Tharparkar** are **cattle** breeds, not sheep.
- CECRI = **Karaikudi**; NBRI = **Lucknow**; vegetable research = **Varanasi**, not Gorakhpur.
- Quinine is an antimalarial, not an antibiotic; aspirin is a painkiller, not an antiseptic.
- Frog = 3-chambered heart; fish = 2; birds and mammals = 4.

UP LINK

UP link: Lucknow alone hosts **CDRI** (Saheli pill), **CIMAP**, **NBRI**, **IITR**, **BSIP** and **NBFGR**; add **IVRI** (Bareilly), **IIVR** (Varanasi), **CIRG** (Makhdoom), **ALIMCO** and **NSI** (Kanpur), and **JALMA** (Agra).

LAST-MINUTE 10

1. AIDS Day **1 Dec**; TB Day **24 Mar**; Anti-Leprosy **30 Jan**
2. Mission Indradhanush = child immunisation
3. WHO traditional medicine centre = **Jamnagar**
4. Cholesterol = steroid; trans fat lowers HDL; urochrome = urine colour
5. EEG brain, ECG heart, EMG muscle, EOG eyes
6. Endoscope = TIR; Barnard = first heart transplant; Jarvik-7 = artificial heart
7. Saheli = CDRI pill; Gambusia eats larvae
8. Frog = 3-chambered heart; Angora = rabbit; Holstein = high-yield cow
9. 2,4-D, butachlor = herbicides; Rafflesia = largest flower
10. CDRI, CIMAP = Lucknow; IVRI = Bareilly; IIVR = Varanasi; ALIMCO = Kanpur

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